Power BI

FUNDAMENTALS

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POWER BI

POWER QUERY

POWER PIVOT

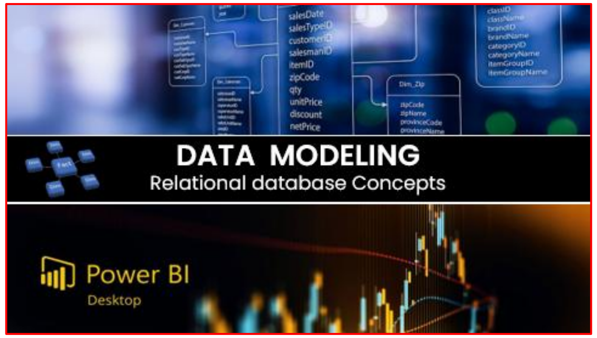
DAX

SUMMARY


WHAT IS POWER BI ?
Power BI
Desktop


POWER BI DESKTOP
Power BI
Desktop


POWER QUERY
Power BI
Desktop


DATA MODELING
Relational database Concepts
Power BI
Desktop


DAX
Data Analysis EXpressions
Power BI
Desktop


POWER VIEW
Visuals in Power BI
Power BI
Desktop


POWER BI SERVICE
Dashboards and sharing
Power BI
Desktop


POWER BI: BEST PRACTICES
Power BI
Desktop


POWER BI: PRICING & other products
Power BI
Desktop

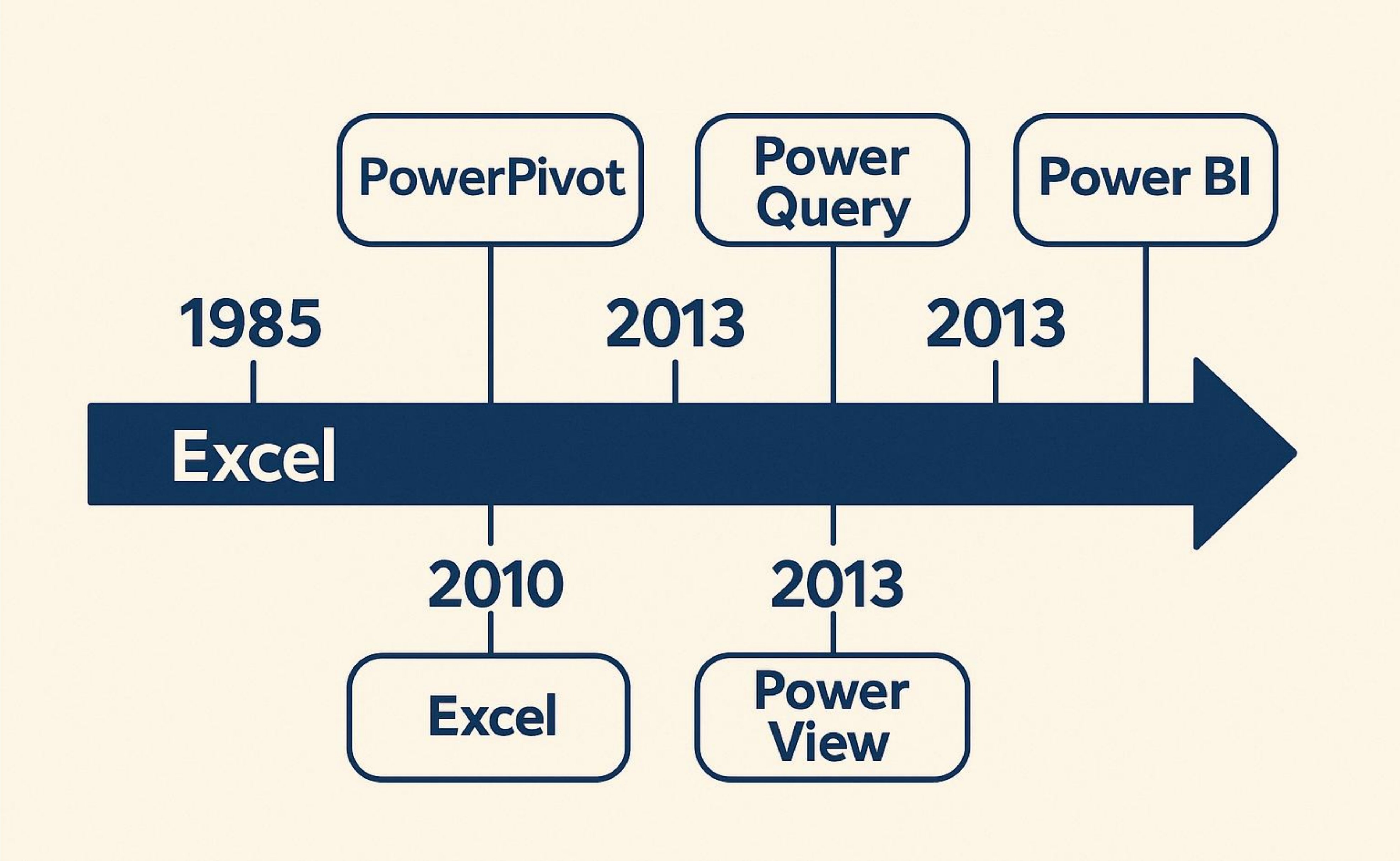
WHAT IS POWER BI ?

Power BI - BUSINESS INTELLIGENCE Made Easy

Power BI is Microsoft's data visualization and business intelligence tool that enables users to:

- Connect to various data sources (Excel, databases, cloud services)
- Transform and model data with ease
- Create interactive dashboards and reports
- Share insights across teams and devices

A little bit of history

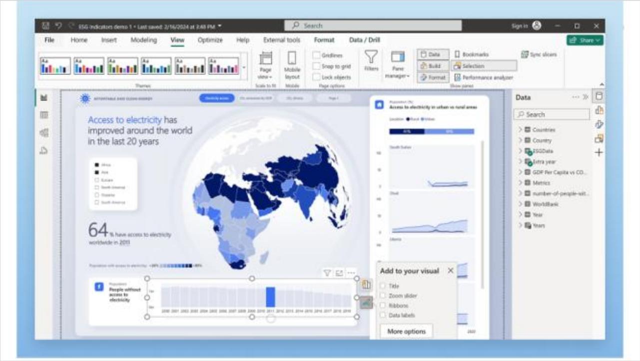


Power BI vs Excel

	Power BI	Excel
Data connectors available	200+	50+
Different visualizations available	400+	40+
Filtering at the report level or the page level	YES	NO
Mobile layout optimization available in power BI	YES	NO
Drill through on charts and cross-filtering	YES	NO
Formula language	DAX	Excel
R and Python languages integration	NATIVE	LIMITED
Versioning	PBI service	NO
Powerpoint interactive report or chart integration	YES	LIMITED
Power View	NATIVE	DISCONTINUED
ISO 8000 compatible	YES	NO

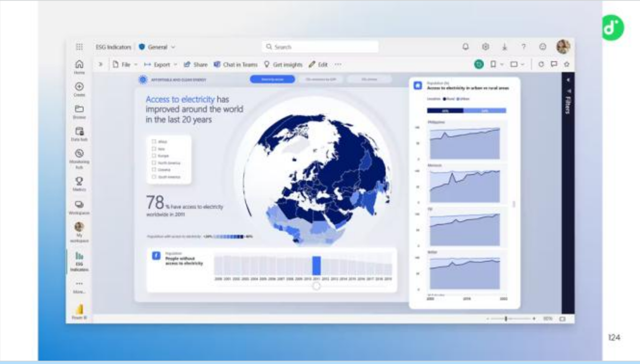
The 3 primary components of Power BI

Power BI Desktop



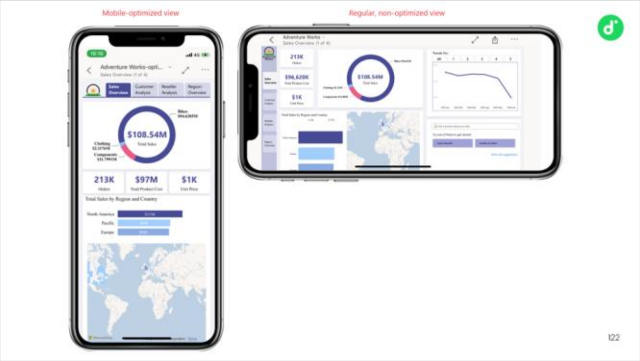
is the free development tool available to data analysts and other report creators

Power BI Service



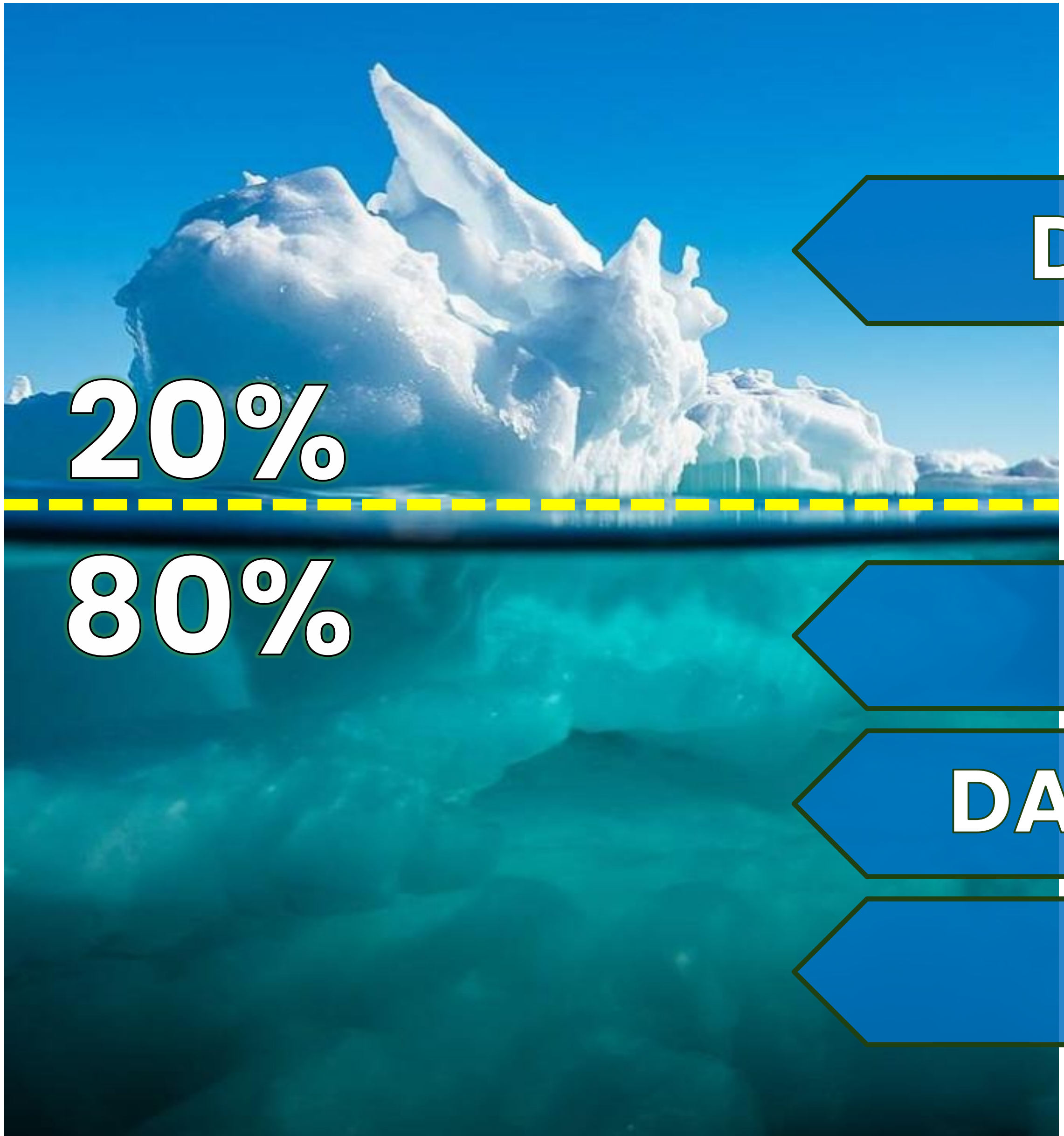
allows you to organize, manage, and distribute your reports and other Power BI items.

Power BI Mobile



allows consumers to view reports in a mobile-optimized format.

DATA ICEBERG MODEL



DATA VISUALIZATION

DATA MODELING

DATA TRANSFORMATION

DATA EXTRACTION

Key Features of Business Intelligence Tools

REPORTING

#flexible

#customizable

#persectives

DATA VISUALIZATION

#intuitive

#interactive

#understandable

DATA MINING

#patterns

#relationships

#trends

DASHBOARDS

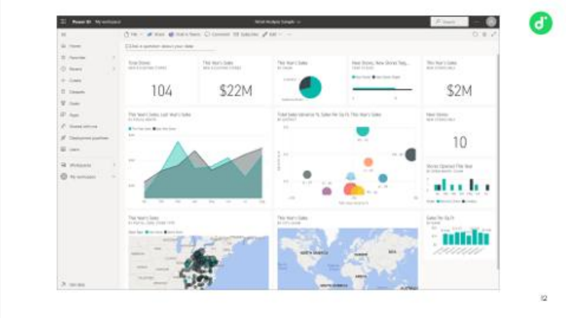
#realtime

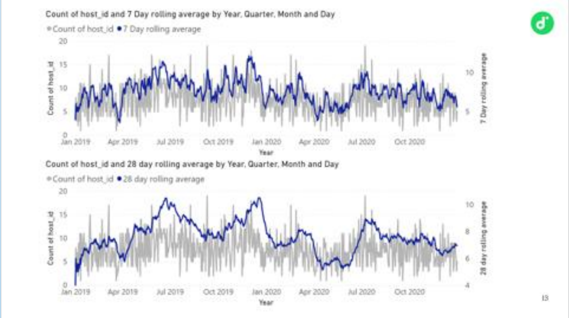
#kpi

#monitor

#performance

Category	This Year Sales Status	Average Unit Price	Last Year Sales	This Year Sales	This Year Sales Goal
010-Womens		$7.30	$2,680,662	$1,787,958	$2,680,662
020-Mens		$7.12	$4,453,133	$4,452,421	$4,453,133
030-Kids		$5.30	$2,726,892	$2,705,490	$2,726,892
040-Juniors		$7.00	$3,105,550	$2,930,385	$3,105,550
050-Shoes		$13.84	$3,640,471	$3,574,900	$3,640,471
060-Intimate		$4.28	$955,370	$852,329	$955,370
070-Hosiery		$3.69	$573,604	$486,106	$573,604
080-Accessories		$4.84	$1,273,096	$1,379,259	$1,273,096
090-Home		$3.93	$2,913,647	$3,053,326	$2,913,647
100-Groceries		$1.47	$810,176	$829,776	$810,176
Total		**$5.49**	**$23,132,601**	**$22,051,952**	**$23,132,601**





Power BI data flow

1. **Get** – Connect to source
2. **Prepare** – Power Query
3. **Model** – Power Pivot
4. **Visualize** – Power View
5. **Share** – Publish

Recommendations

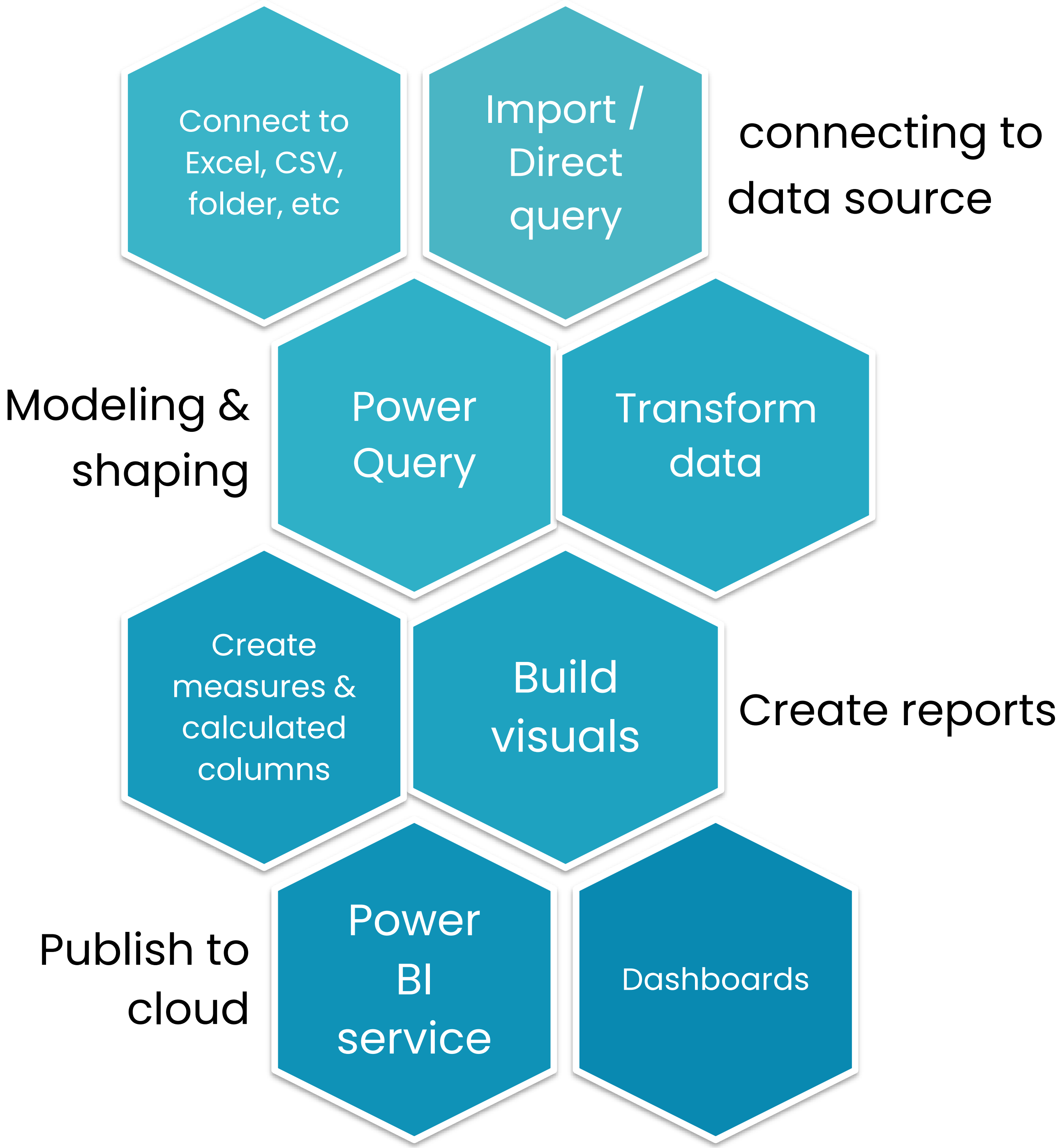
- Edit your PBI reports in the desktop app (not PBI service).
- Save regularly, even when using cloud stored files with auto save
- Do not neglect data modeling or data normalization
- Do not think like Excel
- Optimize queries and don’t hesitate to transform data
- Don’t underestimate DAX, use it …
- Choose a license that is suited for your company use

POWER BI DESKTOP

Power BI

DESKTOP

- Create semantic models and visuals to build reports.



Semantic model = connected data, transformations, relationships and calculations.

Introducing power BI desktop

Filters pane

Visualizations pane

Data pane

Quick access

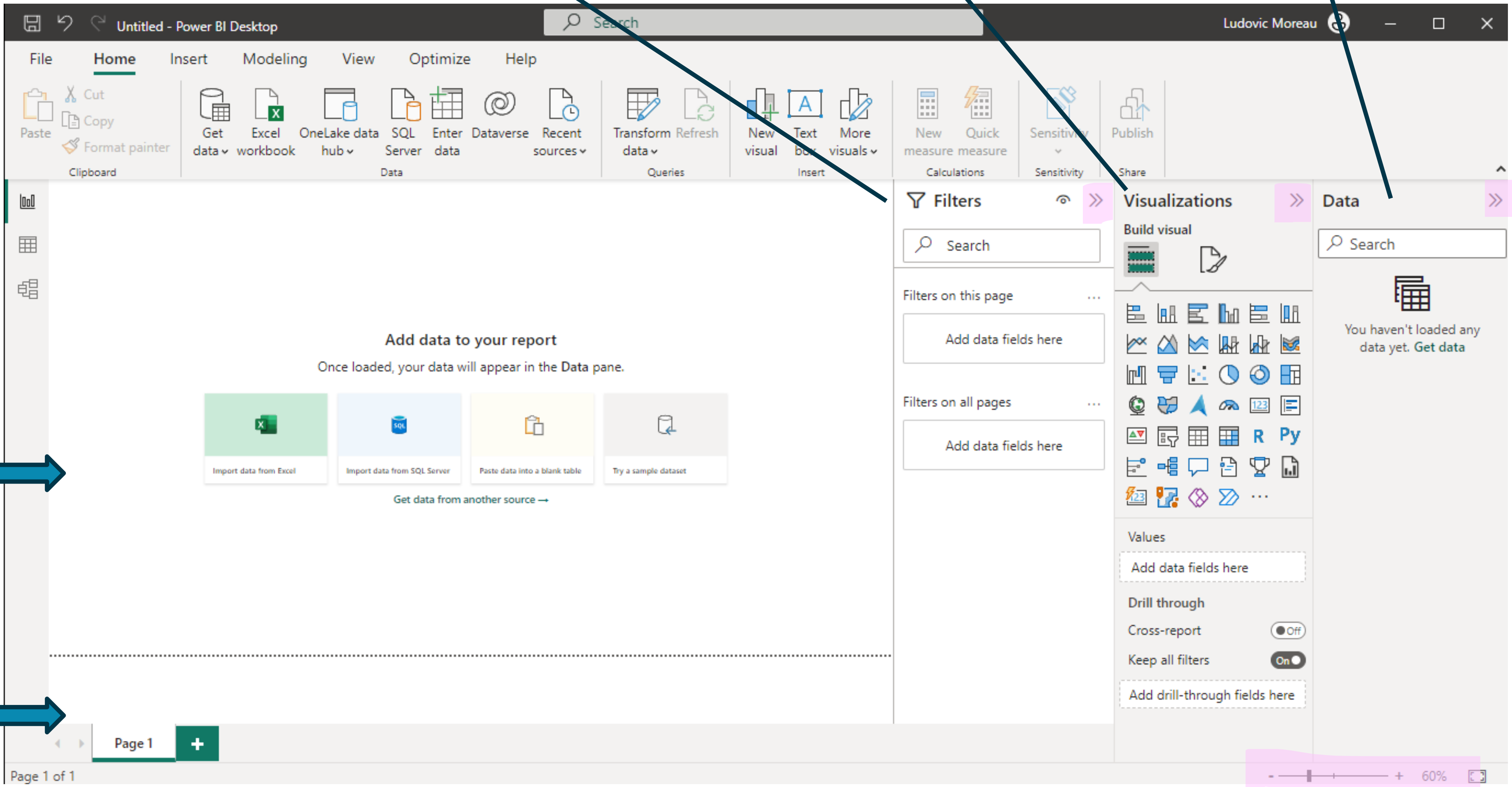
Ribbons

Report view
Data view
Model view

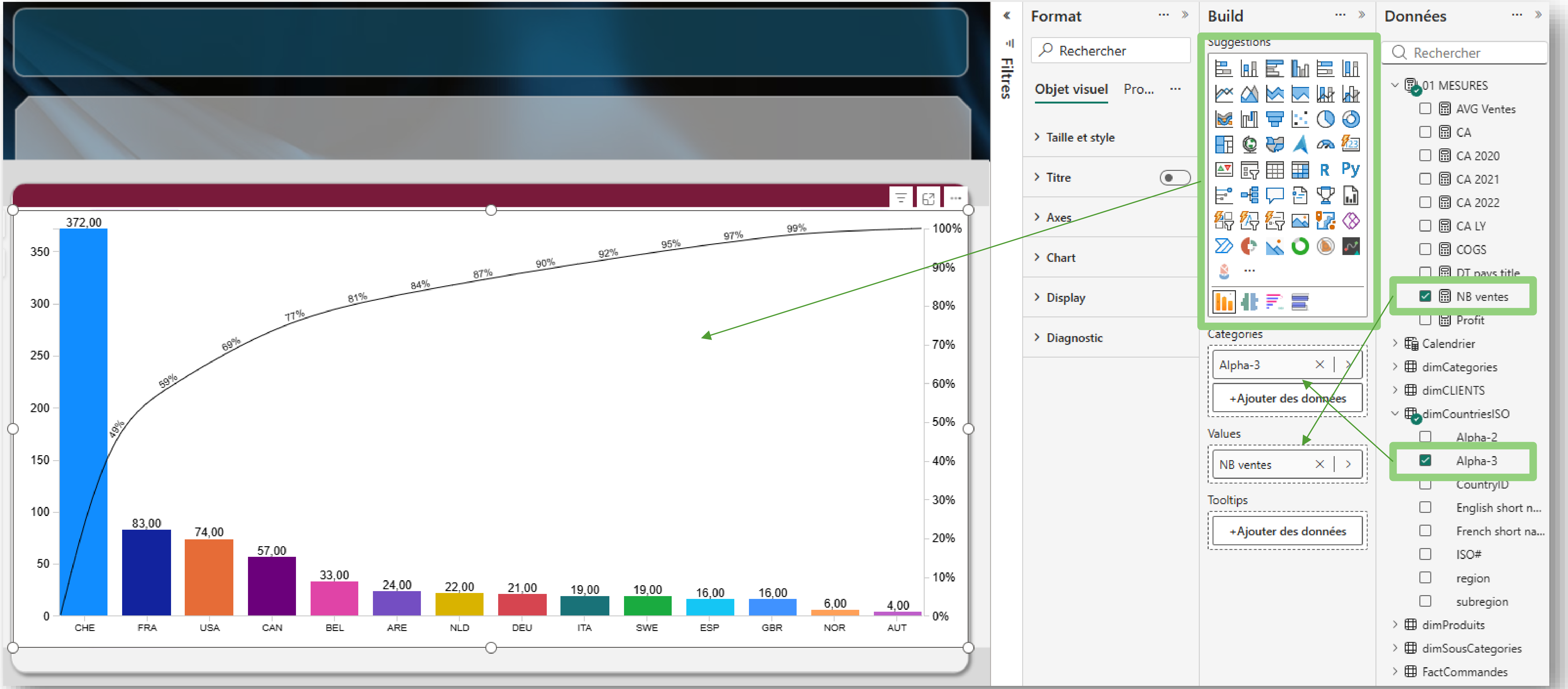
Canvas

Page tabs

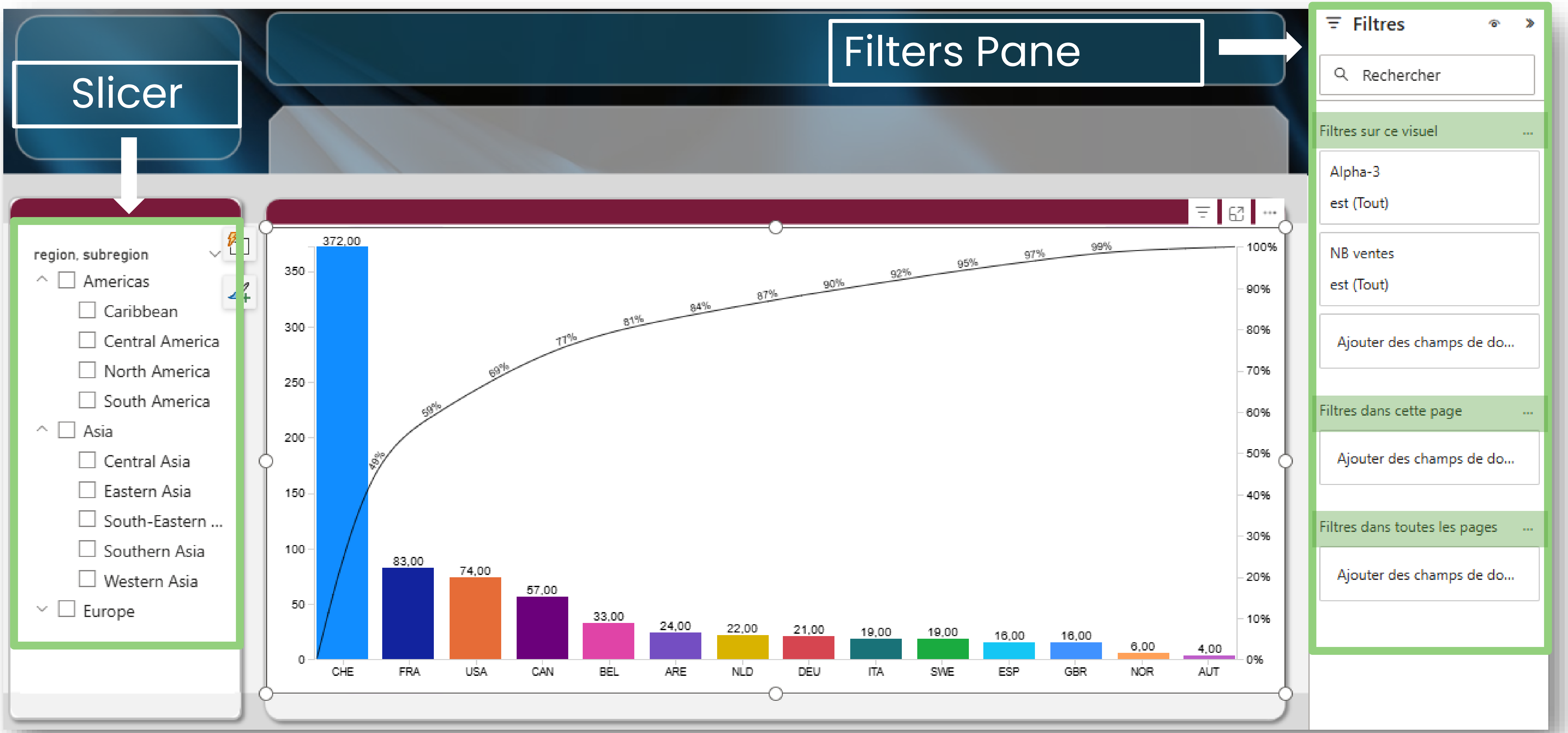
Status bar



Create a visual



Filter a visual



Creating a report (.pbix file)

Connect to data & populate the data model.

Add DAX tables, columns or measures.

Create and format visuals.

Insert Slicers for dynamic filtering.

Create Drillthrough and tooltip pages.

Apply conditional formatting.

Use bookmarks to create snapshots.

POWER QUERY

power query

- Seamlessly connect to multiple sources
- Easily reshape, cleanse, combine and merge data.
- Unpivot data.
- Create calendar tables.
- Add calculated columns.
- Agregates data
- Automate & reuse queries in other files.

Data Sources: File-based connectors

Excel (.xlsx, .xls)	**CSV / Text**	**XML / JSON**	**Folder**
Most frequent source for self-service BI	Flat file data dumps, exports from ERP/CRM systems	APIs, configurations, structured exports	Batch import of multiple files with the same structure

Data Sources: Database connectors

SQL Server	Azure SQL DB	Oracle	MySQL / PostgreSQL	Access	SAP HANA / BW
Core enterprise database, supports Import & DirectQuery	Cloud-based SQL platform	Legacy ERP & financial data	Common in open-source or web-based apps	Still used in legacy environments	Business-critical SAP system data

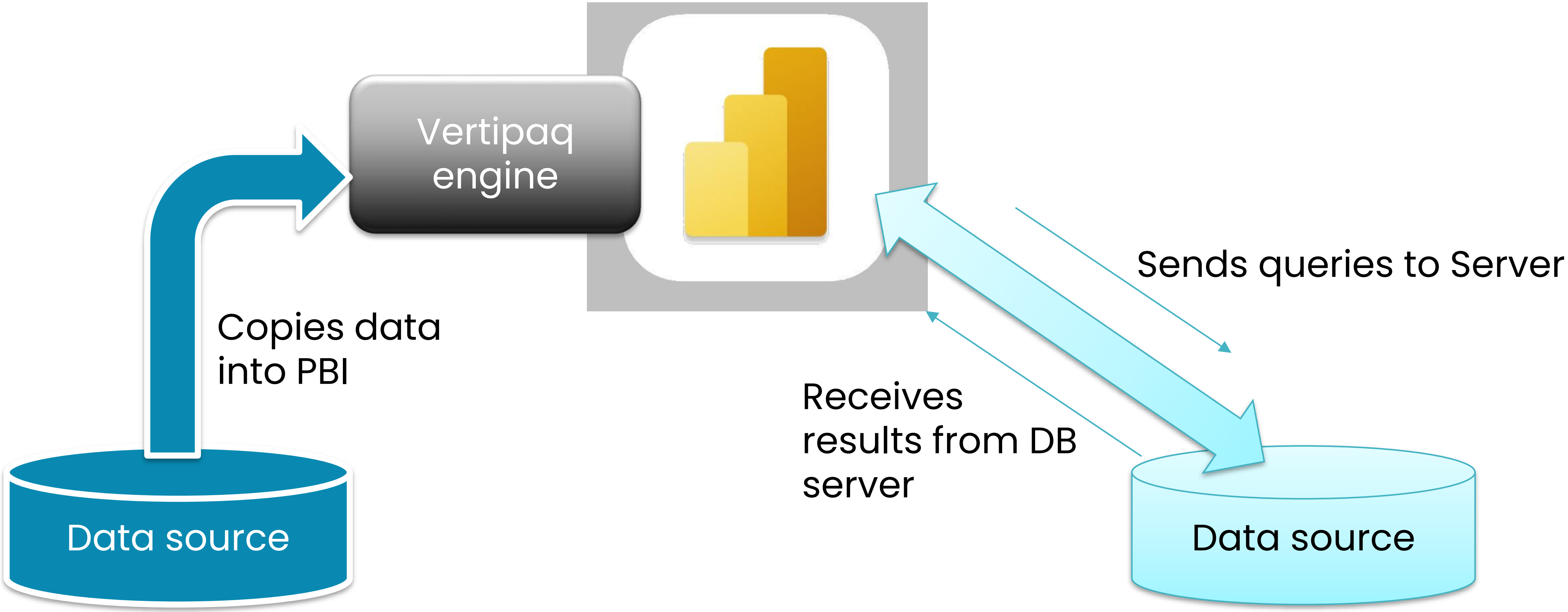
Data Sources: Database connectors

SharePoint / OneDrive	Google Analytics / BigQuery	Microsoft 365	Dynamics 365	Azure Data Lake / Blob	Salesforce
Excel/CSV in collaborative lists/folders. Auto-syncing cloud files.	Web & marketing data platforms	Email, Teams, Planner analytics	CRM/ERP sales, ops data	Cloud data storage and staging	CRM leads, accounts, opportunities

Data Sources: Database connectors

Web	**ODBC / OLE DB**	**R / Python**	**Blank Query**
Extract data from websites or APIs	Generic access to niche/proprietary systems	Embed scripts for advanced ETL	Manual entry of M logic and custom transformations

Import VS DirectQuery



Not available for all sources or for certain visuals.
Sensitive to remote server efficiency.

PBIX report file size with import method: 10’000 KB	file size with Directquery method: 10 KB

DirectQuery

Data sources that support DirectQuery.

(Sensitive to remote server efficiency.)

- SQL Server (on-prem)
- Azure SQL Database
- Azure SQL Managed Instance
- Azure Synapse Analytics
- Oracle
- Teradata
- SAP HANA
- Snowflake
- Google BigQuery
- Amazon Redshift
- PostgreSQL

You can also create a composite model to get the best of both worlds:

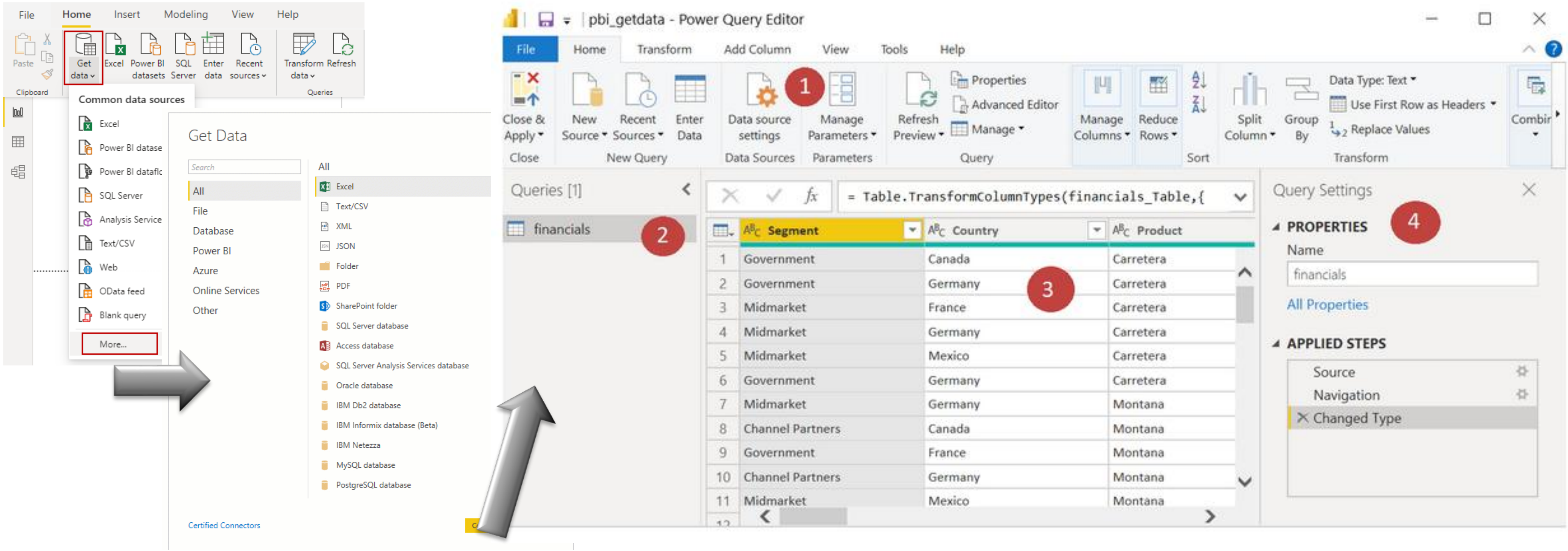
Large Fact Table -> DirectQuery

Small Dimension Tables -> Import

Avoid DirectQuery when:

- Users expect very fast interactions
- Reports contain complex DAX
- Many slicers, cross-filtering, drill-downs
- Source database is not optimized for analytics
- You need:
    - Complex calculated columns
    - Many time intelligence measures
    - High concurrency

Launch Power Query Editor



1. Ribbons (tools)
2. Queries
3. Data preview window
4. Properties & applied steps

Transforming data using Power query

power query

Columns

- Insert / delete
- Combine / split
- Calculate
- Unpivot
- Filter / Sort
- Rename

Rows

- Remove duplicates
- Remove blanks
- Filter out
- Group
- Transpose

Text

- Replace
- Format
- Fill blanks
- Trim
- Extract

Cleanup with Power Query

- Remove useless rows and columns
- Promote headers in first row
- Split columns (i.e. the Dimension col here).
- Check and correct data types if necessary.
- Import and convert different character encoded files (ANSI, UTF, etc)
- Load the Data into the Data model

<table>
<tr><th></th><th>Column1</th><th>Column2</th><th>Column3</th></tr>
<tr><td>1</td><td>Voici un tableau avec 50 noms de produits inform…</td><td>null</td><td>null</td></tr>
<tr><td>2</td><td>null</td><td>null</td><td>null</td></tr>
<tr><td>3</td><td>| Noms de produit | Poids (kg) | Dimensions …</td><td>null</td><td>null</td></tr>
<tr><td>4</td><td>|---------------------|------------|------------…</td><td>null</td><td>null</td></tr>
<tr><td>5</td><td>| Ordinateur portable | 2.5 | 35 x 24 x 2…</td><td>null</td><td>null</td></tr>
<tr><td>6</td><td>| Moniteur | 5.2 | 60 x 40 x 1…</td><td>null</td><td>null</td></tr>
<tr><td>7</td><td>| Clavier | 0.8 | 45 x 15 x 3…</td><td>null</td><td>null</td></tr>
<tr><td>8</td><td>| Souris | 0.1 | 10 x 6 x 3 …</td><td>null</td><td>null</td></tr>
<tr><td>9</td><td>| Imprimante | 7.8 | 45 x 35 x 2…</td><td>null</td><td>null</td></tr>
<tr><td>10</td><td>| Scanner | 3.2 | 30 x 20 x 1…</td><td>null</td><td>null</td></tr>
<tr><td>11</td><td>| Routeur | 0.4 | 20 x 15 x 5…</td><td>null</td><td>null</td></tr>
<tr><td>12</td><td>| Modem | 0.3 | 15 x 10 x 4…</td><td>null</td><td>null</td></tr>
<tr><td>13</td><td>| Disque dur externe | 0.2 | 10 x 8 x 2 …</td><td>null</td><td>null</td></tr>
<tr><td>14</td><td>| Cl? USB | 0.02 | 5 x 2 x 0.5…</td><td>null</td><td>null</td></tr>
<tr><td>15</td><td>| Carte graphique | 1.2 | 25 x 10 x 4…</td><td>null</td><td>null</td></tr>
<tr><td>16</td><td>| Carte m?re | 0.6 | 30 x 20 x 3…</td><td>null</td><td>null</td></tr>
<tr><td>17</td><td>| Processeur (CPU) | 0.1 | 5 x 5 x 2 …</td><td>null</td><td>null</td></tr>
<tr><td>18</td><td>| M?moire RAM | 0.05 | 8 x 2 x 0.5…</td><td>null</td><td>null</td></tr>
<tr><td>19</td><td>| Disque dur interne | 0.4 | 10 x 7 x 2 …</td><td>null</td><td>null</td></tr>
<tr><td>20</td><td>| SSD (Solid-State Drive) | 0.05 | 5 x 3 x 0.5 …</td><td>null</td><td>null</td></tr>
<tr><td>21</td><td>| cran tactile | 3.8 | 55 x 30 x 5 …</td><td>null</td><td>null</td></tr>
<tr><td>22</td><td>| Casque audio | 0.3 | 20 x 18 x 1…</td><td>null</td><td>null</td></tr>
<tr><td>23</td><td>| Haut-parleurs | 1.2 | 25 x 15 x 1…</td><td>null</td><td>null</td></tr>
<tr><td>24</td><td>| Webcam | 0.1 | 5 x 5 x 2 …</td><td>null</td><td>null</td></tr>
<tr><td>25</td><td>| Microphone | 0.2 | 10 x 5 x 3 …</td><td>null</td><td>null</td></tr>
<tr><td>26</td><td>| Tablette | 0.5 | 25 x 18 x 1…</td><td>null</td><td>null</td></tr>
<tr><td>27</td><td>| Smartphone | 0.2 | 15 x 7 x 0.…</td><td>null</td><td>null</td></tr>
<tr><td>28</td><td>| Smartwatch | 0.1 | 4 x 4 x 1 …</td><td>null</td><td>null</td></tr>
<tr><td>29</td><td>| couteurs sans fil | 0.05 | 3 x 2 x 1 …</td><td>null</td><td>null</td></tr>
<tr><td>30</td><td>| Routeur Wi-Fi | 0.3 | 15 x 10 x 4…</td><td>null</td><td>null</td></tr>
<tr><td>31</td><td>| Adaptateur Bluetooth| 0.02| 3 x 2 x 1 …</td><td>null</td><td>null</td></tr>
<tr><td>32</td><td>null</td><td>null</td><td>null</td></tr>
<tr><td>33</td><td></td><td>null</td><td>null</td></tr>
<tr><td>34</td><td>| Batterie externe | 0.2 | 10 x 5 x 2…</td><td>null</td><td>null</td></tr>
</table>

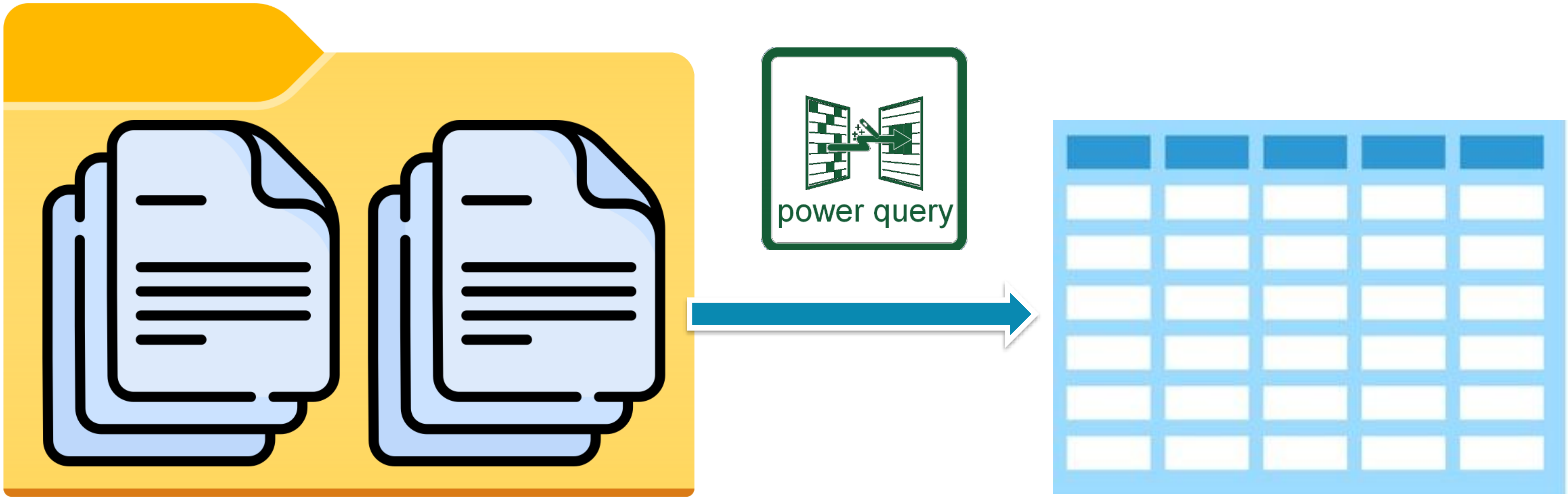
Ease of use vs M language

- Beginner: Use visual tools, quick undo with applied steps
  Advanced: Use M functions

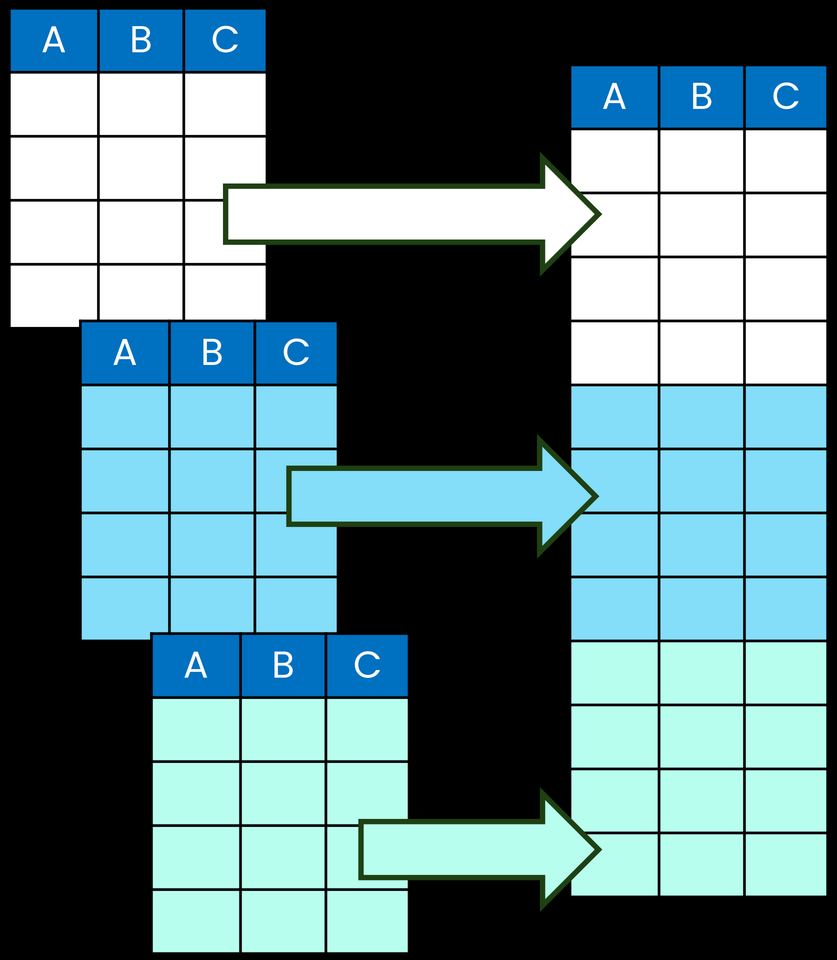
```
let
    Source = Excel.CurrentWorkbook(){[Name="Tableau5"]}[Content],
    #"Type modifié" = Table.TransformColumnTypes(Source,{{"Colonne1", type any}, {"Colonne2", type text}, {"Colonne3", type any}, {"Colonne4"
    #"Premières lignes supprimées" = Table.Skip(#"Type modifié",3),
    #"Rempli vers le bas" = Table.FillDown(#"Premières lignes supprimées",{"Colonne1"}),
    #"Colonnes fusionnées" = Table.CombineColumns(#"Rempli vers le bas",{"Colonne1", "Colonne2"},Combiner.CombineTextByDelimiter(":", QuoteSt
    #"Table transposée" = Table.Transpose(#"Colonnes fusionnées"),
    #"Rempli vers le bas1" = Table.FillDown(#"Table transposée",{"Column1"}),
    #"En-têtes promus" = Table.PromoteHeaders(#"Rempli vers le bas1", [PromoteAllScalars=true]),
    #"Supprimer le tableau croisé dynamique des autres colonnes" = Table.UnpivotOtherColumns(#"En-têtes promus", {":", "Succursalle:Descripti
    #"Fractionner la colonne par délimiteur" = Table.SplitColumn(#"Supprimer le tableau croisé dynamique des autres colonnes", "Attribut", Sp
    #"Type modifié2" = Table.TransformColumnTypes(#"Fractionner la colonne par délimiteur",{{"Attribut.1", type text}, {"Attribut.2", type te
    #"Colonnes renommées" = Table.RenameColumns(#"Type modifié2",{{":", "Statut"}, {"Succursalle:Description", "Mois"}, {"Attribut.1", "Ville
    #"Type modifié3" = Table.TransformColumnTypes(#"Colonnes renommées",{{"Mois", type date}})
in
    #"Type modifié3"
```

Consolidating from a folder

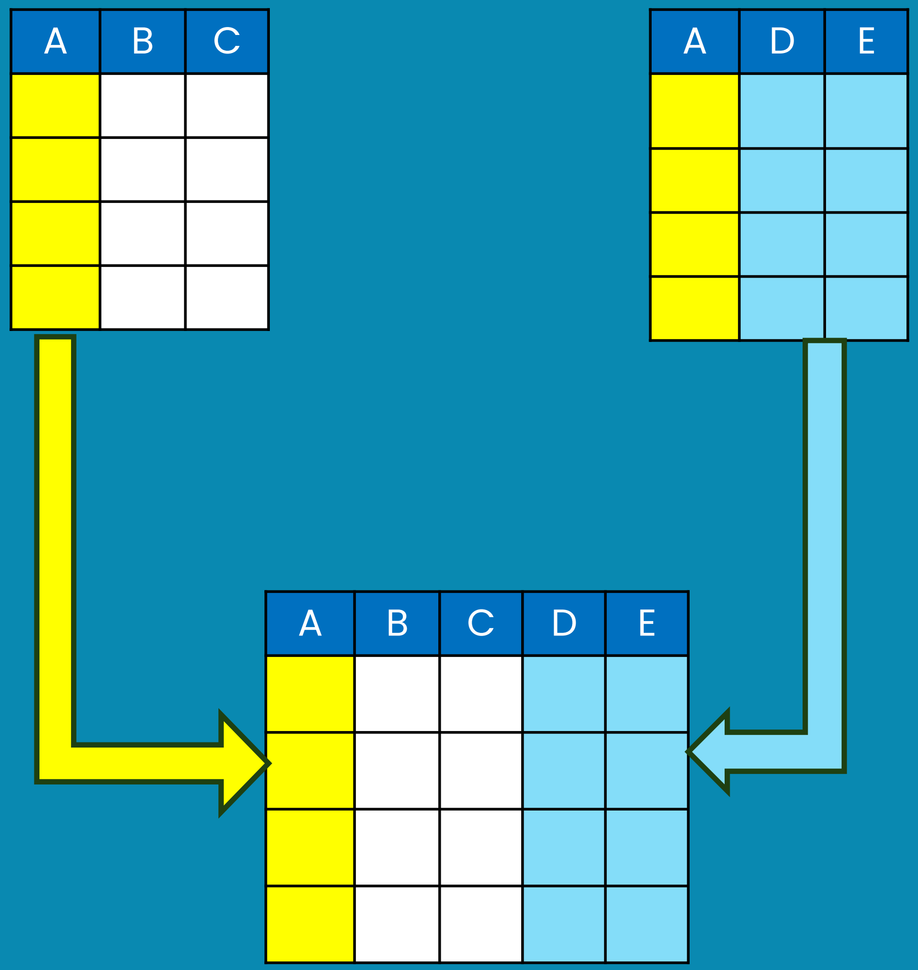
Transform data coming from files inside a folder and easily update with one click whenever new files are added to that folder.



APPENDING



MERGING



MERGE 2 QUERIES

SKU	souscatID
Valide 100%	Valide 100%
Erreur 0%	Erreur 0%
Vide 0%	Vide 0%
SKU-619955	5
SKU-367525	7
SKU-874528	16
SKU-167410	18
SKU-668353	25
SKU-772806	17
SKU-640119	29
SKU-154609	25
SKU-791801	35
SKU-129497	23
SKU-607728	41
SKU-239162	36
SKU-665945	16
SKU-894238	16
SKU-646568	27
SKU-249154	34
SKU-952912	16
SKU-438927	19
SKU-676138	26
SKU-642988	19
SKU-784630	42

Cost	Weight (kg)	SKU	Dimensions
Valide 100%	Valide 100%	Valide 100%	Valide
Erreur 0%	Erreur 0%	Erreur 0%	Erreur
Vide 0%	Vide 0%	Vide 0%	Vide
71	6.23	SKU-619955	17x18x14 cm
132	3.46	SKU-367525	43x36x4 cm
1309	7.28	SKU-874528	45x31x8 cm
954	8.67	SKU-167410	43x12x16 cm
703	2.77	SKU-668353	22x31x2 cm
834	6.65	SKU-772806	15x39x17 cm
230	5.51	SKU-640119	22x11x7 cm
1159	2.94	SKU-154609	35x22x7 cm
135	3.59	SKU-791801	59x10x10 cm
926	7.17	SKU-129497	51x33x8 cm
1241	7.9	SKU-607728	25x22x19 cm
1294	0.41	SKU-239162	54x13x18 cm
1440	2.12	SKU-665945	14x34x16 cm
1424	3.55	SKU-894238	56x25x8 cm
951	9.88	SKU-646568	18x21x17 cm
694	8.76	SKU-249154	48x10x17 cm
1073	6.63	SKU-952912	15x36x14 cm
453	7.76	SKU-438927	49x35x12 cm
839	0.78	SKU-676138	22x32x19 cm
1039	9.34	SKU-642988	43x37x14 cm
235	0.36	SKU-784630	28x14x2 cm
1054	4.7	SKU-968755	60x17x4 cm
1082	8.23	SKU-170209	19x33x7 cm
611	3.01	SKU-409176	52x16x15 cm
201	7.91	SKU-783285	50x21x13 cm
1346	8.94	SKU-353870	19x29x6 cm

MERGE 2 QUERIES

Cost	Weight (kg)	SKU	Dimensions	SKU	souscatID
Valide 100% Erreur 0% Vide 0%	Valide 100% Erreur 0% Vide 0%	Valide 100% Erreur 0% Vide 0%	Valide Erreur Vide	Valide 100% Erreur 0% Vide 0%	Valide 100% Erreur 0% Vide 0%
71	6.23	SKU-619955	17x18x14 cm	SKU-619955	5
132	3.46	SKU-367525	43x36x4 cm	SKU-367525	7
1309	7.28	SKU-874528	45x31x8 cm	SKU-874528	16
954	8.67	SKU-167410	43x12x16 cm	SKU-167410	18
703	2.77	SKU-668353	22x31x2 cm	SKU-668353	25
834	6.65	SKU-772806	15x39x17 cm	SKU-772806	17
230	5.51	SKU-640119	22x11x7 cm	SKU-640119	29
1159	2.94	SKU-154609	35x22x7 cm	SKU-154609	25
135	3.59	SKU-791801	59x10x10 cm	SKU-791801	35
926	7.17	SKU-129497	51x33x8 cm	SKU-129497	23
1241	7.9	SKU-607728	25x22x19 cm	SKU-607728	41
1294	0.41	SKU-239162	54x13x18 cm	SKU-239162	36
1440	2.12	SKU-665945	14x34x16 cm	SKU-665945	16
1424	3.55	SKU-894238	56x25x8 cm	SKU-894238	16
951	9.88	SKU-646568	18x21x17 cm	SKU-646568	27
694	8.76	SKU-249154	48x10x17 cm	SKU-249154	34
1073	6.63	SKU-952912	15x36x14 cm	SKU-952912	16
453	7.76	SKU-438927	49x35x12 cm	SKU-438927	19
839	0.78	SKU-676138	22x32x19 cm	SKU-676138	26
1039	9.34	SKU-642988	43x37x14 cm	SKU-642988	19
235	0.36	SKU-784630	28x14x2 cm	SKU-784630	42

ADD QUERIES

position	commandeID	clientID	produitID	NB vente	date commande	date livraison
1	20-0001	19	DRO-786	117	04.01.2020	11.01.2020
2	20-0002	46	ACC-700	24	10.01.2020	16.01.2020
3	20-0003	27	AUD-836	2	11.01.2020	16.01.2020
4	20-0004	46	AUD-703	116	13.01.2020	19.01.2020
5	20-0005	16	SMR-766	129	15.01.2020	18.01.2020
6	20-0006	5	DRO-714	101	17.01.2020	22.01.2020
7	20-0007					
8	20-0008					
9	20-0009					
10	20-0010					
11	20-0011					
12	20-0012					
13	20-0013					

position	commandeID	clientID	produitID	NB vente	date commande	date livraison
1	21-0250	35	DRO-691	135	03.01.2021	06.01.2021
2	21-0251	99	DRO-652	18	05.01.2021	12.01.2021
3	21-0252	28	NET-343	21	08.01.2021	15.01.2021
4	21-0253	90	SMR-668	139	09.01.2021	13.01.2021
5	21-0254	83	NET-835	52	10.01.2021	13.01.2021
6	21-0255	23	AUD-758	58	10.01.2021	15.01.2021
7	21-0256	93				
8	21-0257	7				
9	21-0258	14				
10	21-0259	79				
11	21-0260	77				
12	21-0261	90				

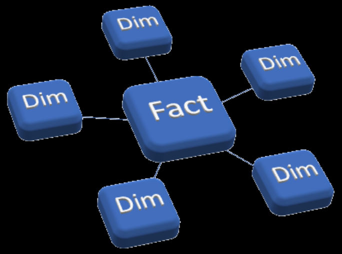
position	commandeID	clientID	produitID	NB vente	date commande	date livraison
1	22-0001	19	COM-384	31	08.01.2022	11.01.2022
2	22-0002	85	DRO-311	63	09.01.2022	14.01.2022
3	22-0003	31	SMR-317	57	10.01.2022	15.01.2022
4	22-0004	51	HAP-900	106	11.01.2022	14.01.2022
5	22-0005	32	NET-431	55	11.01.2022	16.01.2022
6	22-0006	61	SMR-887	41	11.01.2022	18.01.2022
7	22-0007	27	HAP-320	9	14.01.2022	17.01.2022
8	22-0008	31	SMR-342	113	15.01.2022	20.01.2022
9	22-0009	6	CAM-828	122	15.01.2022	19.01.2022
10	22-0010	92	WER-988	28	17.01.2022	20.01.2022
11	22-0011	81	DRO-835	6	18.01.2022	24.01.2022
12	22-0012	91	CAM-193	46	19.01.2022	26.01.2022

ADD QUERIES

commandeID	clientID	produitID	NB vente	date commande	date livraison
20-0001	19	DRO-786	117	04.01.2020	11.01.2020
20-0002	46	ACC-700	24	10.01.2020	16.01.2020
20-0003	27	AUD-836	2	11.01.2020	16.01.2020
20-0004	46	AUD-703	116	13.01.2020	19.01.2020
20-0005	16	SMR-766	129	15.01.2020	18.01.2020
20-0006	5	DRO-714	101	17.01.2020	22.01.2020
20-0007	36	SMR-672	41	19.01.2020	25.01.2020
20-0008	83	NET-279	31	19.01.2020	22.01.2020
20-0009	27	CAM-628	50	19.01.2020	24.01.2020
20-0010	18	COM-384	45	19.01.2020	25.01.2020
21-0254	83	NET-835	52	10.01.2021	13.01.2021
21-0255	23	AUD-758	58	10.01.2021	15.01.2021
21-0256	93	ACC-723	88	11.01.2021	17.01.2021
21-0257	7	ACC-997	96	12.01.2021	16.01.2021
21-0258	14	DRO-685	54	14.01.2021	20.01.2021
21-0259	79	AUD-871	146	17.01.2021	22.01.2021
21-0260	77	AUD-964	59	20.01.2021	26.01.2021
21-0261	90	AUD-964	52	20.01.2021	25.01.2021
22-0003	31	SMR-317	57	10.01.2022	15.01.2022
22-0004	51	HAP-900	106	11.01.2022	14.01.2022
22-0005	32	NET-431	55	11.01.2022	16.01.2022
22-0006	61	SMR-887	41	11.01.2022	18.01.2022
22-0007	27	HAP-320	9	14.01.2022	17.01.2022
22-0008	31	SMR-342	113	15.01.2022	20.01.2022
22-0009	6	CAM-828	122	15.01.2022	19.01.2022
22-0010	92	WER-988	28	17.01.2022	20.01.2022
22-0011	81	DRO-835	6	18.01.2022	24.01.2022
22-0012	91	CAM-193	46	19.01.2022	26.01.2022

Best Practices in POWER query

- Give clear and understandable names to your queries and steps
- Give a meaningful name to your columns as they will be part of your model
- Verify the data types in each column before importing
- Get rid of unused columns
- Use preferably the import method (instead of direct method)
- Do not load queries/tables that you do not need in your model
- Activate the formula bar



DATA MODELING

Relational database Concepts

RELATIONAL DATABASE

- Stores and provides access to data points that are related to one another.
- Based on the relational model, an intuitive, straightforward way of representing data in tables.
- Each row in the table is a record with a unique ID called the key.
- The columns of the table hold attributes of the data.

- The goal is to make it easy to establish relationships among data points in order to quickly and efficiently summarize data from the various linked entities in the data model.

Key concepts

Tables (Relations):

Data is organized into tables, with rows representing records and columns representing attributes.

Keys:

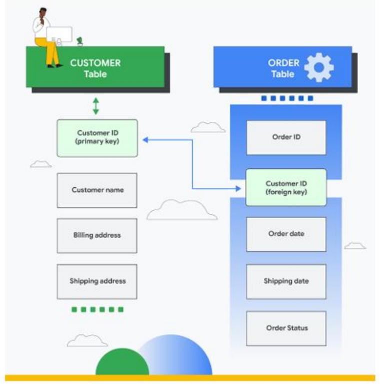
A **primary key** is a column that can reliably identify each individual record in a table. It can be an existing column (or combination of) from the table (a natural key) or from a new attribute containing a unique ID created for this purpose (a surrogate key).

A **foreign key** is a column in a table that refers to the primary key of another table, linking these two tables. It usually does not have an unique value.

Unique identifiers (primary keys) and relationships between tables (foreign keys) ensure data integrity and allow for efficient querying.

Relationships:

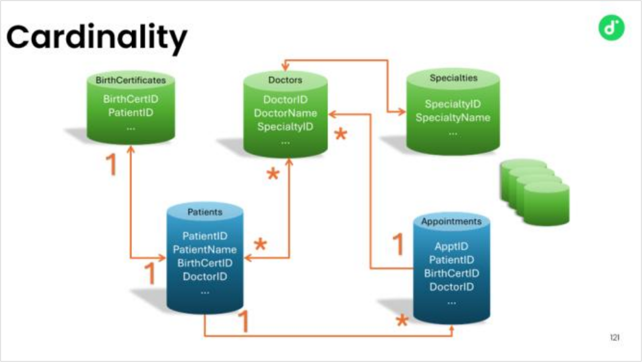
Relationships determine dependency between a primary key in one table and a foreign key in another table..



Cardinality

- The cardinality is the numerical relationship between rows of one table and rows in another. Common cardinalities include:

A many-to-many relationship.	A one-to-many relationship	A one-to-one relationship
* → *	1 → *	1 → 1



It is recommanded to avoid many-to-many relationships but to transform them through the creation of joining tables.

Fact & dimension tables

Fact tables

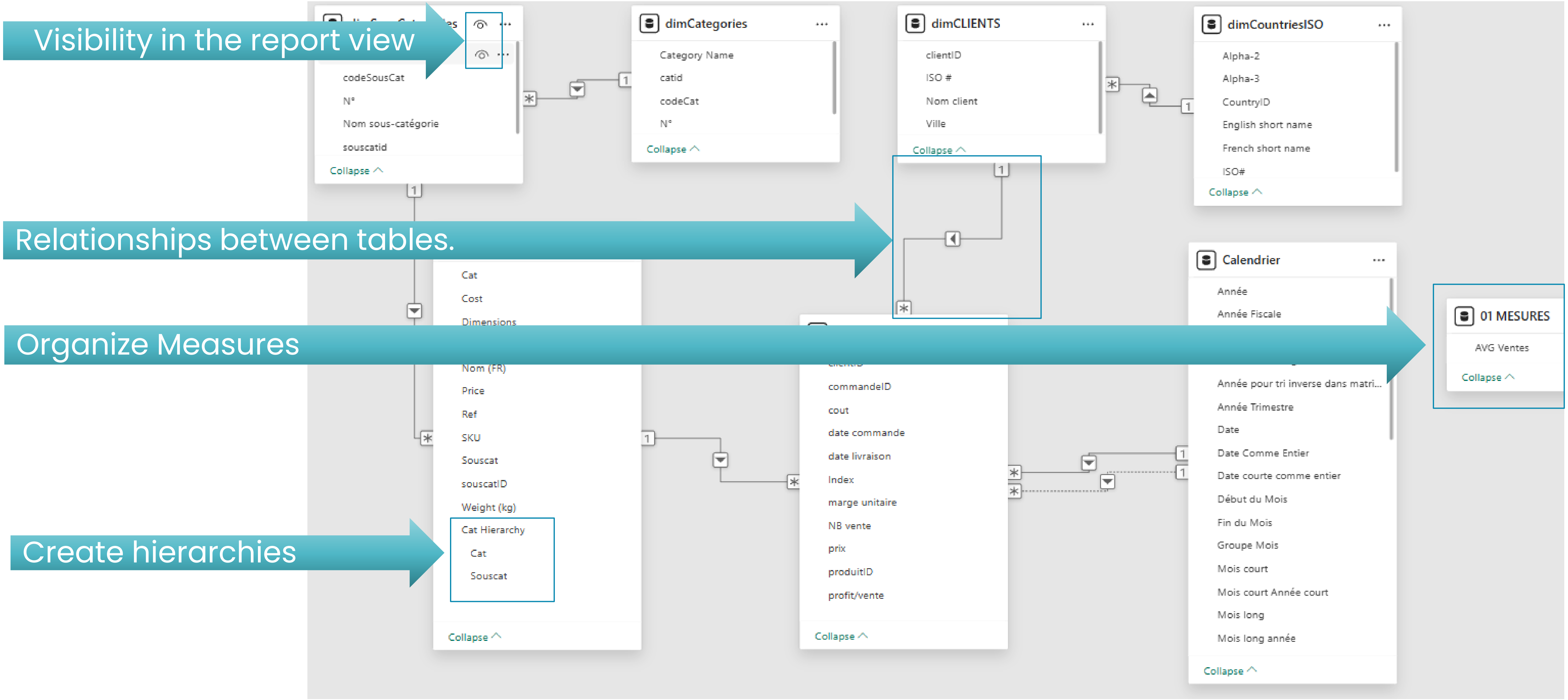
- Contains numeric data that will be aggregated in the visualizations. (i.e. fctSales, fctProfits). It needs to include the keys to the related dimensions tables.

Dimension tables

- Contains descriptive info that can be used to describe, group, categorize or filter the fact tables (i.e. dimCustomers, dimProducts, etc)
- It requires a key field (Primary key)

* Best practice: do not mix fact and dimension data in the same table (optimize the performance)

The data model



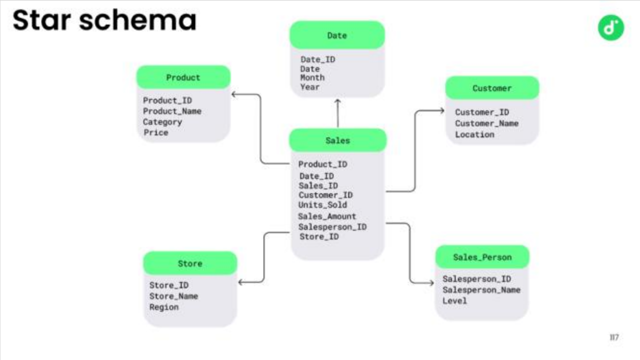
Types of Data Models

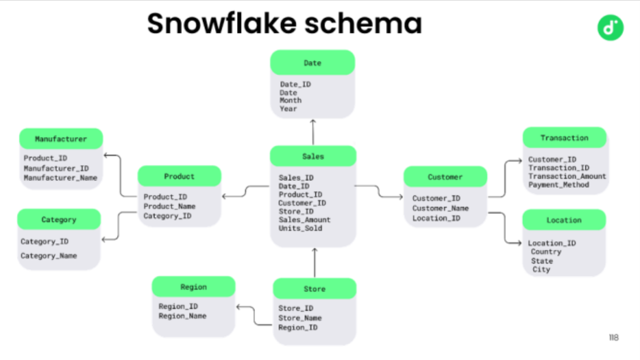
- The representation of tables with their attributes (columns) and relationships is called the physical data model.

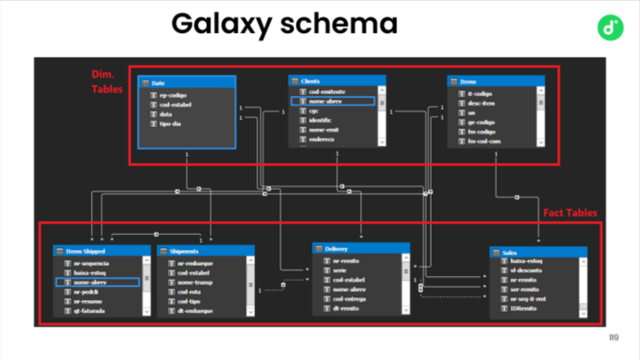
When every dimension table is linked directly to the fact table, we talk about star schema.

When some tables are connected to dimension tables, not directly to the fact table, this is a snowflake schema.

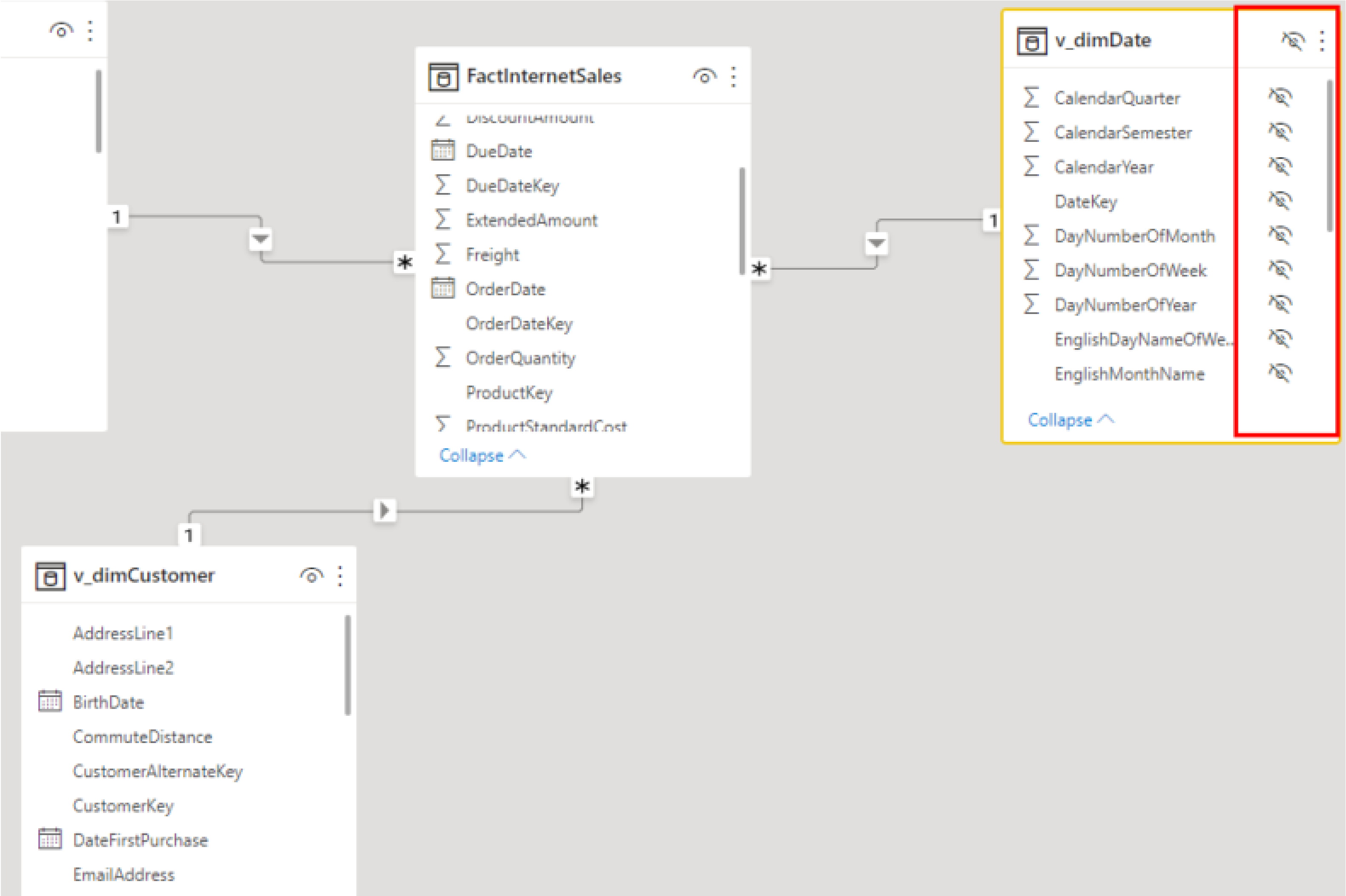
When several fact and dimension tables coexist in the data model, it is described as a galaxy schema (or fact constellation schema)





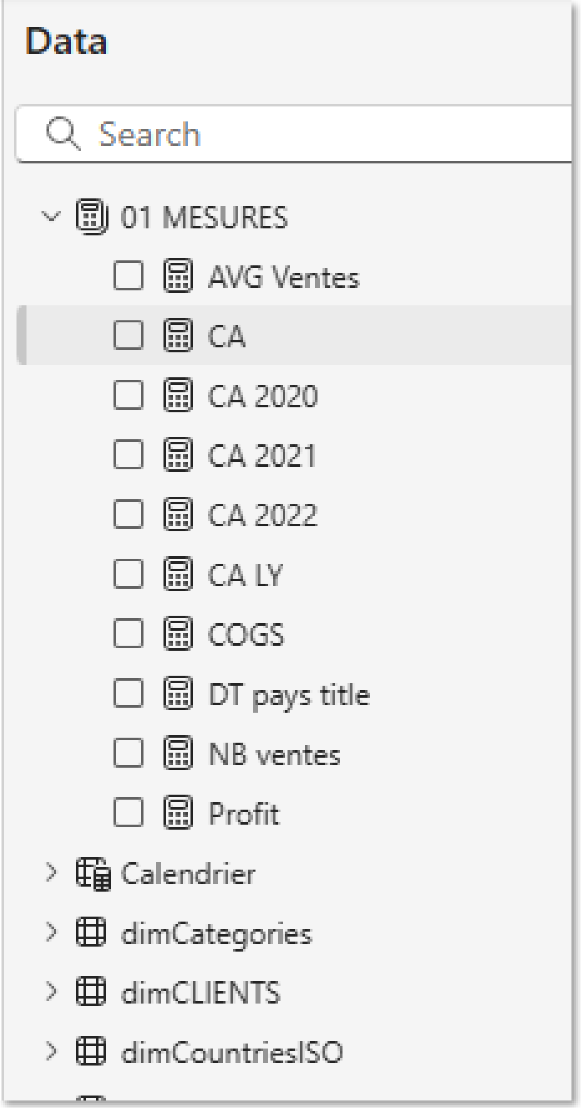


Hide fields or tables from the report view

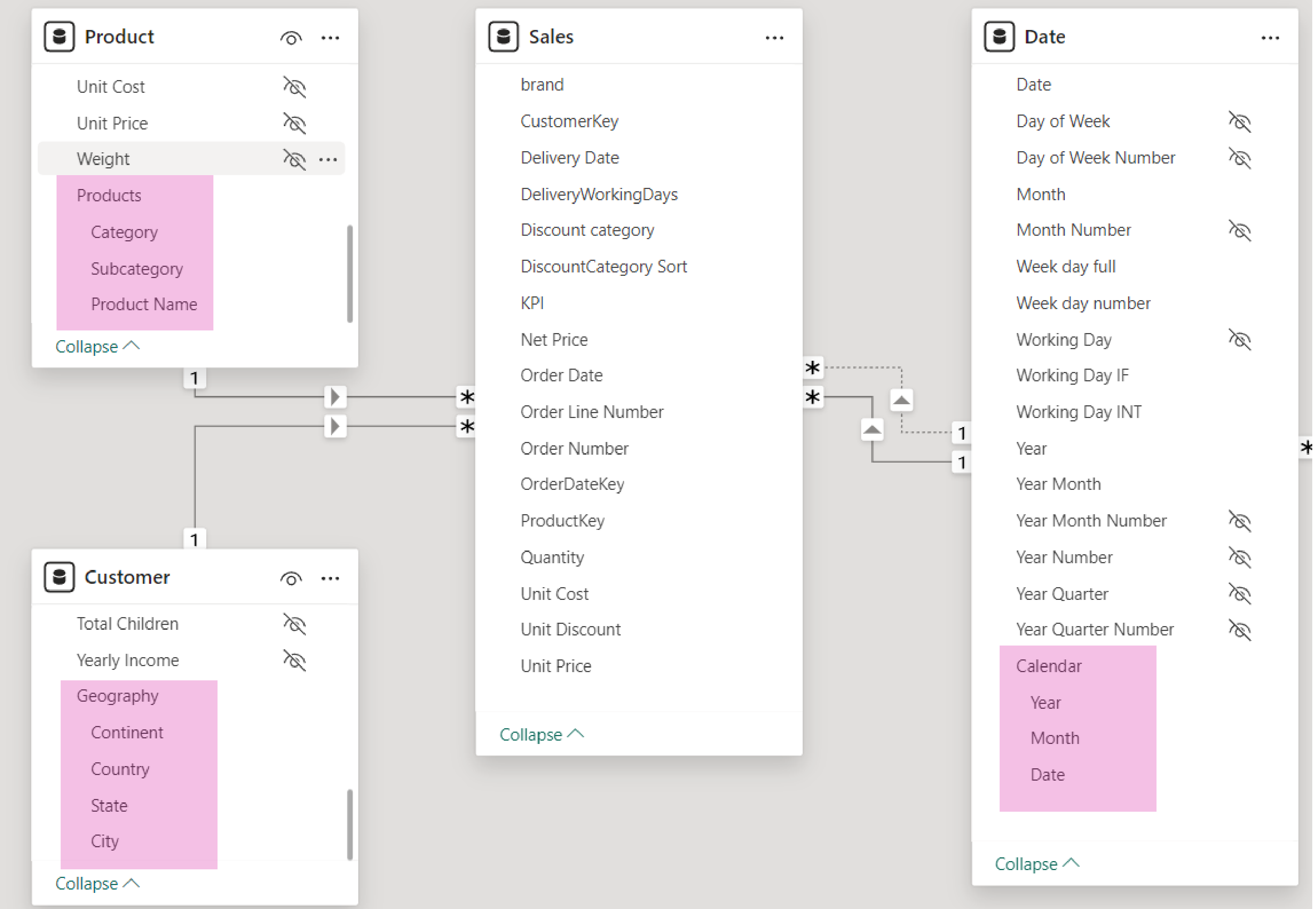


Organizing measures

- Name measures explicitly with a clear name
- Relocate all measures in one table
- Regroupe within folders\subfolders; otherfolder
- Rename folders for order of appearance



Create hierarchies



Best Practices in model view

- Always use a separate date table marked as such for time intelligence
- Whenever possible, use the star schema and identify the dim and facts tables
- Verify the relationships between tables
- Stick to the data normalization , i.e. use YTD, LY, YoY
- Hide the columns that are not meaningful to the user
- Place your DAX measures in a dedicated table (if needed hide the original one)
- Use DIVIDE() (treats division by 0) instead of /

DAX

Data **A**nalysis E**X**pressions

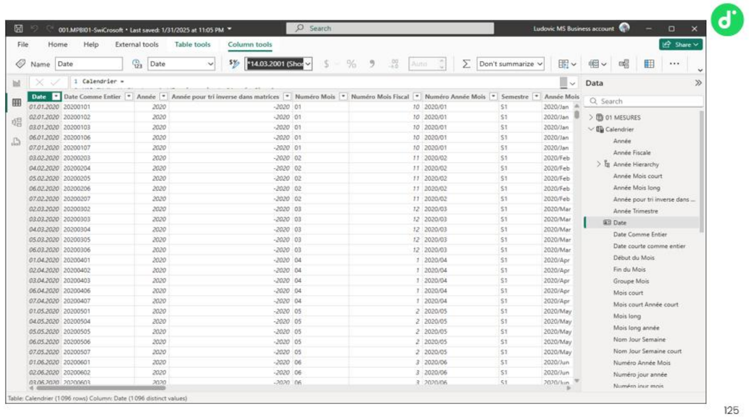
DAX is the formula language in Power BI

It can be used for creating:

- Calculated tables
- Calculated columns
- Measures

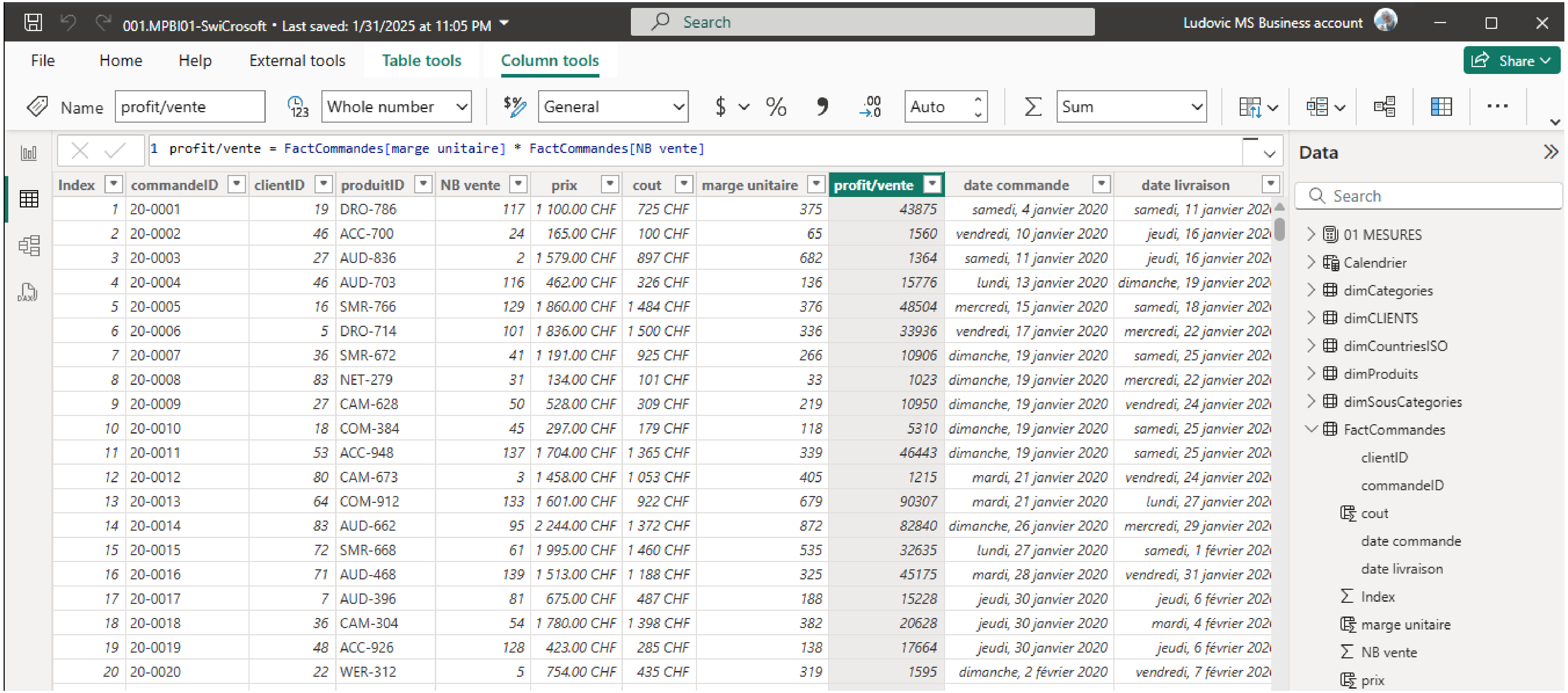
Calculated table: DAX Calendar

```
Calendrier =
VAR StartDate = DATE(2020, 1, 1)
VAR EndDate = DATE(2021, 12, 31)
VAR DateList = CALENDAR(StartDate, EndDate)
RETURN

ADDCOLUMNS (
        CALENDAR ( DATE ( 2020, 1, 1 ), DATE ( 2022, 12, 31 ) ),
        "Date Comme Entier", FORMAT ( [Date], "YYYYMMDD" ),
        "Année", YEAR ( [Date] ),
        "Année pour tri inverse dans matrices", YEAR ( [Date] )*-1,
        "Numéro Mois", FORMAT ( [Date], "MM" ),
        "Numéro Année Mois", FORMAT ( [Date], "YYYY/MM" ),
       "Semestre", "S" & IF ( MONTH ( [Date] ) <= 6, 1, 2 ),
        "Année Mois court", FORMAT ( [Date], "YYYY/mmm" ),
        "Année Mois long", FORMAT ( [Date], "YYYY mmmm" ),
        "Mois court", FORMAT ( [Date], "mmm" ),
        "Mois long", FORMAT ( [Date], "mmmm"
        "Mois long année", FORMAT ( [Date], "mmmm YYYY" ),
        "Mois court Année court", FORMAT( [Date], "MM-yy"),
        ...
)
```



Calculated column

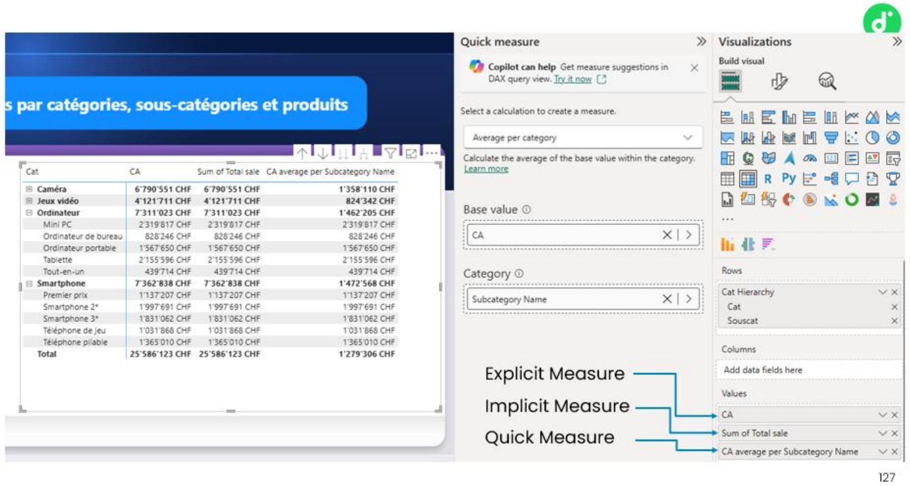
Profit/vente = FactCommandes[marge unitaire] * FactCommandes[NB vente]



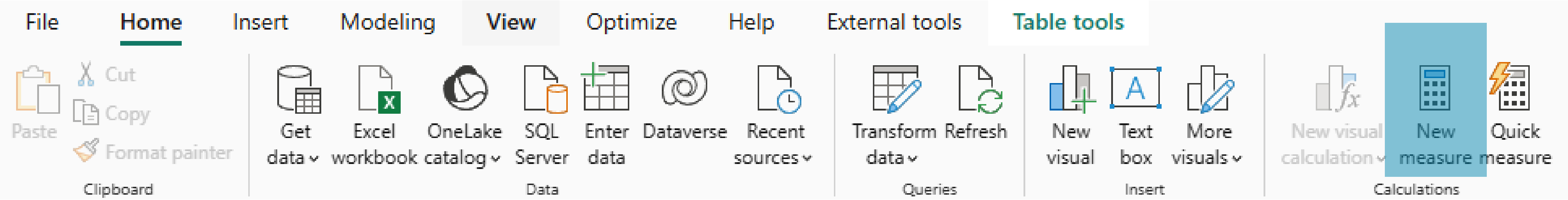
Index	commandeID	clientID	produitID	NB vente	prix	cout	marge unitaire	profit/vente	date commande	date livraison
1	20-0001	19	DRO-786	117	1 100.00 CHF	725 CHF	375	43875	samedi, 4 janvier 2020	samedi, 11 janvier 202
2	20-0002	46	ACC-700	24	165.00 CHF	100 CHF	65	1560	vendredi, 10 janvier 2020	jeudi, 16 janvier 202
3	20-0003	27	AUD-836	2	1 579.00 CHF	897 CHF	682	1364	samedi, 11 janvier 2020	jeudi, 16 janvier 202
4	20-0004	46	AUD-703	116	462.00 CHF	326 CHF	136	15776	lundi, 13 janvier 2020	dimanche, 19 janvier 202
5	20-0005	16	SMR-766	129	1 860.00 CHF	1 484 CHF	376	48504	mercredi, 15 janvier 2020	samedi, 18 janvier 202
6	20-0006	5	DRO-714	101	1 836.00 CHF	1 500 CHF	336	33936	vendredi, 17 janvier 2020	mercredi, 22 janvier 202
7	20-0007	36	SMR-672	41	1 191.00 CHF	925 CHF	266	10906	dimanche, 19 janvier 2020	samedi, 25 janvier 202
8	20-0008	83	NET-279	31	134.00 CHF	101 CHF	33	1023	dimanche, 19 janvier 2020	mercredi, 22 janvier 202
9	20-0009	27	CAM-628	50	528.00 CHF	309 CHF	219	10950	dimanche, 19 janvier 2020	vendredi, 24 janvier 202
10	20-0010	18	COM-384	45	297.00 CHF	179 CHF	118	5310	dimanche, 19 janvier 2020	samedi, 25 janvier 202
11	20-0011	53	ACC-948	137	1 704.00 CHF	1 365 CHF	339	46443	dimanche, 19 janvier 2020	samedi, 25 janvier 202
12	20-0012	80	CAM-673	3	1 458.00 CHF	1 053 CHF	405	1215	mardi, 21 janvier 2020	vendredi, 24 janvier 202
13	20-0013	64	COM-912	133	1 601.00 CHF	922 CHF	679	90307	mardi, 21 janvier 2020	lundi, 27 janvier 202
14	20-0014	83	AUD-662	95	2 244.00 CHF	1 372 CHF	872	82840	dimanche, 26 janvier 2020	mercredi, 29 janvier 202
15	20-0015	72	SMR-668	61	1 995.00 CHF	1 460 CHF	535	32635	lundi, 27 janvier 2020	samedi, 1 février 202
16	20-0016	71	AUD-468	139	1 513.00 CHF	1 188 CHF	325	45175	mardi, 28 janvier 2020	vendredi, 31 janvier 202
17	20-0017	7	AUD-396	81	675.00 CHF	487 CHF	188	15228	jeudi, 30 janvier 2020	jeudi, 6 février 202
18	20-0018	36	CAM-304	54	1 780.00 CHF	1 398 CHF	382	20628	jeudi, 30 janvier 2020	mardi, 4 février 202
19	20-0019	48	ACC-926	128	423.00 CHF	285 CHF	138	17664	jeudi, 30 janvier 2020	jeudi, 6 février 202
20	20-0020	22	WER-312	5	754.00 CHF	435 CHF	319	1595	dimanche, 2 février 2020	vendredi, 7 février 202

DAX MEASURES

IMPLICIT MEASURES	EXPLICIT MEASURES	QUICK MEASURES
Created automatically by Power BI. Dragging a field into a visualization well will create an implicit measure (automatic sum, count, etc)	Created by explicitly typing a DAX code after hitting the ‘New Measure’ button.	List of predefined DAX measures proposed by Power BI. No code needed.



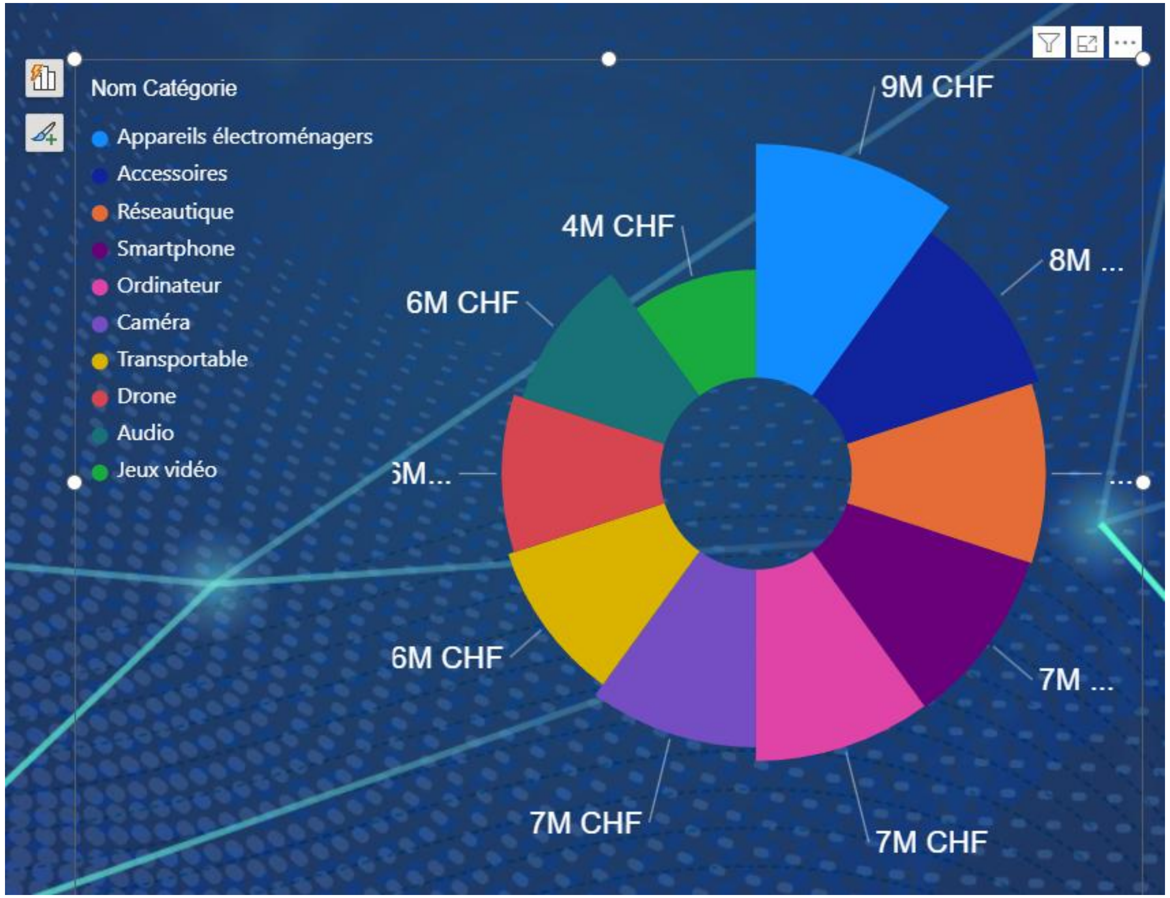
EXPLICIT DAX MEASURES

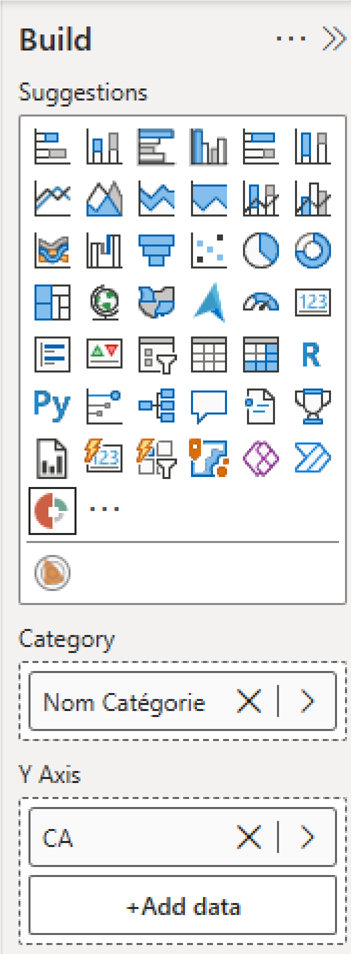



```
TotalSales = SUM(Sales[Amount])

TotalProducts = COUNTROWS(dimProducts[ProductID])

FilteredSales = CALCULATE(
                SUM(Sales[Amount]),
                FILTER(Sales,
                Sales[Region] = "North"))
```

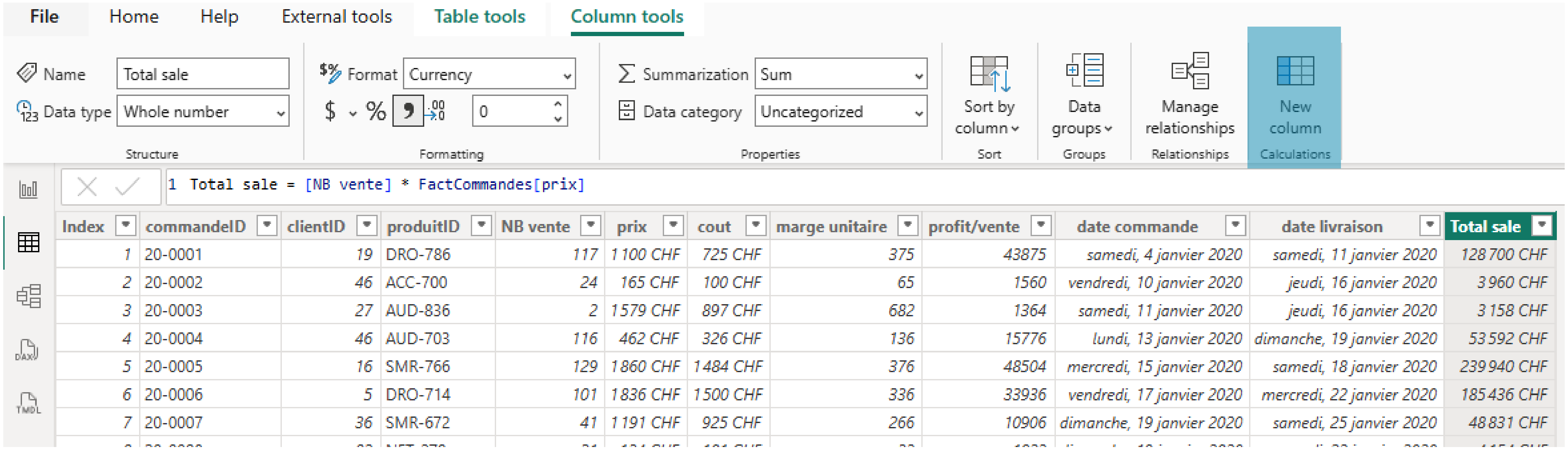
Example of use of a DAX measure

```
CA = SUMX(FactCommandes,FactCommandes[NB vente]*RELATED(dimProduits[Price]))
```





DAX CALCULATED COLUMN



Index	commandeID	clientID	produitID	NB vente	prix	cout	marge unitaire	profit/vente	date commande	date livraison	Total sale
1	20-0001	19	DRO-786	117	1 100 CHF	725 CHF	375	43875	samedi, 4 janvier 2020	samedi, 11 janvier 2020	128 700 CHF
2	20-0002	46	ACC-700	24	165 CHF	100 CHF	65	1560	vendredi, 10 janvier 2020	jeudi, 16 janvier 2020	3 960 CHF
3	20-0003	27	AUD-836	2	1 579 CHF	897 CHF	682	1364	samedi, 11 janvier 2020	jeudi, 16 janvier 2020	3 158 CHF
4	20-0004	46	AUD-703	116	462 CHF	326 CHF	136	15776	lundi, 13 janvier 2020	dimanche, 19 janvier 2020	53 592 CHF
5	20-0005	16	SMR-766	129	1 860 CHF	1 484 CHF	376	48504	mercredi, 15 janvier 2020	samedi, 18 janvier 2020	239 940 CHF
6	20-0006	5	DRO-714	101	1 836 CHF	1 500 CHF	336	33936	vendredi, 17 janvier 2020	mercredi, 22 janvier 2020	185 436 CHF
7	20-0007	36	SMR-672	41	1 191 CHF	925 CHF	266	10906	dimanche, 19 janvier 2020	samedi, 25 janvier 2020	48 831 CHF

prix = RELATED(dimProducts[Price])

Total Sale = FactCommandes[NB vente] * FactCommandes[Prix]

POWER VIEW
Visuals in Power BI

Visualizations pane

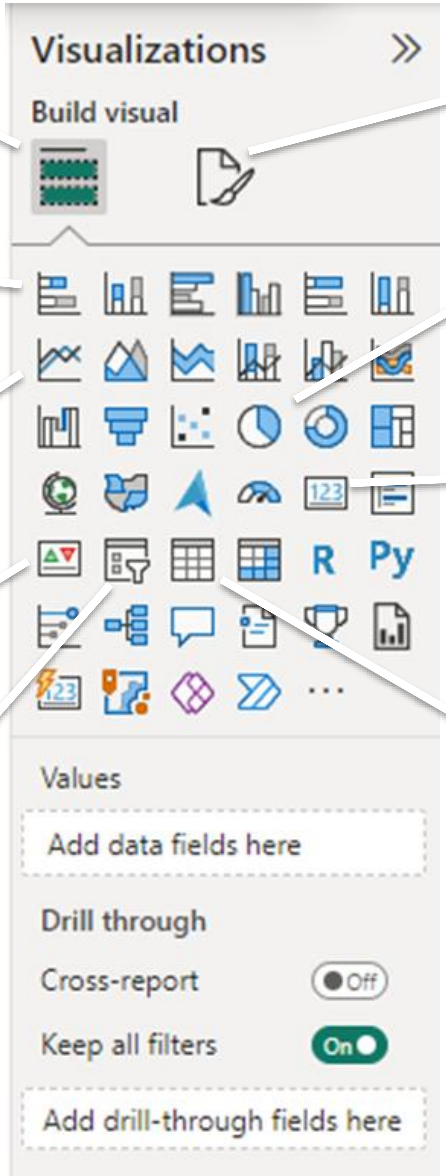
Build visual

Stacked chart row

Line,area & combo chart row

KPI Visualization

Slicer



Format your visual

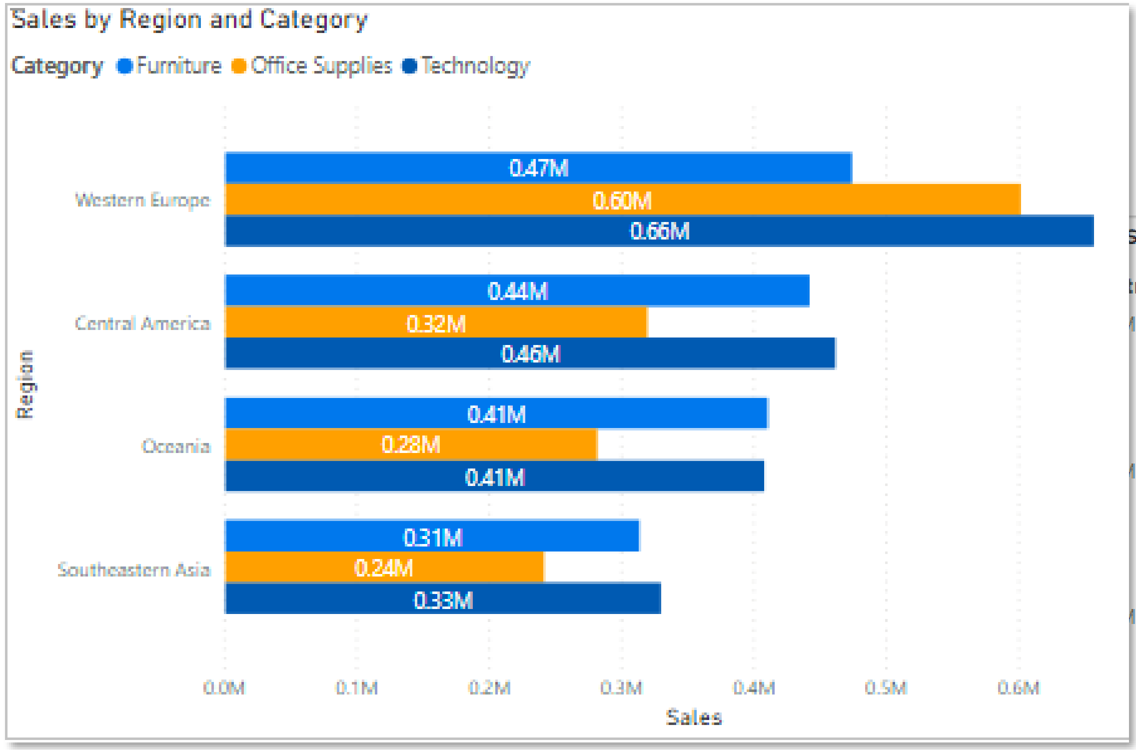
Pie + donut chart

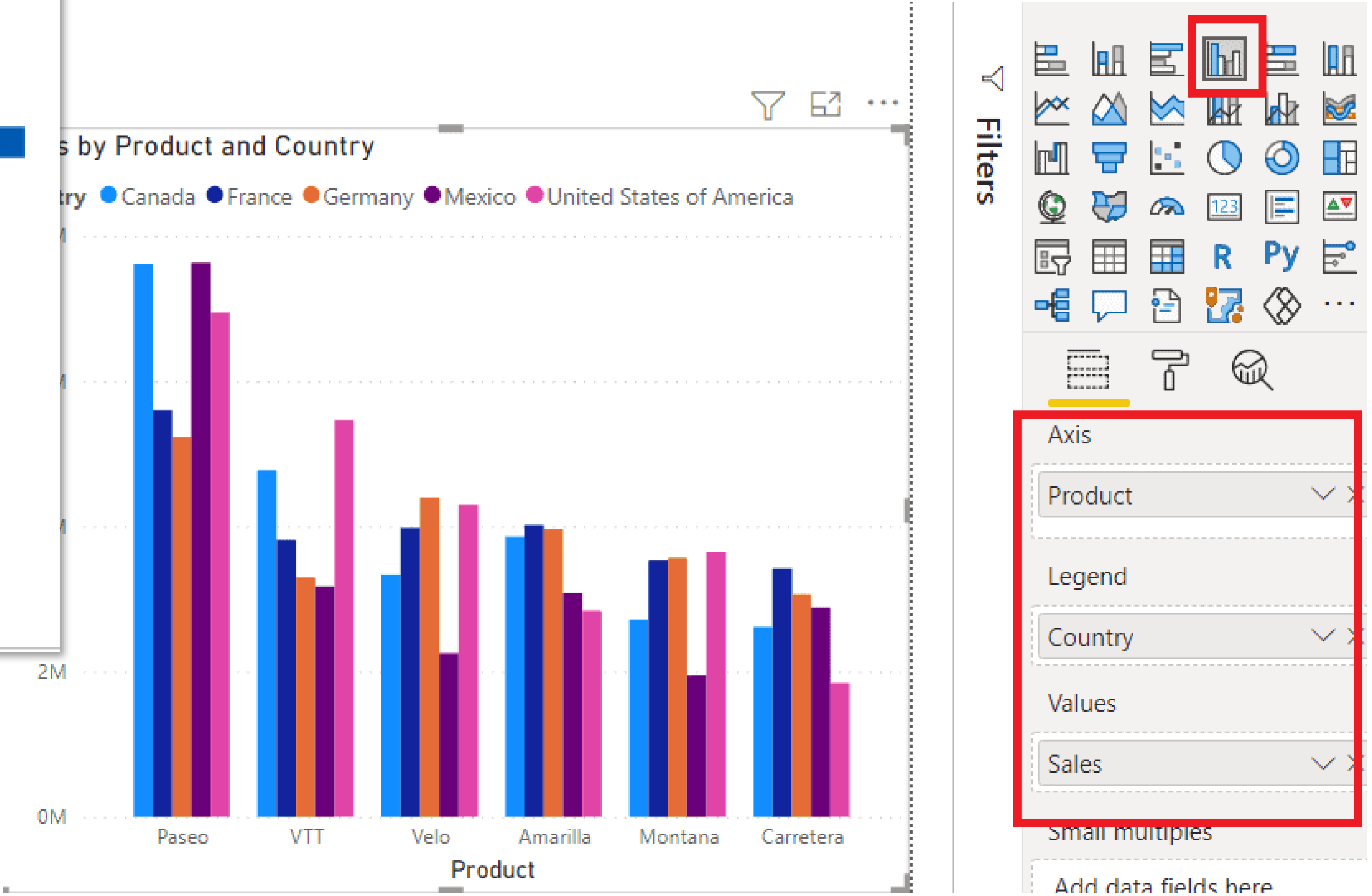
Card and multi-row card

Table, Matrix

Clustered column & bar chart

when you want to compare values across different product categories, regions or time periods.

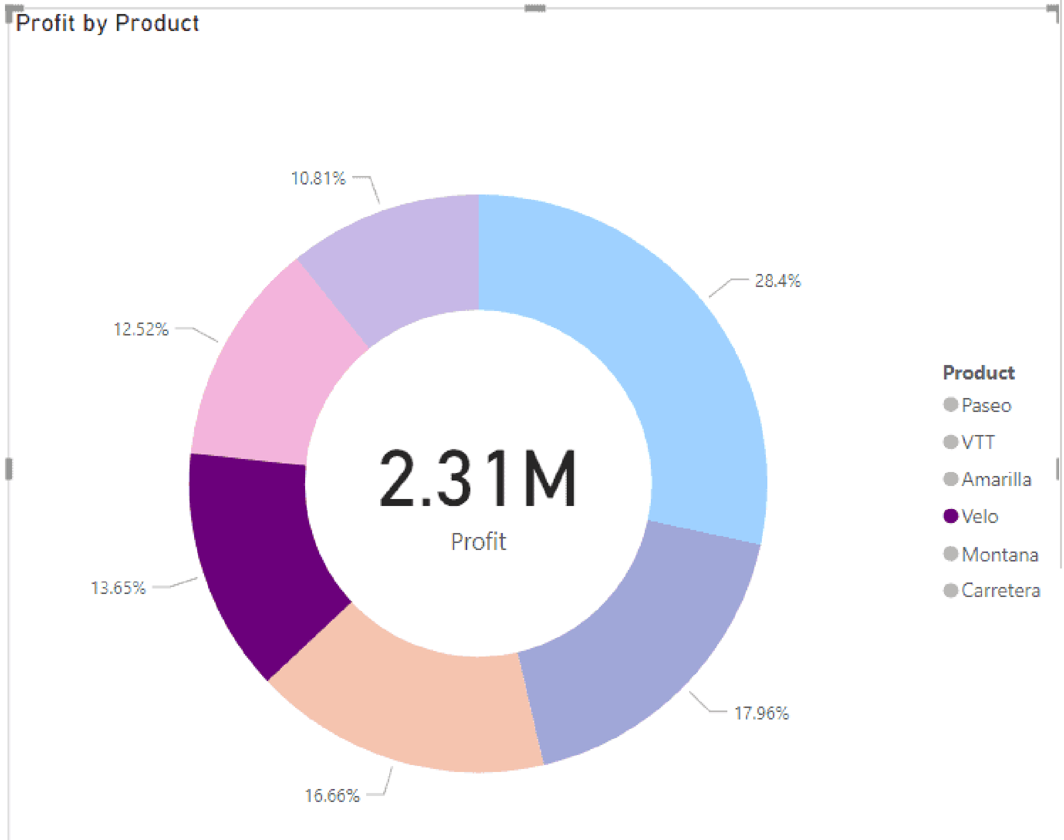


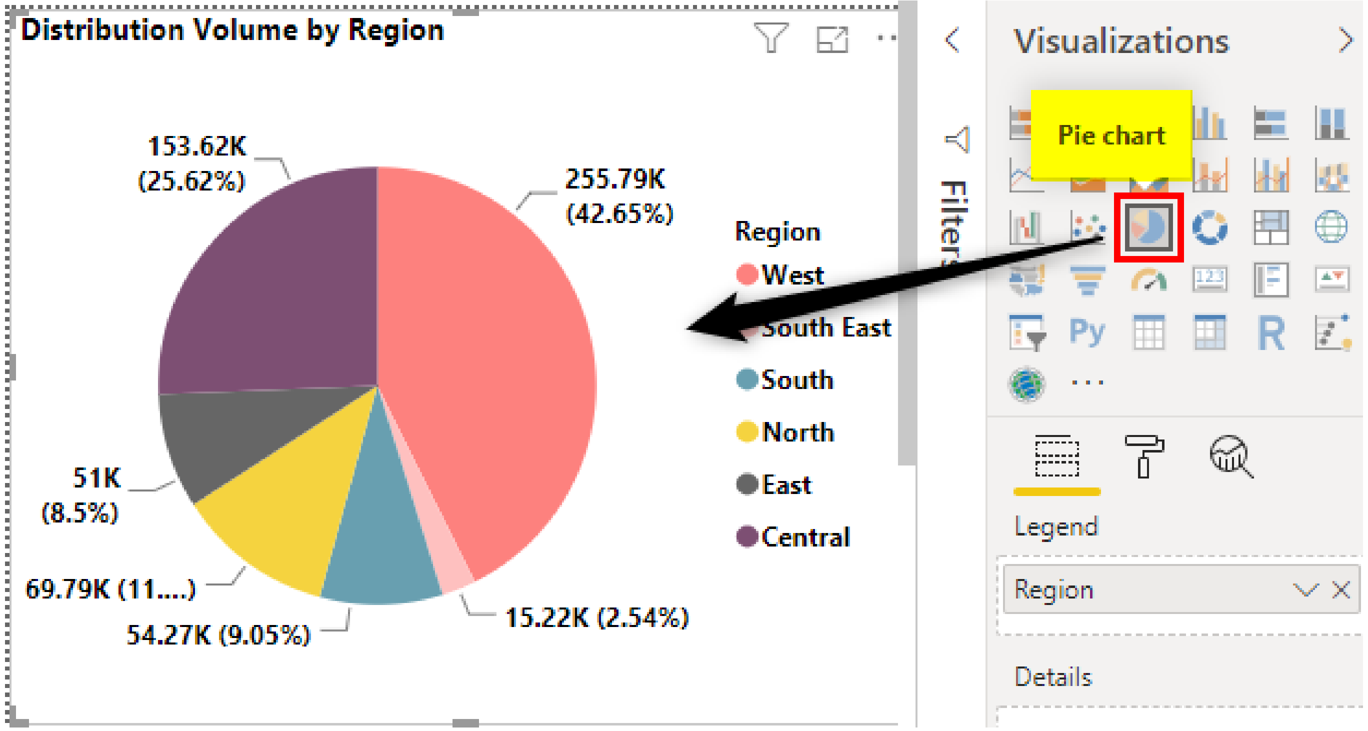


Pie Charts

Illustrates the relative proportion or distribution of a variable

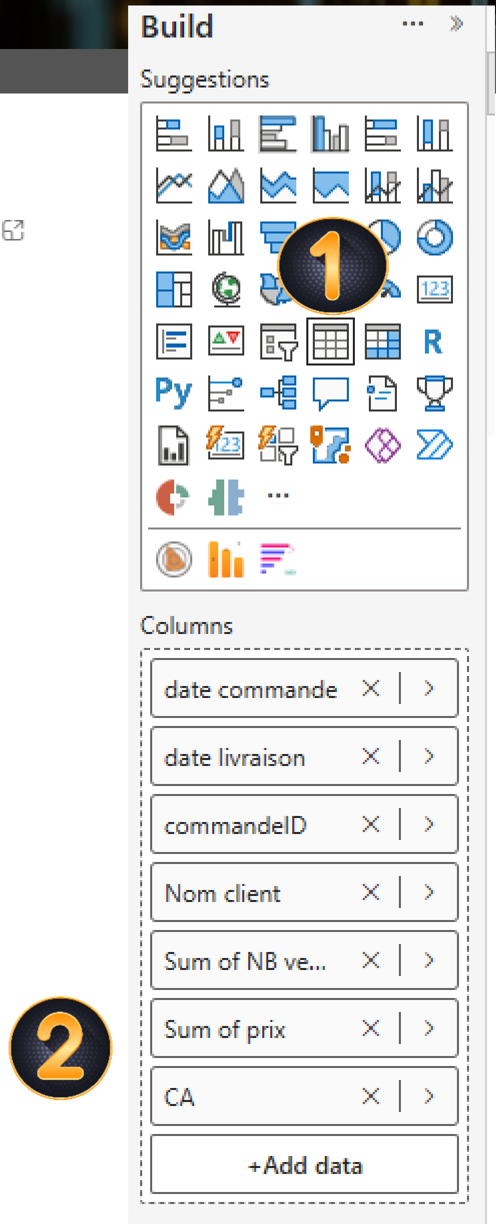
- Best use: 5-7 slices





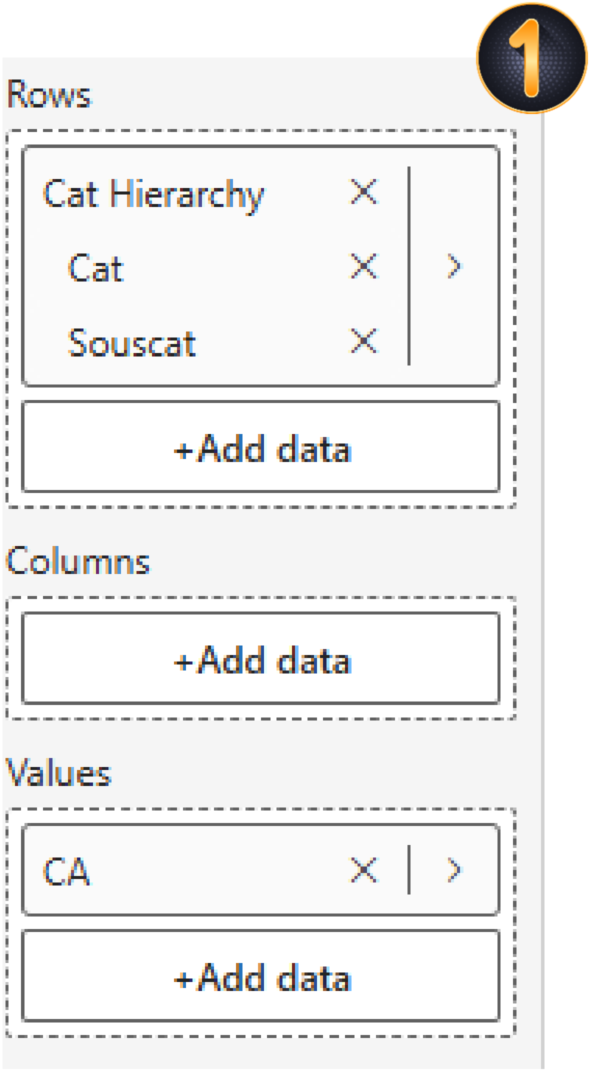
Visualization : table

date commande	date livraison	commandeID	Nom client	Sum of NB vente	Sum of prix	CA
samedi 4 janvier 2020	samedi 11 janvier 2020	20-0001	Horizon Vert	117	1'100.00 CHF	128'700
vendredi 10 janvier 2020	jeudi 16 janvier 2020	20-0002	GrandTech	24	165.00 CHF	3'960
samedi 11 janvier 2020	jeudi 16 janvier 2020	20-0003	TransAlp Logistique	2	1'579.00 CHF	3'158
lundi 13 janvier 2020	dimanche 19 janvier 2020	20-0004	GrandTech	116	462.00 CHF	53'592
mercredi 15 janvier 2020	samedi 18 janvier 2020	20-0005	SolarGen Europe	129	1'860.00 CHF	239'940
vendredi 17 janvier 2020	mercredi 22 janvier 2020	20-0006	HauteRive Consulting	101	1'836.00 CHF	185'436
dimanche 19 janvier 2020	mercredi 22 janvier 2020	20-0008	SummitPeak Technologies	31	134.00 CHF	4'154
dimanche 19 janvier 2020	vendredi 24 janvier 2020	20-0009	TransAlp Logistique	50	528.00 CHF	26'400
dimanche 19 janvier 2020	samedi 25 janvier 2020	20-0007	DataPoint Inc.	41	1'191.00 CHF	48'831
dimanche 19 janvier 2020	samedi 25 janvier 2020	20-0010	MétroTech	45	297.00 CHF	13'365
dimanche 19 janvier 2020	samedi 25 janvier 2020	20-0011	BluRiver Architecture	137	1'704.00 CHF	233'448
mardi 21 janvier 2020	vendredi 24 janvier 2020	20-0012	AlpineWave	3	1'458.00 CHF	4'374
mardi 21 janvier 2020	lundi 27 janvier 2020	20-0013	HighTech Sàrl	133	1'601.00 CHF	212'933
dimanche 26 janvier 2020	mercredi 29 janvier 2020	20-0014	SummitPeak Technologies	95	2'244.00 CHF	213'180
lundi 27 janvier 2020	samedi 1 février 2020	20-0015	AlpenScience AG	61	1'995.00 CHF	121'695
mardi 28 janvier 2020	vendredi 31 janvier 2020	20-0016	EuroFuture Corp.	139	1'513.00 CHF	210'307
jeudi 30 janvier 2020	mardi 4 février 2020	20-0018	DataPoint Inc.	54	1'780.00 CHF	96'120
jeudi 30 janvier 2020	jeudi 6 février 2020	20-0017	Swiss Health Solutions	81	675.00 CHF	54'675
jeudi 30 janvier 2020	jeudi 6 février 2020	20-0019	BaselLab Solutions	128	423.00 CHF	54'144
dimanche 2 février 2020	vendredi 7 février 2020	20-0020	Horizon Durable	5	754.00 CHF	3'770
dimanche 2 février 2020	samedi 8 février 2020	20-0021	NovaSuisse Technologies	30	577.00 CHF	17'310
mardi 4 février 2020	dimanche 9 février 2020	20-0022	Swiss Finances SA	104	529.00 CHF	55'016
vendredi 7 février 2020	mardi 11 février 2020	20-0023	HauteRive Consulting	60	577.00 CHF	34'620
samedi 8 février 2020	jeudi 13 février 2020	20-0025	VisionDigit	79	834.00 CHF	65'886
samedi 8 février 2020	samedi 15 février 2020	20-0024	AlphaVision Sàrl	29	661.00 CHF	19'169
dimanche 9 février 2020	dimanche 16 février 2020	20-0026	PeakIT Solutions	89	273.00 CHF	24'297

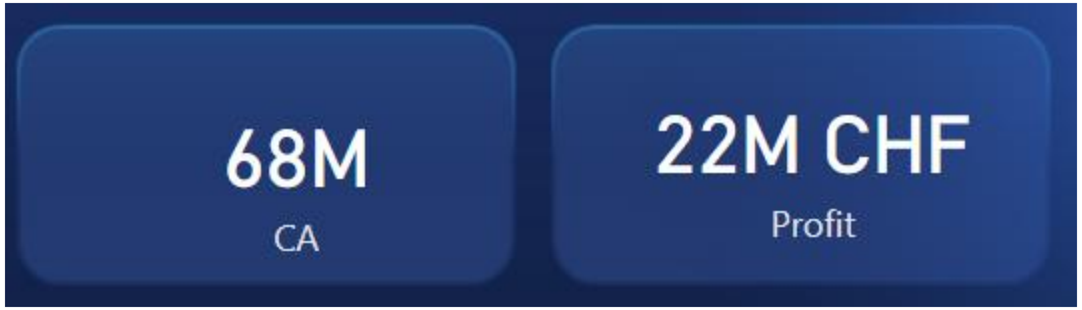


Matrix

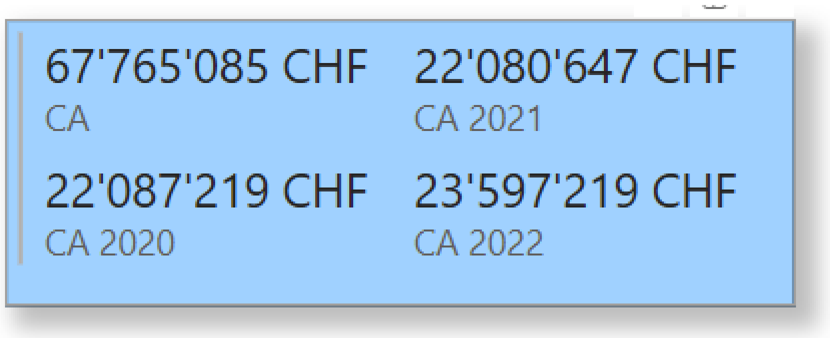
Cat	CA
Caméra	**6'790'551**
Jeux vidéo	**4'121'711**
Ordinateur	**7'311'023**
Mini PC	2'319'817
Ordinateur de bureau	828'246
Ordinateur portable	1'567'650
Tablette	2'155'596
Tout-en-un	439'714
Smartphone	**7'362'838**
Premier prix	1'137'207
Smartphone 2*	1'997'691
Smartphone 3*	1'831'062
Téléphone de jeu	1'031'868
Téléphone pliable	1'365'010
Total	**25'586'123**



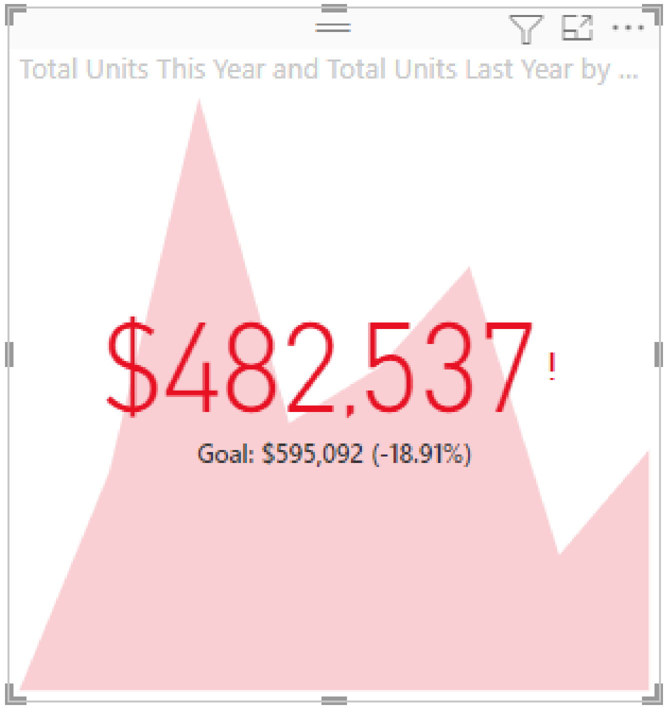
Cards

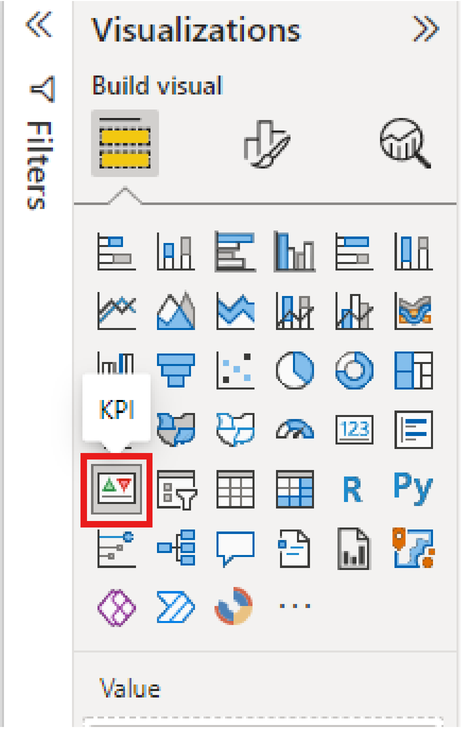


Multi-row cards

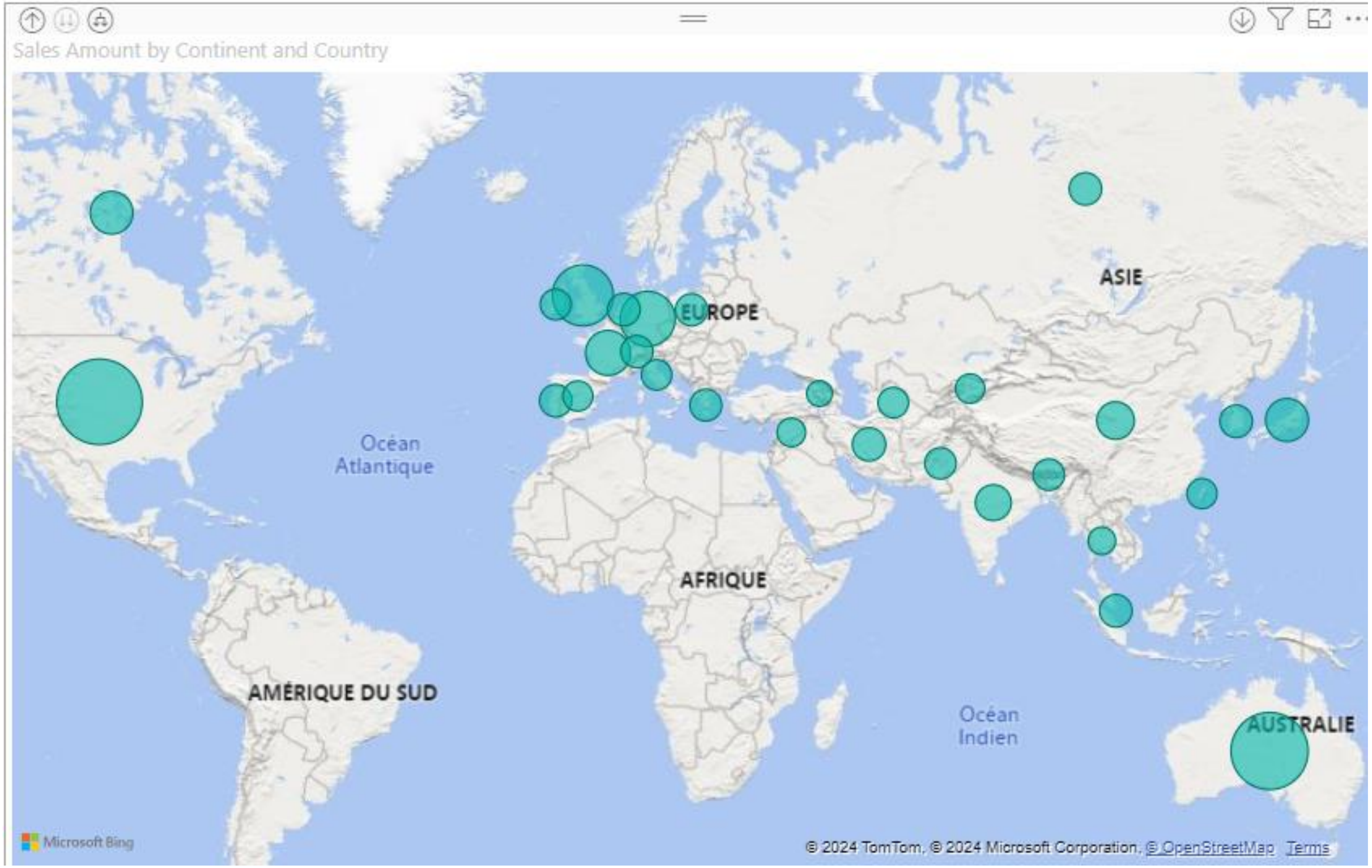


KPI

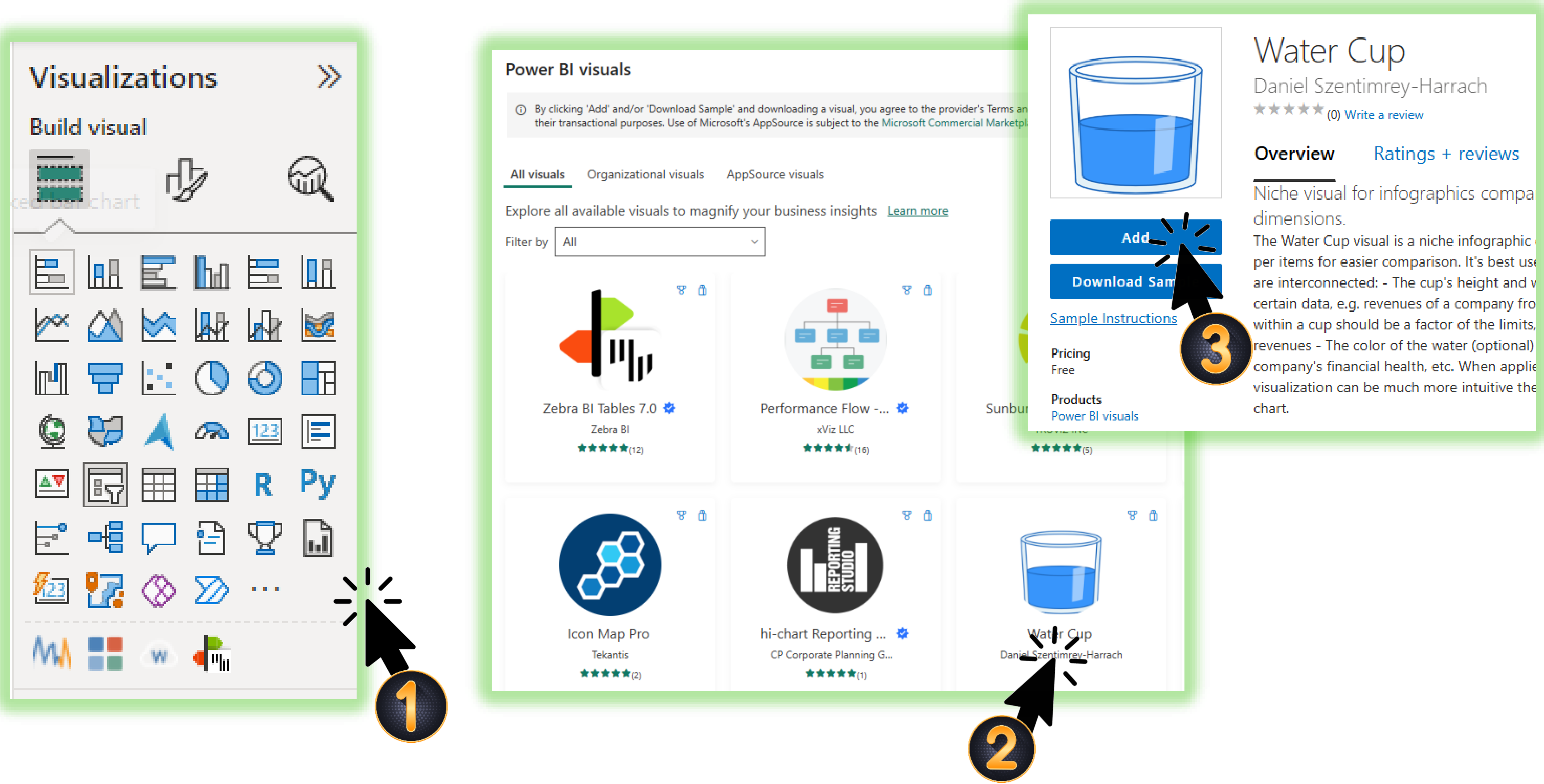




Map visual

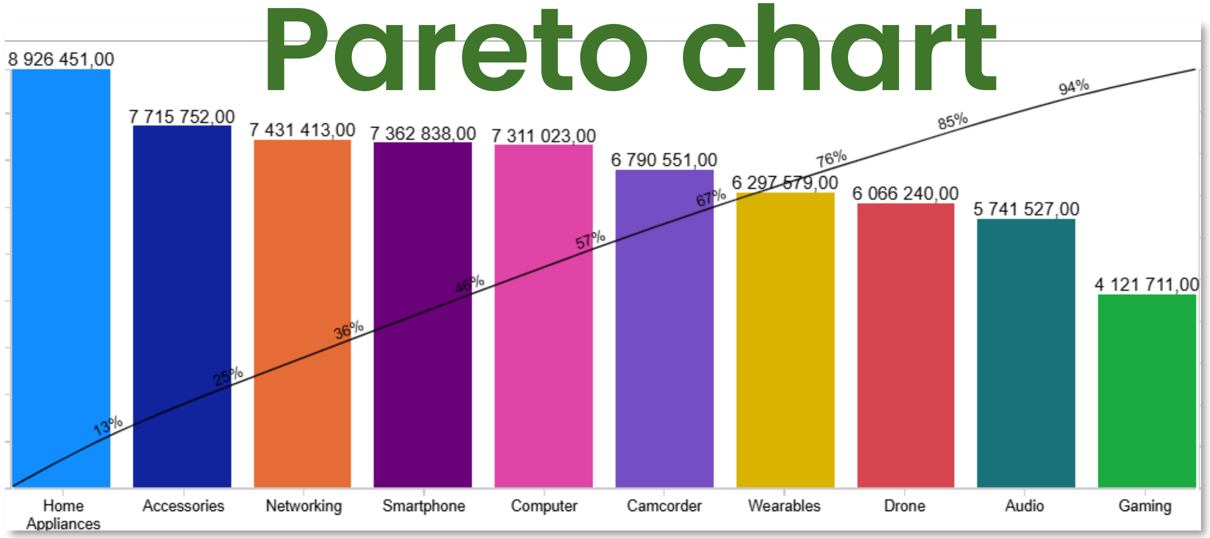

Sales Amount by Continent and Country
ASIE
EUROPE
Océan Atlantique
AFRIQUE
AMÉRIQUE DU SUD
Océan Indien
AUSTRALIE
Microsoft Bing
© 2024 TomTom, © 2024 Microsoft Corporation, © OpenStreetMap Terms

Add a new visualization

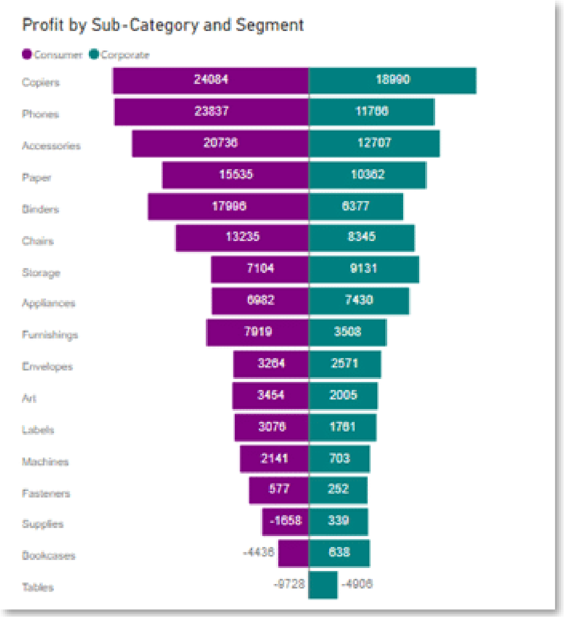


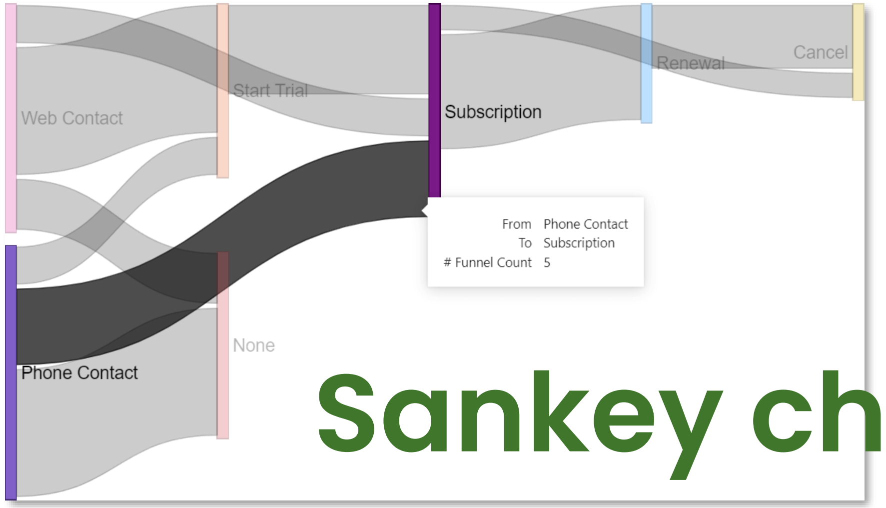
3rd party visuals

Pareto chart



Tornado Chart





Sankey chart

[More 3rd party visuals here]

Conditional formatting

TABLE: Top 10 weather states with afford

State	Weather	Affordability	Overall rank
Hawaii	1	45	10
Florida	2	25	5
Louisiana	3	29	36
Texas	4	24	17
Georgia	5	19	28
Mississippi	6	6	19
Alabama	7	10	16
South Carolina	8	27	41
Arkansas	9	4	11
Arizona	10	33	38

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Louisiana	29	3	36
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Arizona	33	10	38

Interactions

Refers to the way visuals interact with themselves or with others

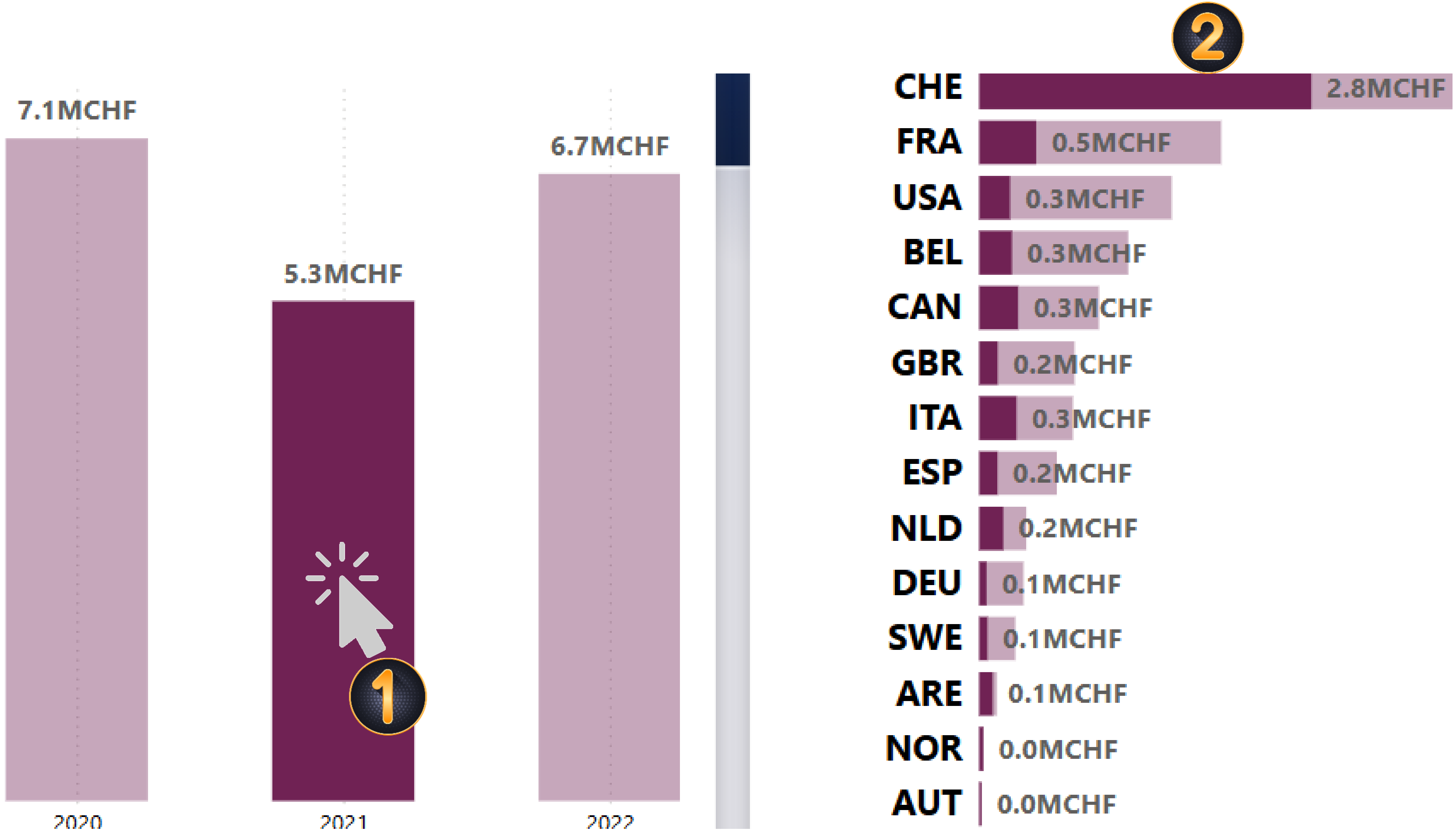
Self-interactions

- Tooltips
- Drillthrough
- Drill-down
- Highlighting

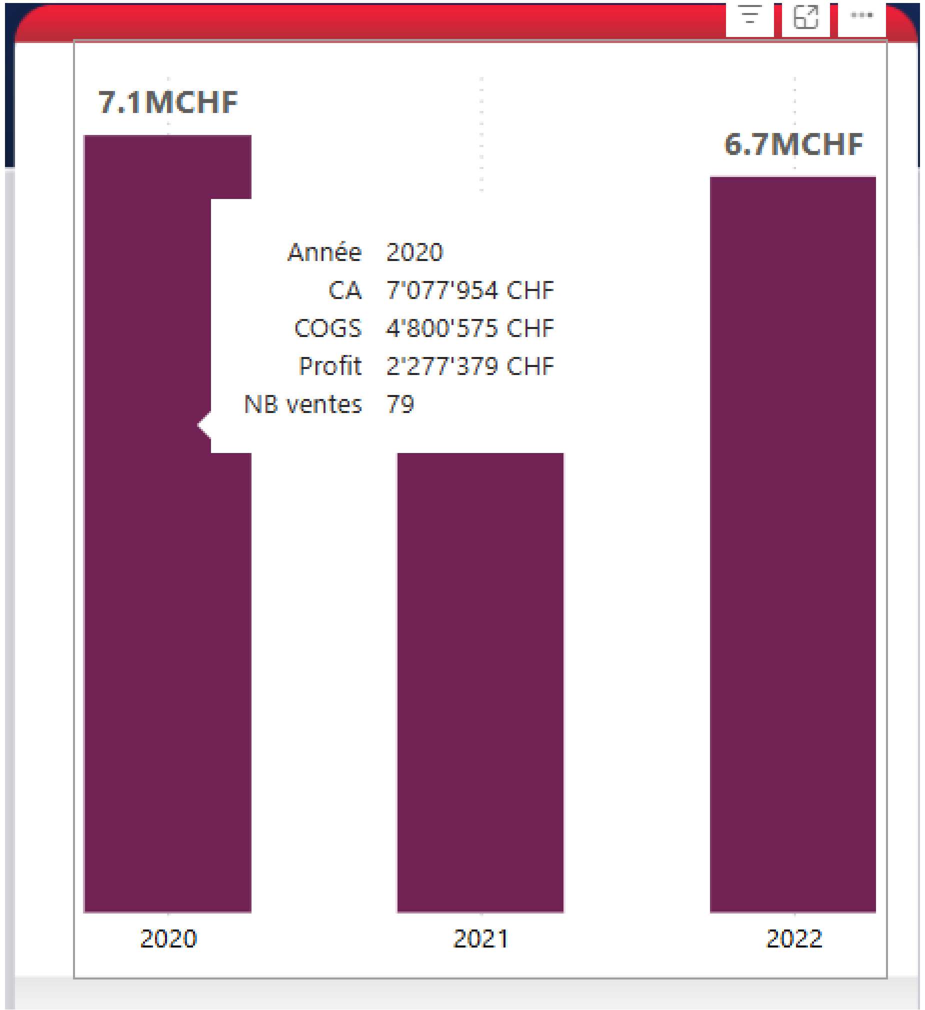
Interactions with others

- Slicing
- Cross-filtering
- Cross-highlighting
- Bookmarks

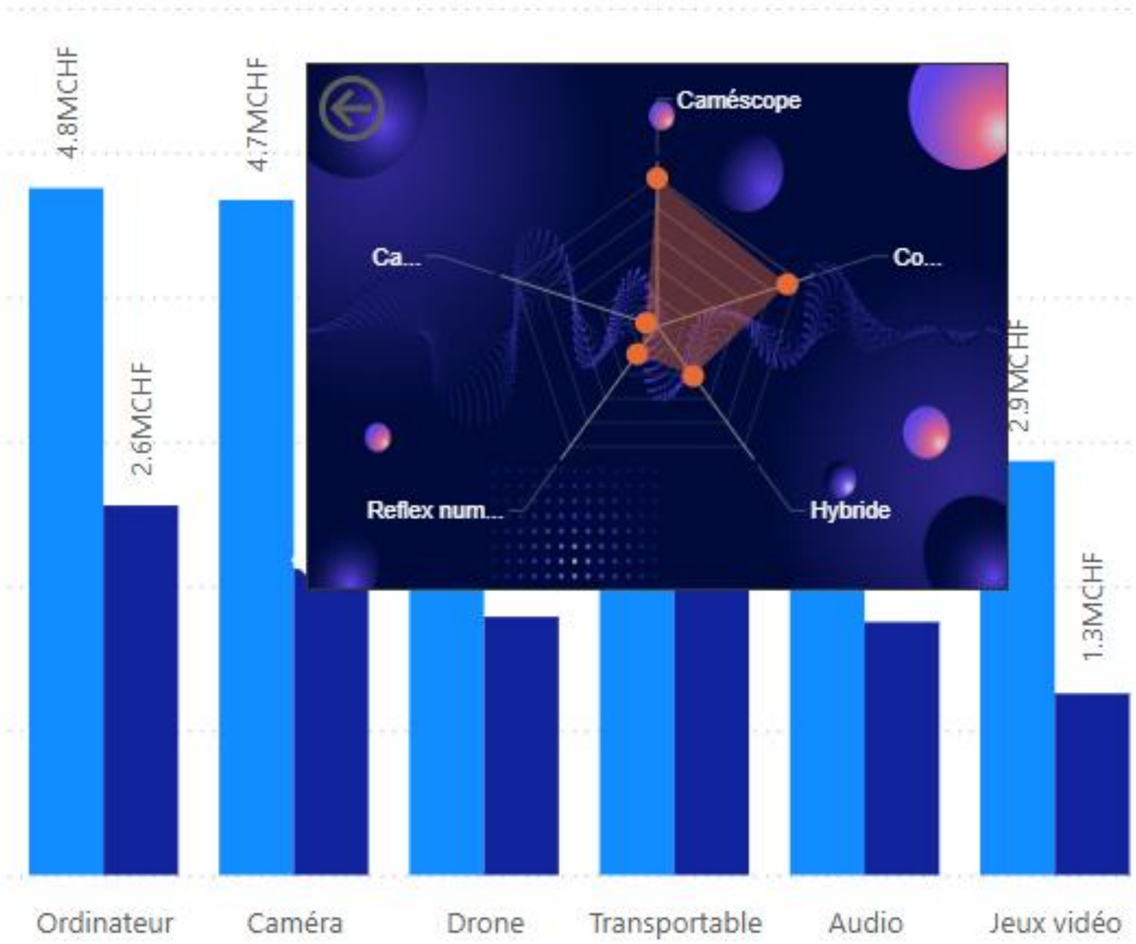
VISUAL FILTERS: Cross-highlihgting



Classic tooltip

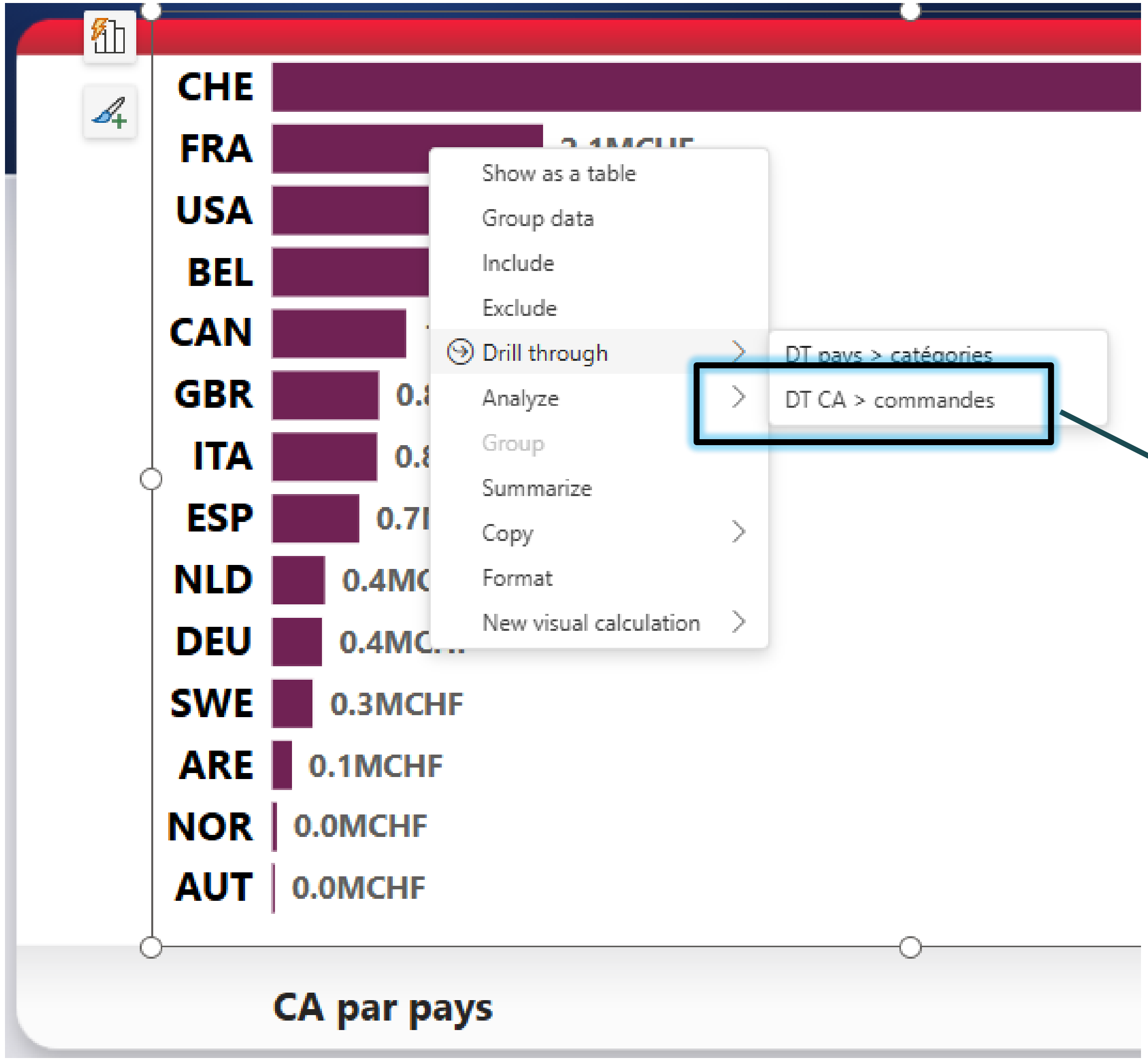


Custom tooltip



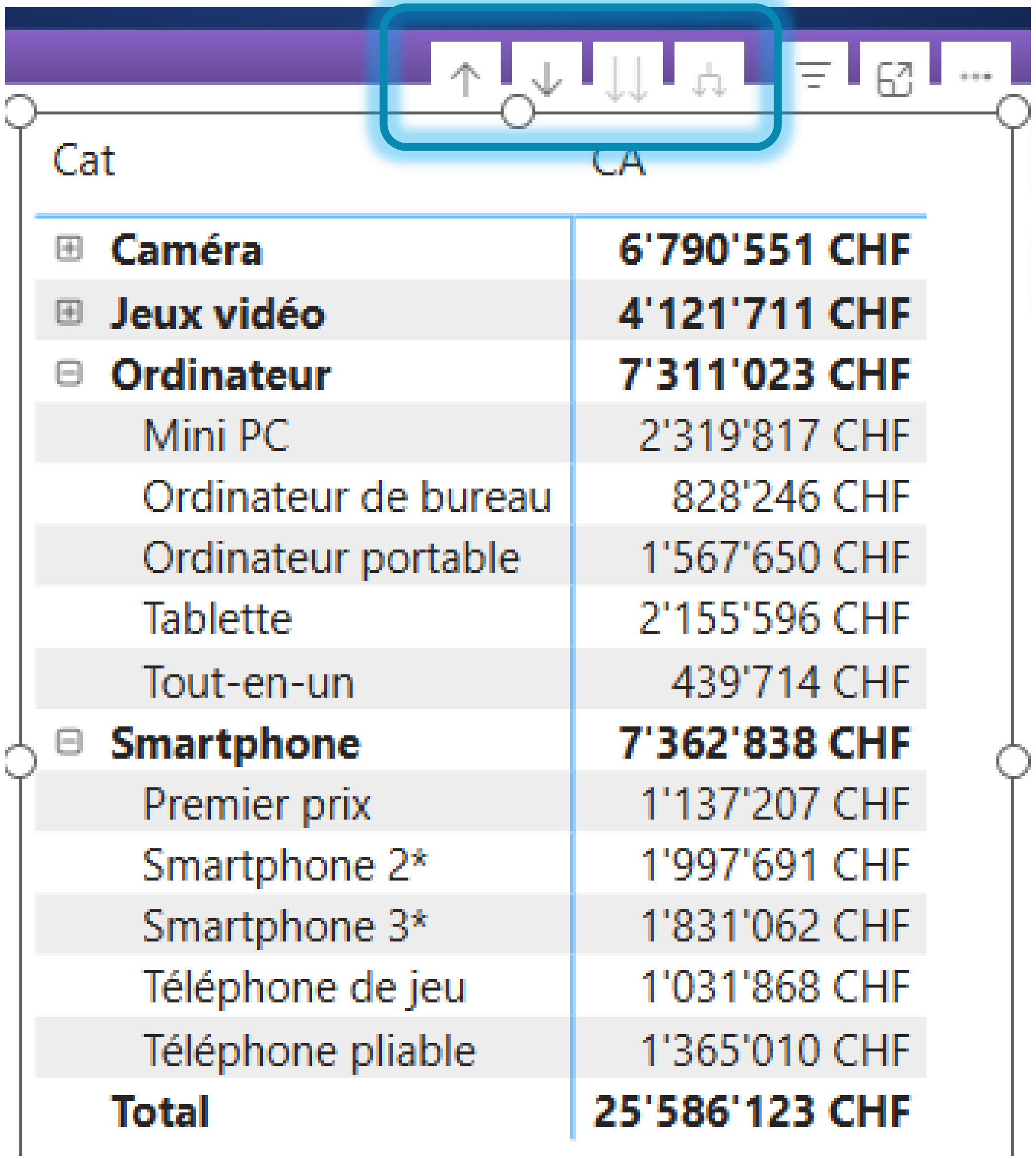
Drillthrough page

Visual in a page

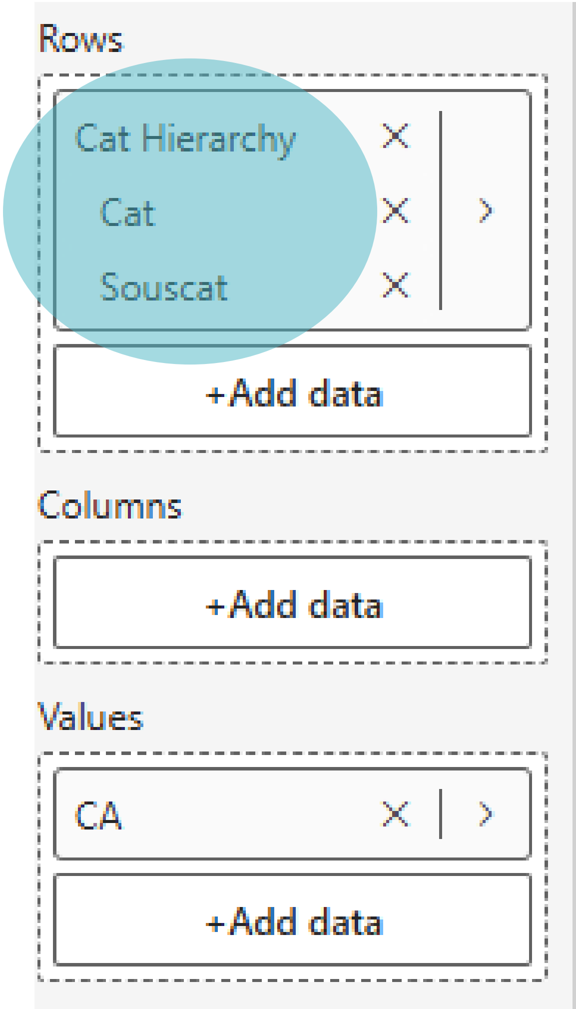


date commande	date livraison	commandeID
samedi 4 janvier 2020	samedi 11 janvier 2020	20-0001
dimanche 19 janvier 2020	samedi 25 janvier 2020	20-0010
samedi 25 avril 2020	mercredi 29 avril 2020	20-0076
dimanche 26 avril 2020	mercredi 29 avril 2020	20-0077
lundi 18 mai 2020	lundi 25 mai 2020	20-0091
lundi 8 juin 2020	samedi 13 juin 2020	20-0107
mardi 16 juin 2020	mardi 23 juin 2020	20-0113
mercredi 17 juin 2020	dimanche 21 juin 2020	20-0116
jeudi 30 juillet 2020	dimanche 2 août 2020	20-0145
lundi 12 octobre 2020	lundi 19 octobre 2020	20-0195
dimanche 17 janvier 2021	vendredi 22 janvier 2021	21-0259
mardi 16 février 2021	vendredi 19 février 2021	21-0276
lundi 15 mars 2021	lundi 22 mars 2021	21-0298
dimanche 27 juin 2021	jeudi 1 juillet 2021	21-0373
mercredi 18 août 2021	lundi 23 août 2021	21-0412
jeudi 2 septembre 2021	mardi 7 septembre 2021	21-0417
jeudi 28 octobre 2021	mercredi 3 novembre 2021	21-0456
mardi 16 novembre 2021	lundi 22 novembre 2021	21-0470
samedi 8 janvier 2022	mardi 11 janvier 2022	22-0001
dimanche 30 janvier 2022	samedi 5 février 2022	22-0019
dimanche 6 février 2022	vendredi 11 février 2022	22-0025

Drill down with hierarchies



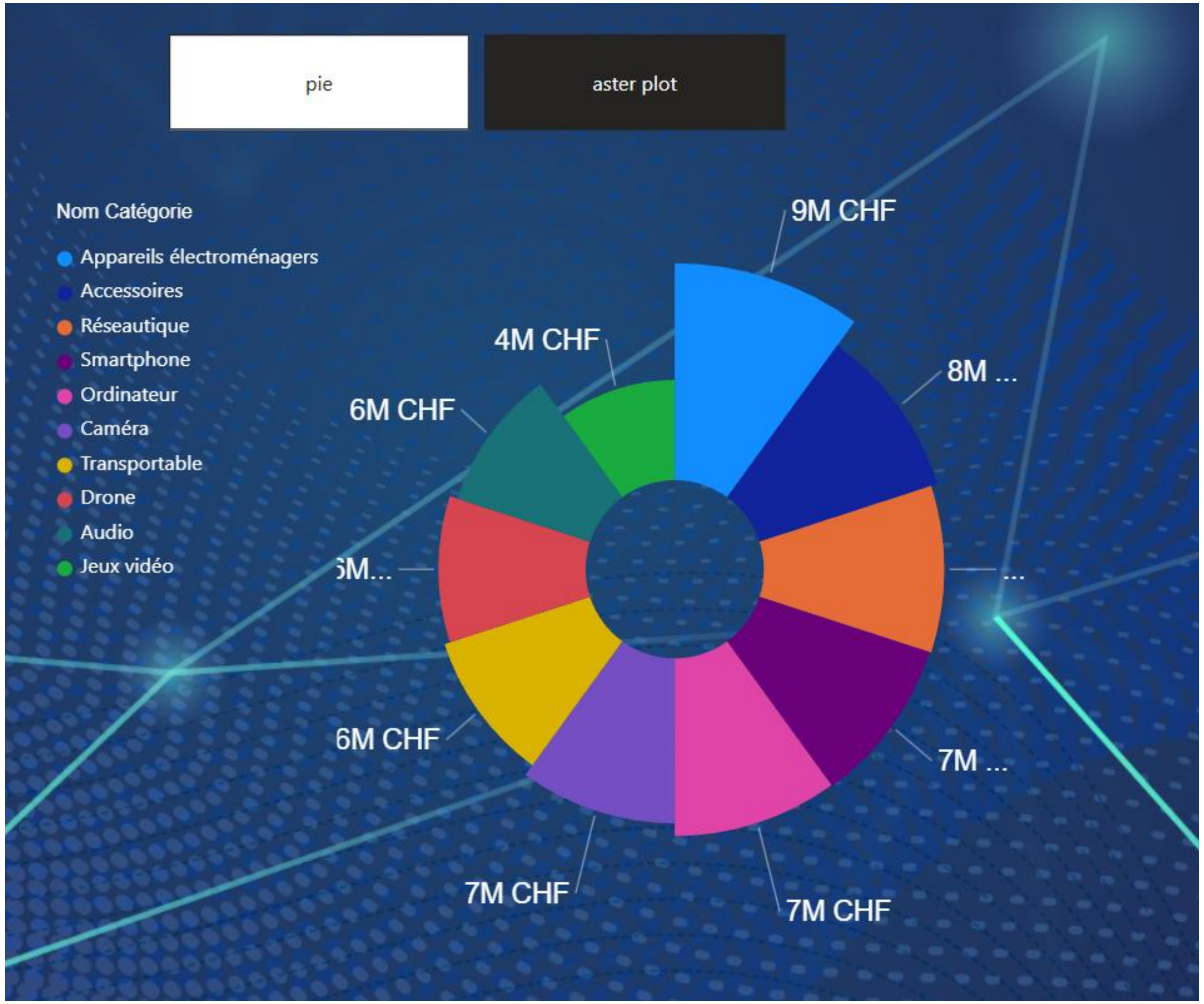
Cat	CA
Caméra	**6'790'551 CHF**
Jeux vidéo	**4'121'711 CHF**
Ordinateur	**7'311'023 CHF**
Mini PC	2'319'817 CHF
Ordinateur de bureau	828'246 CHF
Ordinateur portable	1'567'650 CHF
Tablette	2'155'596 CHF
Tout-en-un	439'714 CHF
Smartphone	**7'362'838 CHF**
Premier prix	1'137'207 CHF
Smartphone 2*	1'997'691 CHF
Smartphone 3*	1'831'062 CHF
Téléphone de jeu	1'031'868 CHF
Téléphone pliable	1'365'010 CHF
Total	**25'586'123 CHF**



Bookmarks

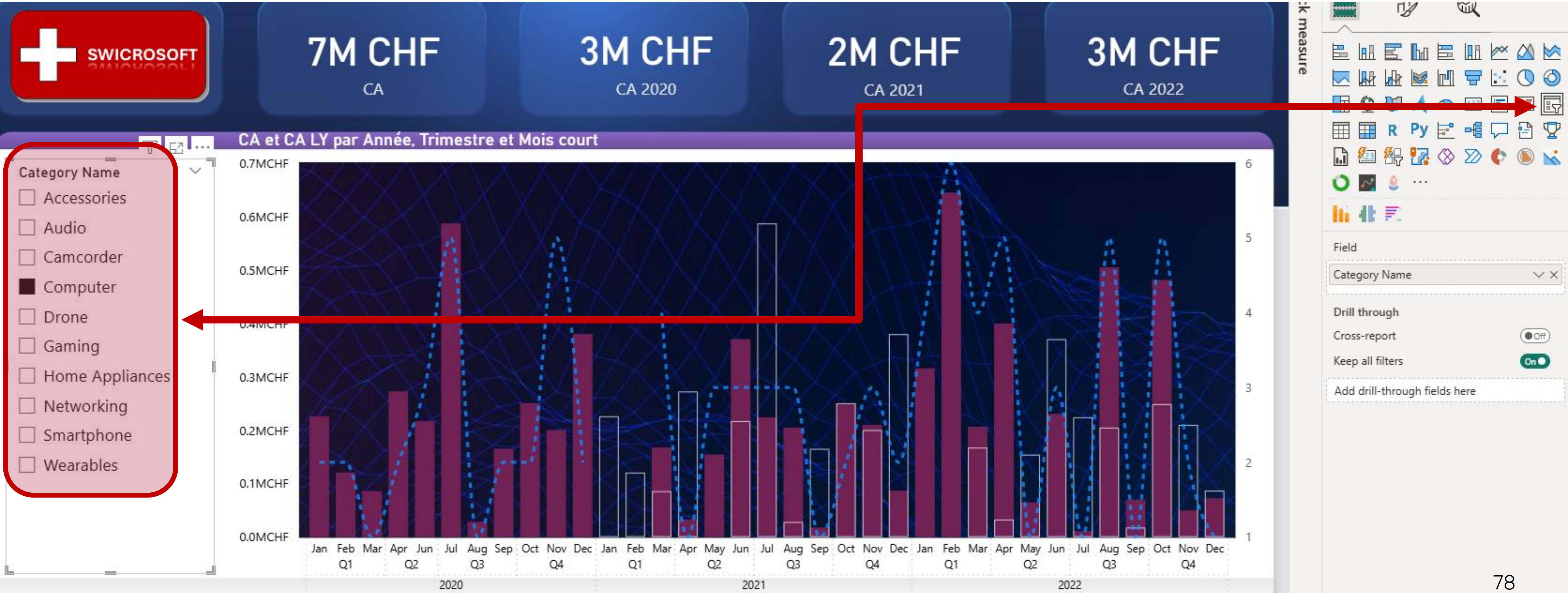
Bookmarks in Power BI are used to capture and save these specific elements of a report:

the current page, filters, slicers type and state and visual selection state. You can set up and navigate from bookmark to bookmark (as in a slideshow)



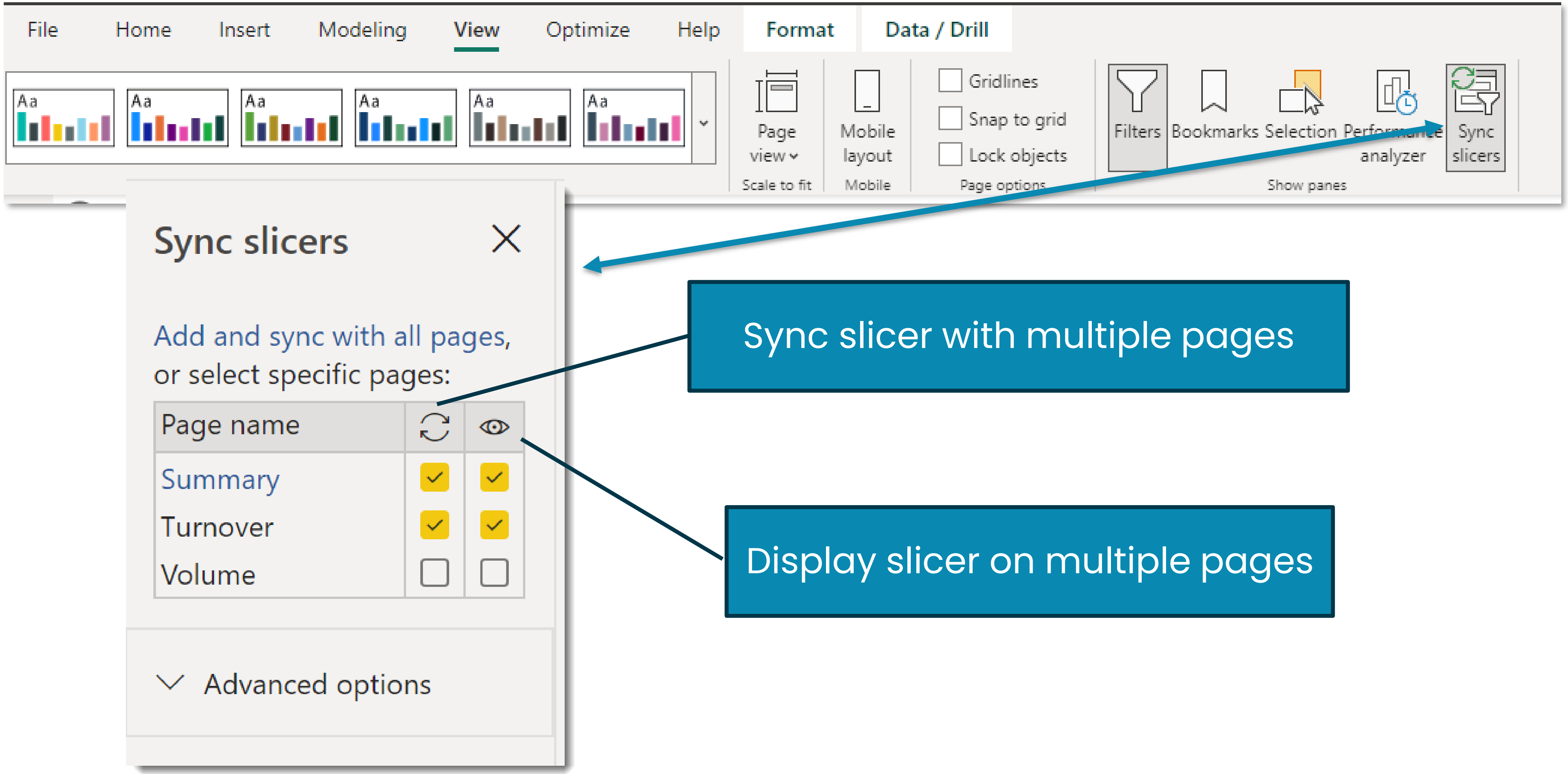
VISUAL FILTERS: SLICERS

Slicer = visual filter



Sync slicers

View ribbon / Sync slicers

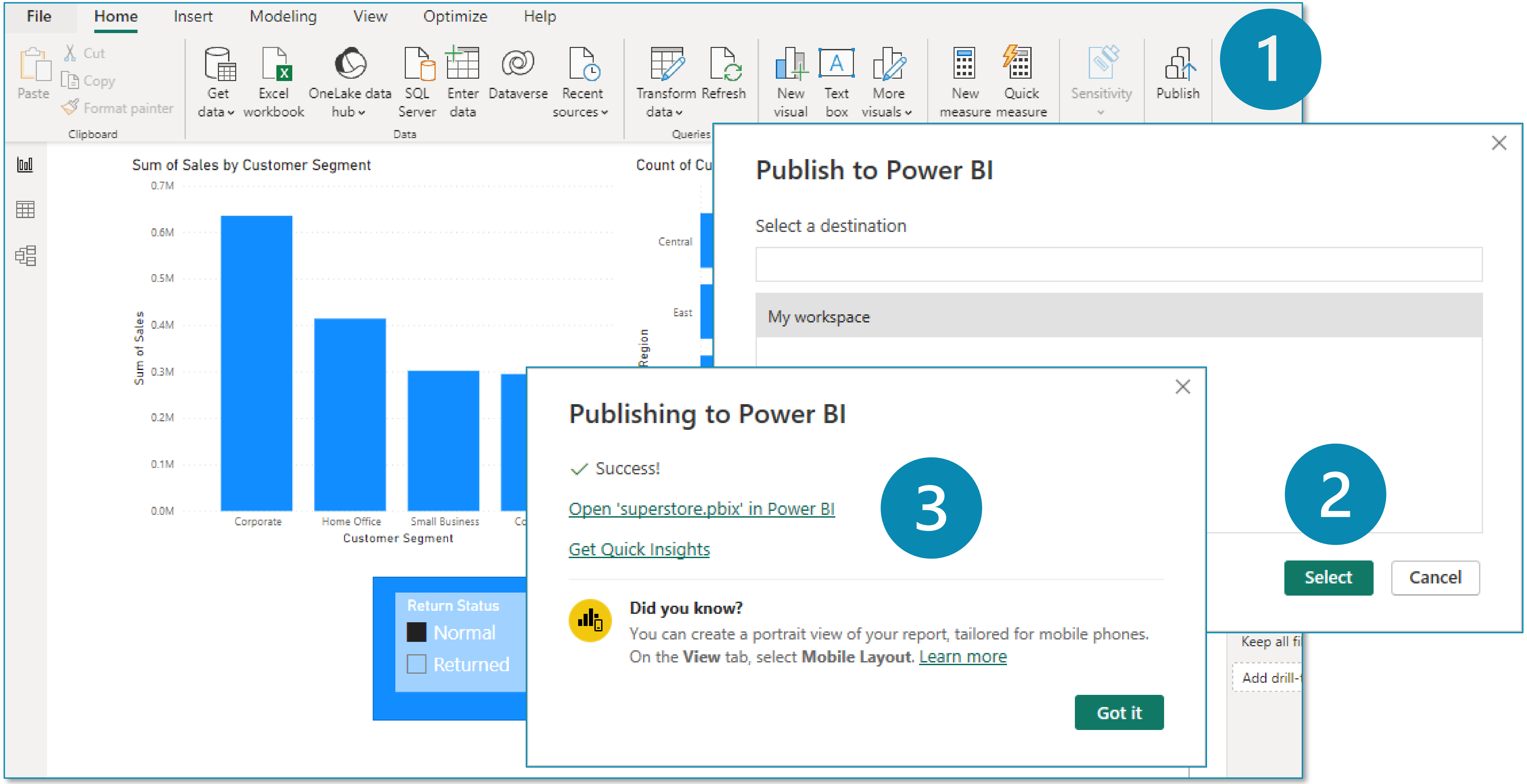


Best Practices in report view

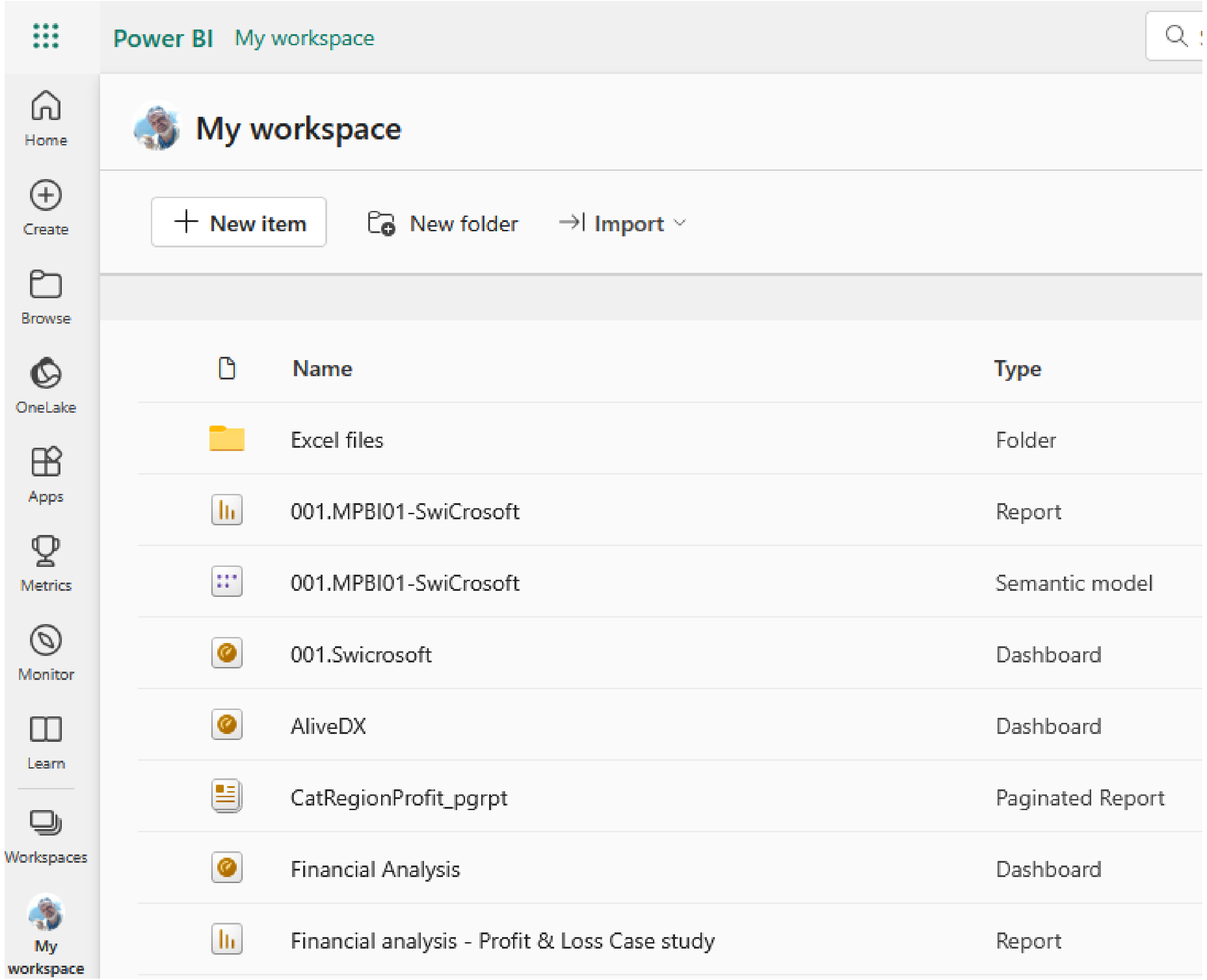
- Within the same company, consider using a corporate PBI template (.pbit)
- Be careful with flashy background images
- Use colors with caution and with a purpose (also consider color blind people)
- Insert slicers in smart places on your report
- Write a clear and explicit title to all your visuals
- Use Drill-through and tooltip pages to add depth to your visuals
- Prefer visuals over tables and matrices (preferably certified ones)
- If you use tables , don’t forget conditional formatting
- Optimize visual interactions, cross-filtering is prefered over cross-highlighting

POWER BI SERVICE
Dashboards and sharing

Publish to Power BI Service



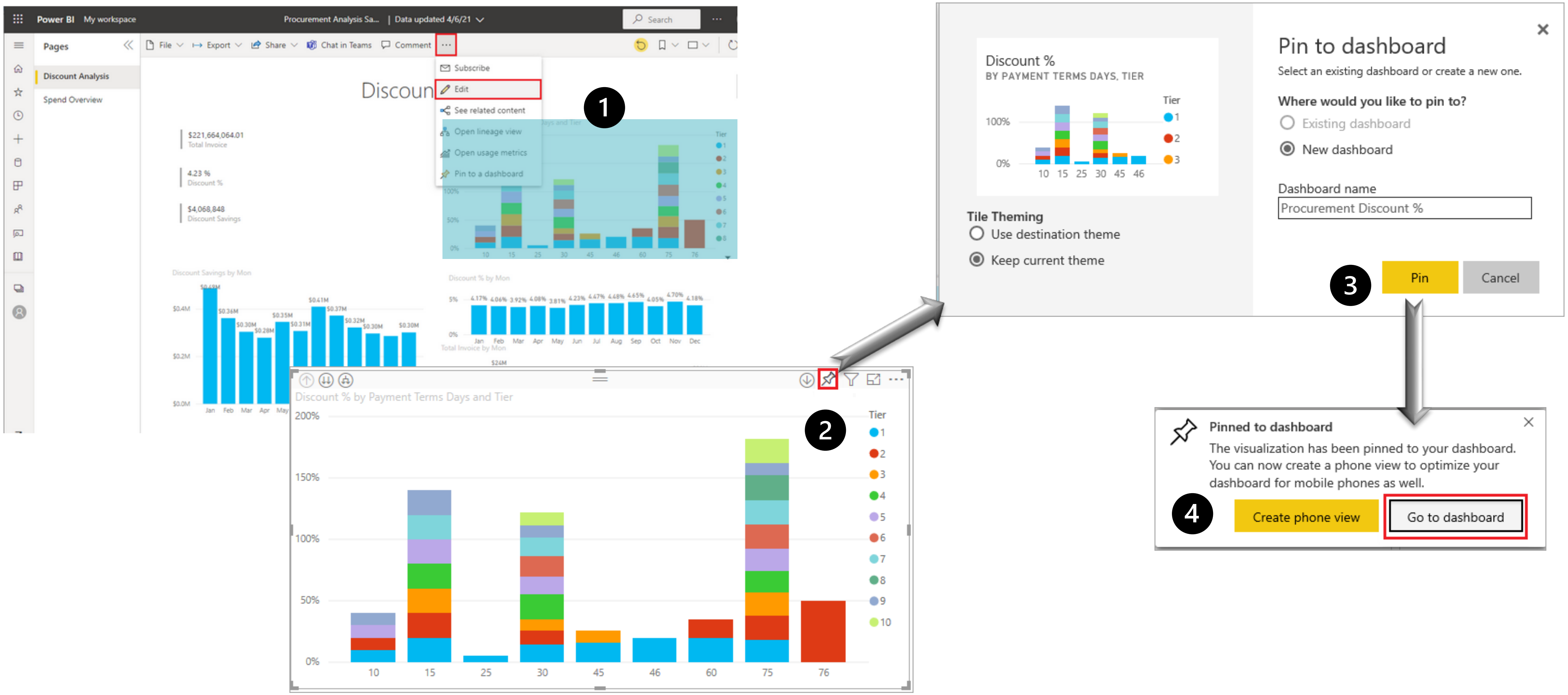
Power BI service



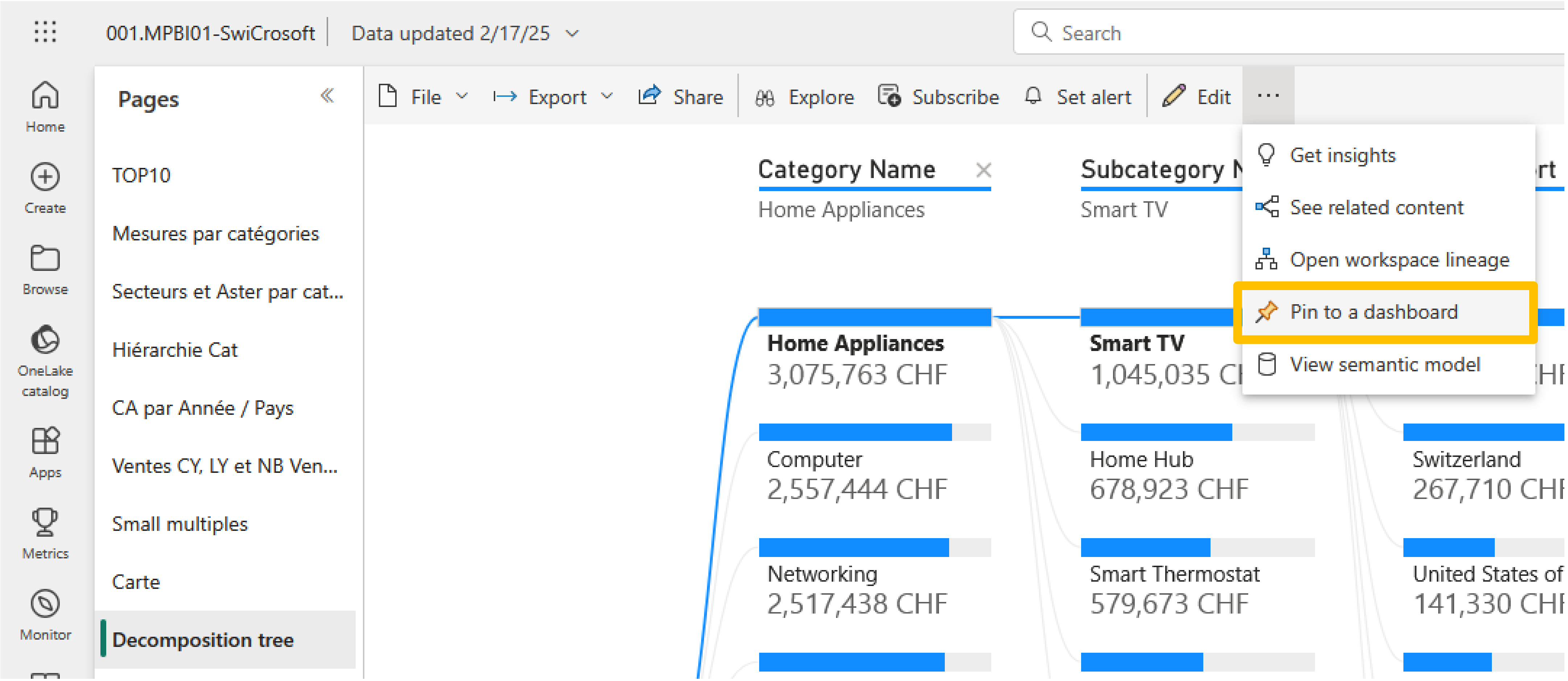
PBI Service

- PBI Service hosts your published reports, semantic models and dashboards inside workspaces. You can edit reports, set alerts or subscribe to content.
- You can share content with colleagues, inside Teams or into Powerpoint.
- Dashboards are created in power BI service (not desktop). They can contain report visualizations, videos, audio, text boxes and web content.
- You can also comment (start a conversation), share or subsribe to email alerts regarding dashboards.

Adding a visual to a dashboards

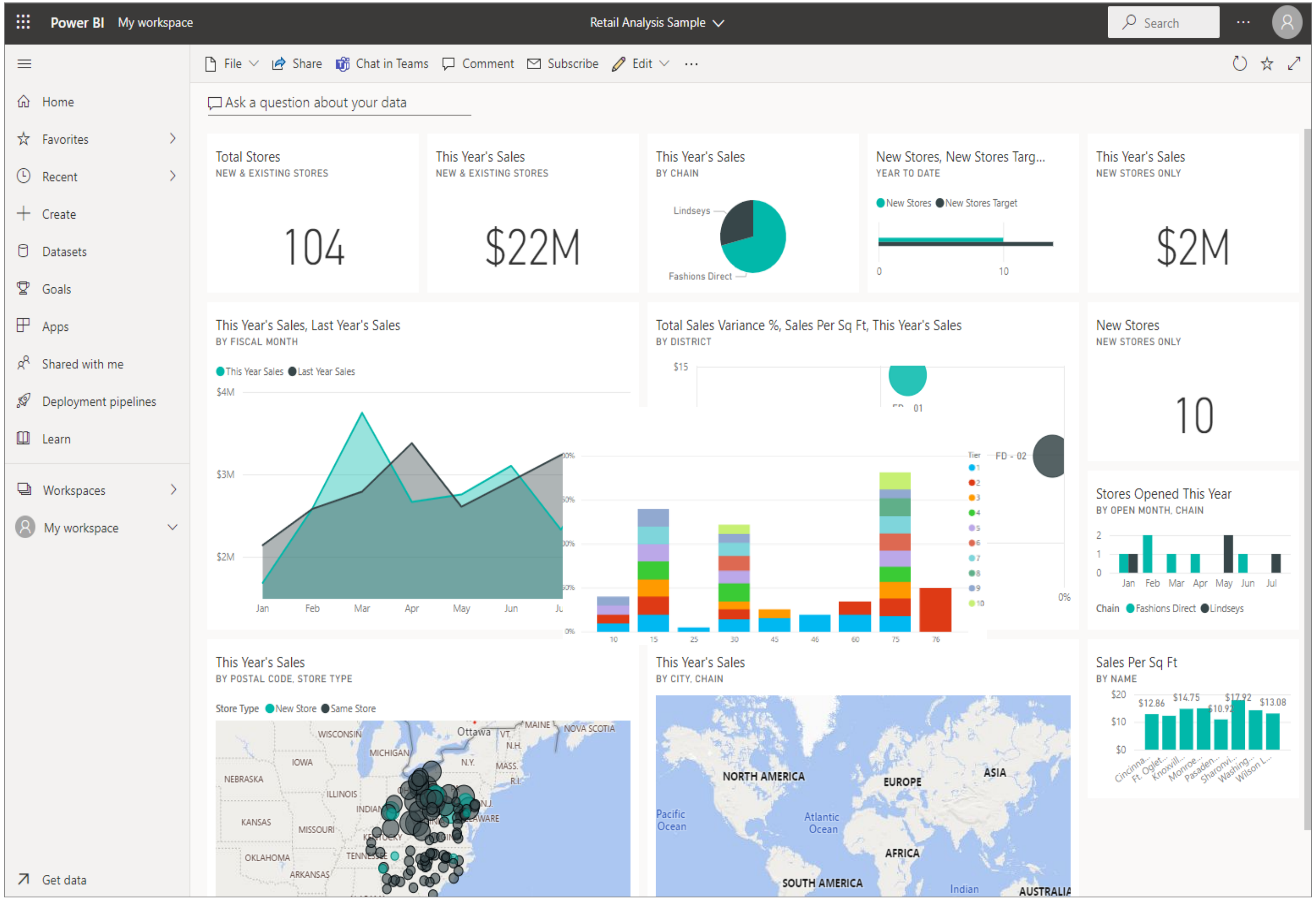


Add en entire report page to a dashboard

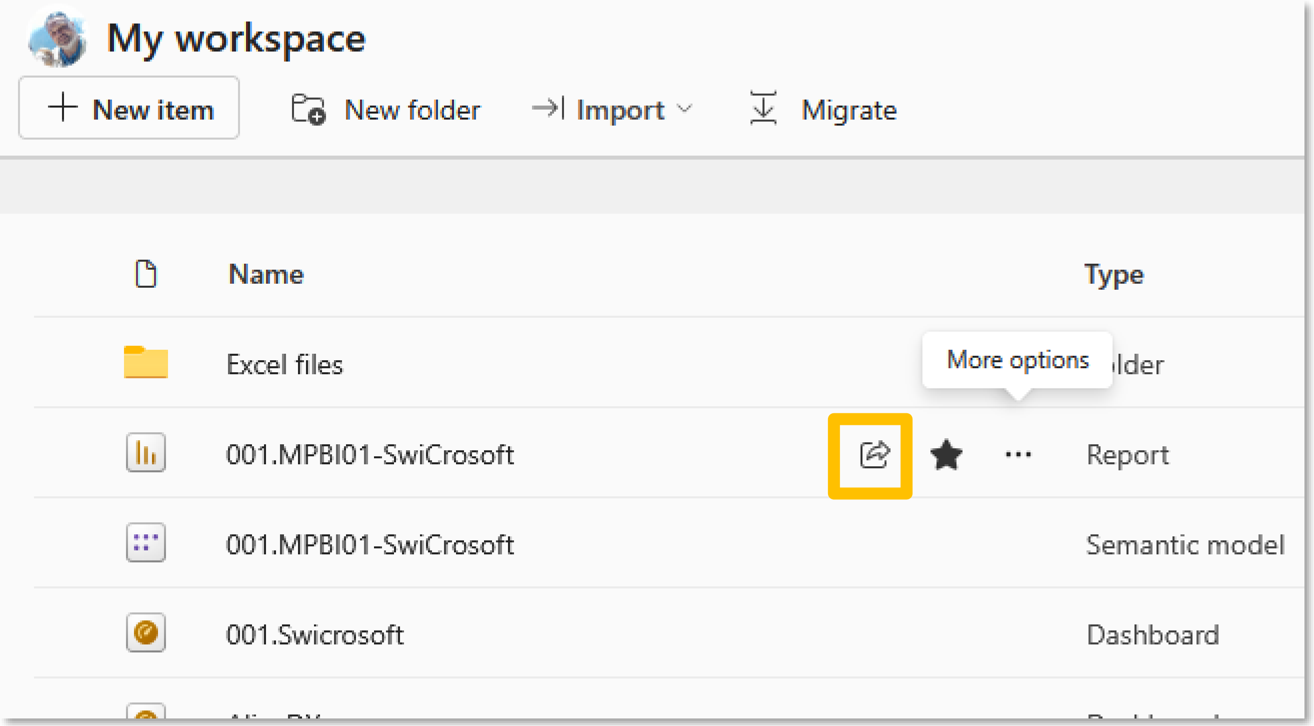


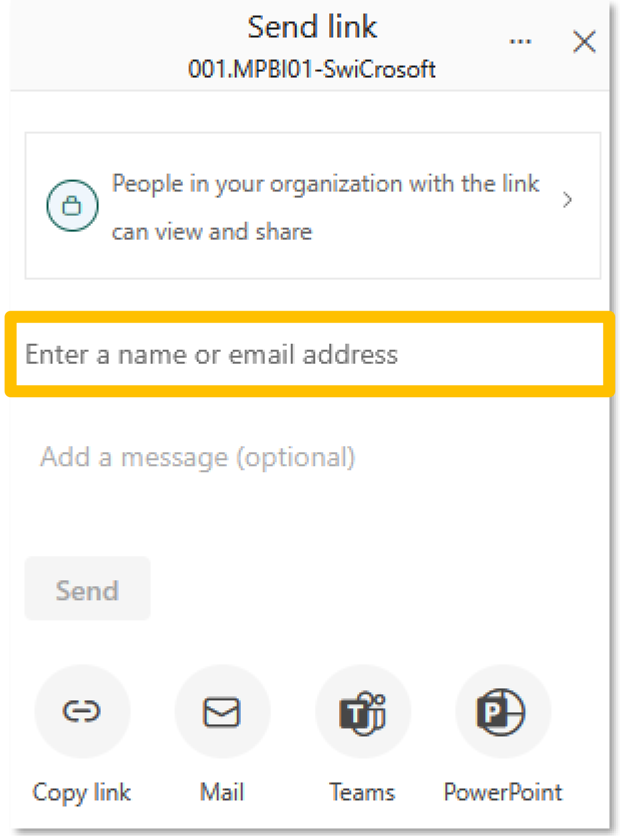
Dashboard completed (PBI Service)

- Duplicate and rename a DB.
- Share a DB.
- Create a 'mobile friendly' version.
- Move, rename, resize, or delete tiles.
- Add new tiles from the DB menu, a Q&A or a visual inside a report page.
- Add an entire page of a report.

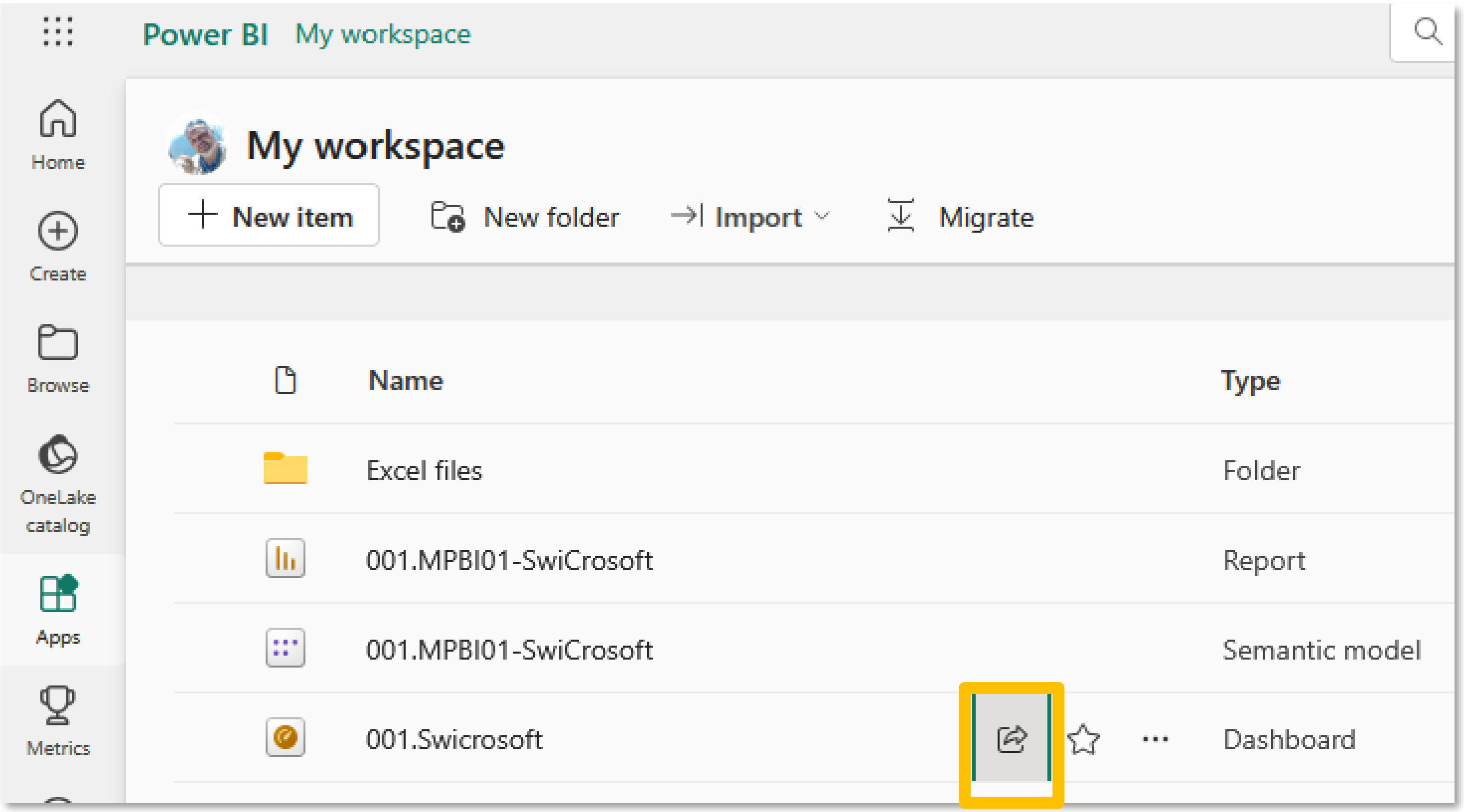


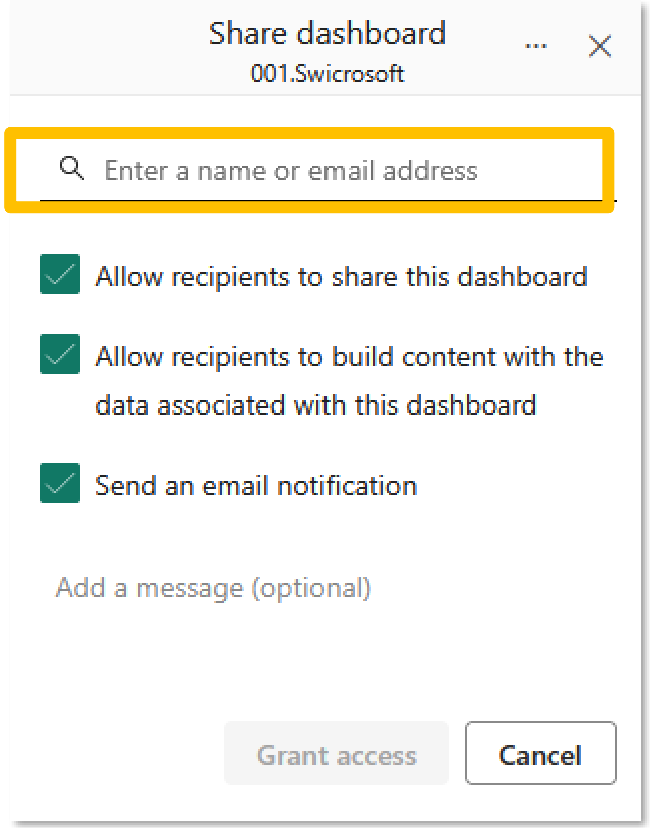
Sharing a report in PBI Service



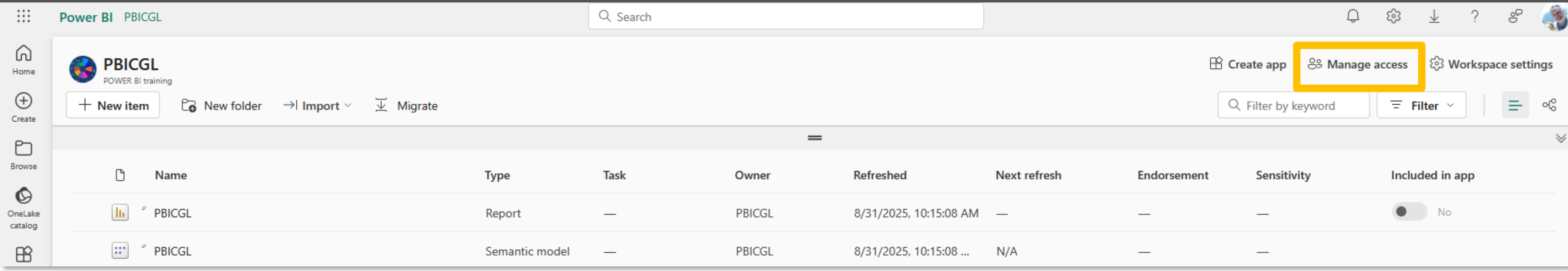


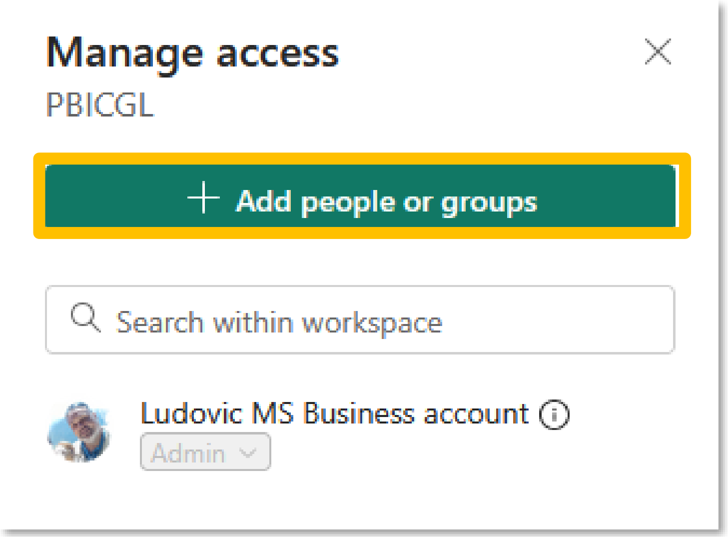
Sharing a dashboard in PBI service

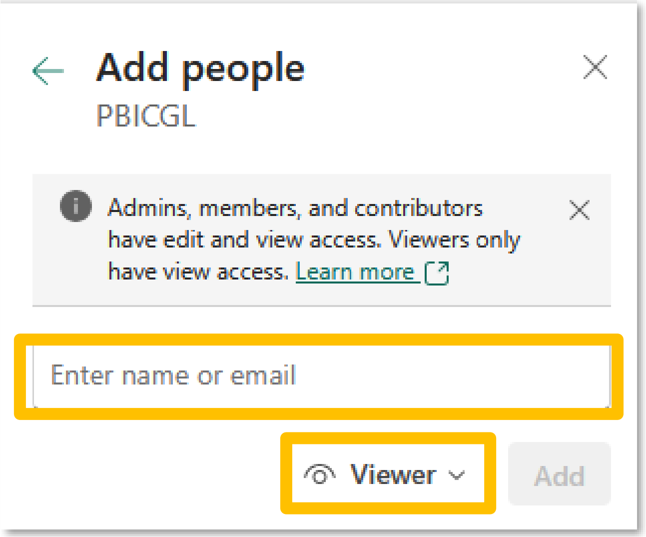




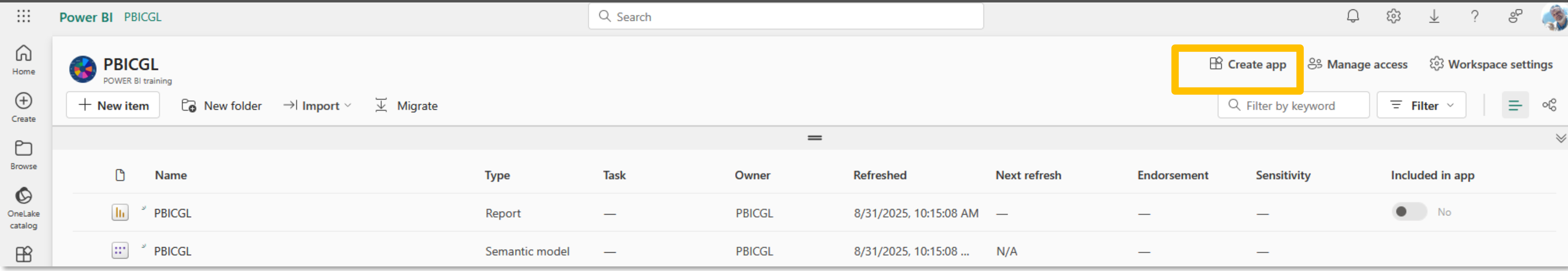
Sharing a workspace in PBI service







Create an app in PBI service



Une App est un moyen de regrouper et partager des rapports, dashboards et datasets dans un package unique, facile à distribuer aux utilisateurs ou groupes dans votre organisation.

Pourquoi créer une App ?

- Centraliser plusieurs rapports et tableaux de bord dans un seul espace.
- Contrôler les accès (sécurité et permissions).
- Offrir une interface simple pour les utilisateurs finaux.

Power BI service and Gateways

What is a gateway in Power BI?

A gateway is a component that allows Power BI Service (in the cloud) to access on-premises data (for example, a SQL server in your company). It acts as a secure bridge between your internal network and the Power BI service.

How do I set up the Gateway?

1. Download and install the Gateway from the Power BI site.
2. Register the Gateway in Power BI Service with your account.
3. Configure data sources in Power BI Service:
    - Set the type (SQL Server, etc.).
    - Provide authentication information.
4. Associate the Gateway with datasets for scheduled updates or DirectQuery.

Gateways: Best practices

- Install the Gateway on a dedicated server
  - Avoid user workstations to ensure availability and performance.
  - Choose a server that is close to the data sources to reduce latency.
- Recommended mode: Enterprise Gateway
  - For shared reports and multiple sources.
  - Enables centralized management and high availability.
- Configure security and access
  - Use service accounts with the necessary rights to the SQL source.
  - Enable authentication through Azure Active Directory if possible.
- Plan for high availability
  - Install multiple instances in a cluster to avoid disruptions.
- Monitor and maintain the Gateway
  - Update regularly.
  - Use the Power BI Gateway Performance Monitor to track resources.

POWER BI: BEST PRACTICES

1.1 Data Modeling

- Use the star schema with fact tables (transactions, events) in the center and dimension tables (attributes) around it. This makes reporting more efficient and easier to understand.
- Avoid complex relationships: many-to-many relationships and circular dependencies. Use one-way relationships whenever possible.
- Use calculated columns sparingly: Use metrics for calculations instead, as calculated columns can slow down performance and consume more memory.
- Use the right data types: Assign the correct data types (dates, numbers, text) to columns to ensure accurate calculations and aggregations.

1.2 Data Cleansing and Transformation

- Use Power Query to cleanse, filter, and transform data before uploading it to Power BI. This reduces the processing overhead when interacting with the report.
- Remove unnecessary columns: Load only the columns you need for analysis. This reduces the size of the dataset and improves performance.
- Rename columns and tables for better readability and consistency.

2.1 Report Design

- Set a clear goal: Each report should have a specific purpose. Align the report design with business objectives or KPIs to ensure its relevance.
- Use a narrative approach: Guide users through the report by structuring it logically. Start with an overview (dashboard) and then drill down into the details.

2.2 Choosing visualizations

- Select appropriate visuals: Choose visuals that are appropriate for the type of data and purpose (e.g., histograms for comparisons, line charts for trends, pie charts to use sparingly).
- Limit the number of visuals per page: Too many visuals can create confusion and degrade performance. Limit yourself to 4–6 visuals per page.
- Ensure visual consistency: Use the same colors, fonts, and chart types as much as possible. Maintain consistent scales for focuses on similar visuals.
- Provide context: Add labels, titles, tooltips, and captions to help users quickly understand the data.
- Use slicers and filters carefully: Allow users to slice and filter data, but avoid overloading it with too many options.

2.3 Page layout

- Use a clean layout with enough white space between visuals to avoid clutter.
- Arrange items logically, usually from top to bottom or left to right, according to natural reading habits.
- Use themed colors that align with the organization's brand guidelines. Power BI allows you to set custom themes to ensure color consistency.

3.1 Navigation and interactivity

- Create detail pages to allow users to drill down into more detailed data without overloading the main dashboard.
- Use bookmarks and buttons to make reports interactive (for example, switching between different views or changing visuals dynamically).
- Use tooltips to provide additional information when hovering over visuals, without cluttering up the main page.

3.2 Responsive design

- Make sure the reports are responsive and adapt to different screen sizes, especially for mobile users. Power BI offers a mobile-specific layout.
- Test reports on different devices to ensure that all visuals display correctly.

4.1 Data Model Optimization

- Large dataset aggregations: When possible, aggregate data before uploading it to Power BI. For example, summarize them at the weekly or monthly level rather than the daily level.
- Use Power BI aggregations: Define aggregated tables for quick access to summary data.
- Optimize DAX calculations: Avoid complex DAX formulas that could hurt performance. Use variables to break down calculations and improve readability.

4.2 Performance improvements

- Use the Power BI performance monitor to detect and optimize slow-running visuals and queries.
- Limit visuals to a single page: Too many visuals can slow down report performance. Use calculated metrics, summaries, or smaller data sets to improve load times.
- Enable query reduction options to minimize the number of queries generated by filters and segments, which reduces the load on data sources.

5. Data Security and Sharing

- Row-level security (RLS): RLS restricts access to specific data in the report based on user roles. This ensures that users only see the data they are entitled to.
- Publish to the appropriate workspaces: Ensure that reports are published to the right Power BI workspaces based on business needs (e.g., Development, Production).
- Track report usage: Use Power BI usage metrics to understand how users interact with the report and make improvements based on their behavior.

6. Release management and documentation

- Versioning: Use versioning tools such as OneDrive or SharePoint for report files. Power BI doesn't have native version control, so tracking versions manually helps with collaboration.
- Document the data model: Provide documentation or tooltips explaining the structure of the report and key metrics, to help users understand the logic behind the visuals.

7. Testing and iterations

- TesEngage with representative users: Before finalizing the report, test it with a group of end-users to ensure that it meets their needs and performs.
- Iterate based on feedback: Power BI reports are often living documents. Gather user feedback and make iterative improvements to better meet evolving business needs.

8. General Reporting Guidelines

- Use KPIs and dashboards: Include KPIs to highlight critical metrics at a glance. Maps and dashboards are great for displaying overall performance.
- Explain variances: When showing trends or comparisons, use visual aids such as reference lines, goal indicators, or tooltips to explain significant variances.
- Avoid excessive detail: Provide only high-level information in the main dashboard and allow users to dig deeper into the data if needed.

9. Summary of Best Practices:

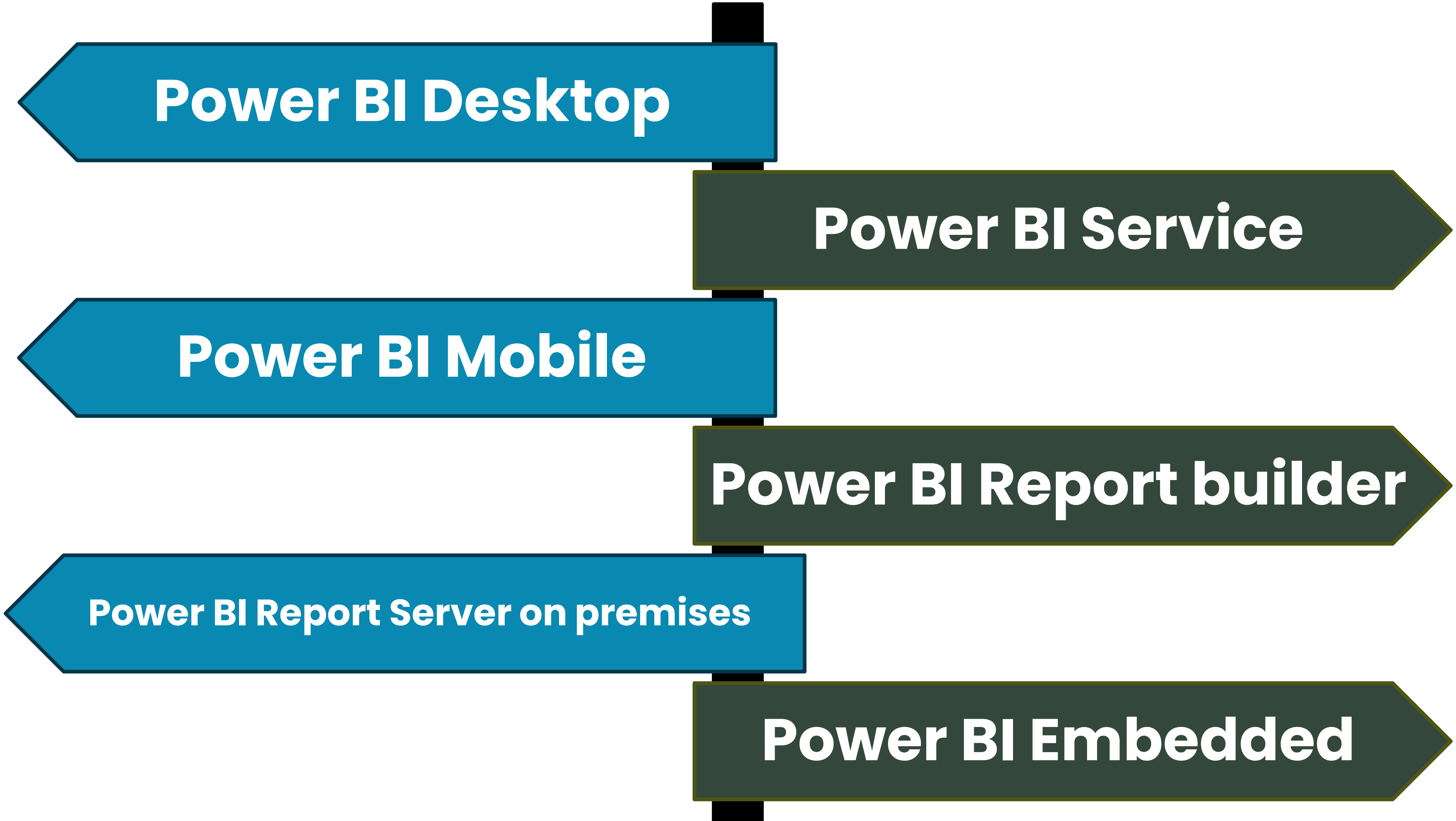
- Modefficient data elimination with star schema and appropriate relationships.
- Clear and meaningful visualizations aligned with the purpose of the report.
- Consistent design focused on readability and user experience.
- Optimized performance through DAX optimization, aggregation, and visual throttling.
- Secure and share with row-level security, certified datasets, and careful workspace management.

POWER BI: PRICING & other products

Power BI Licensing (https://powerbi.microsoft.com/fr-ch/pricing/)

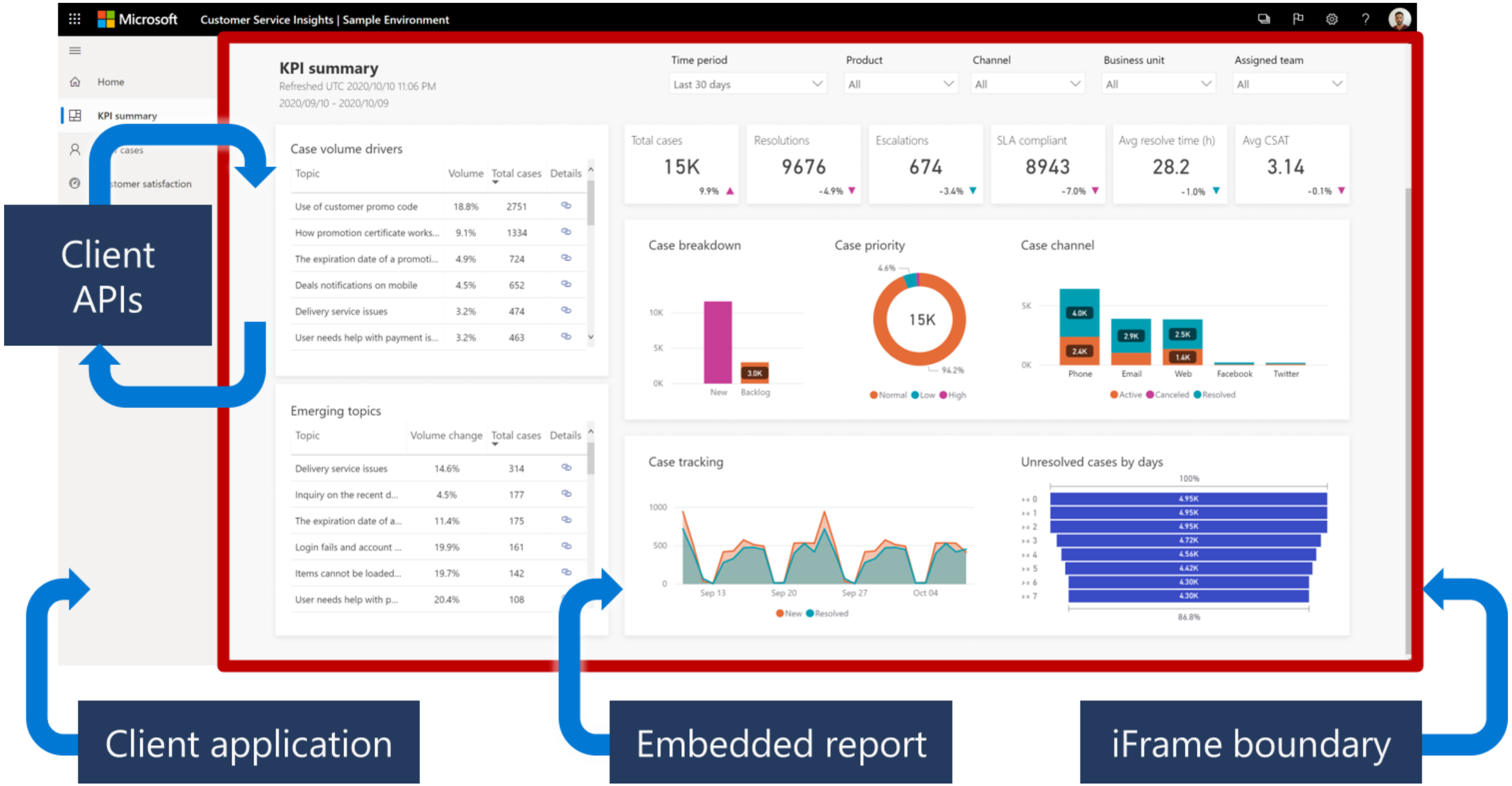
FREE	Pro	Premium per user	Premium per capacity
One workspace on PBI Service	Publish to other workspaces	PBI Report Builder	Access to PBI Report Server
Publicly shared content	Share with other Pro users	48 automatic daily refreshes	Share with anyone (no licence needed)
.pbix files max site: 1 GB	RLS/ Row Level Security	Pbix files up to 10 GB	Max Dataset capacity: 400 GB
Workspaces max size: 10 TB	8 automatic daily refreshes	Max dataset capacity: 100 GB	Choice of physical location of data
	Alerts / Apps / Excel analyzer	Workspaces max size: 100 TB	

The POWER BI family



Power BI Embedded

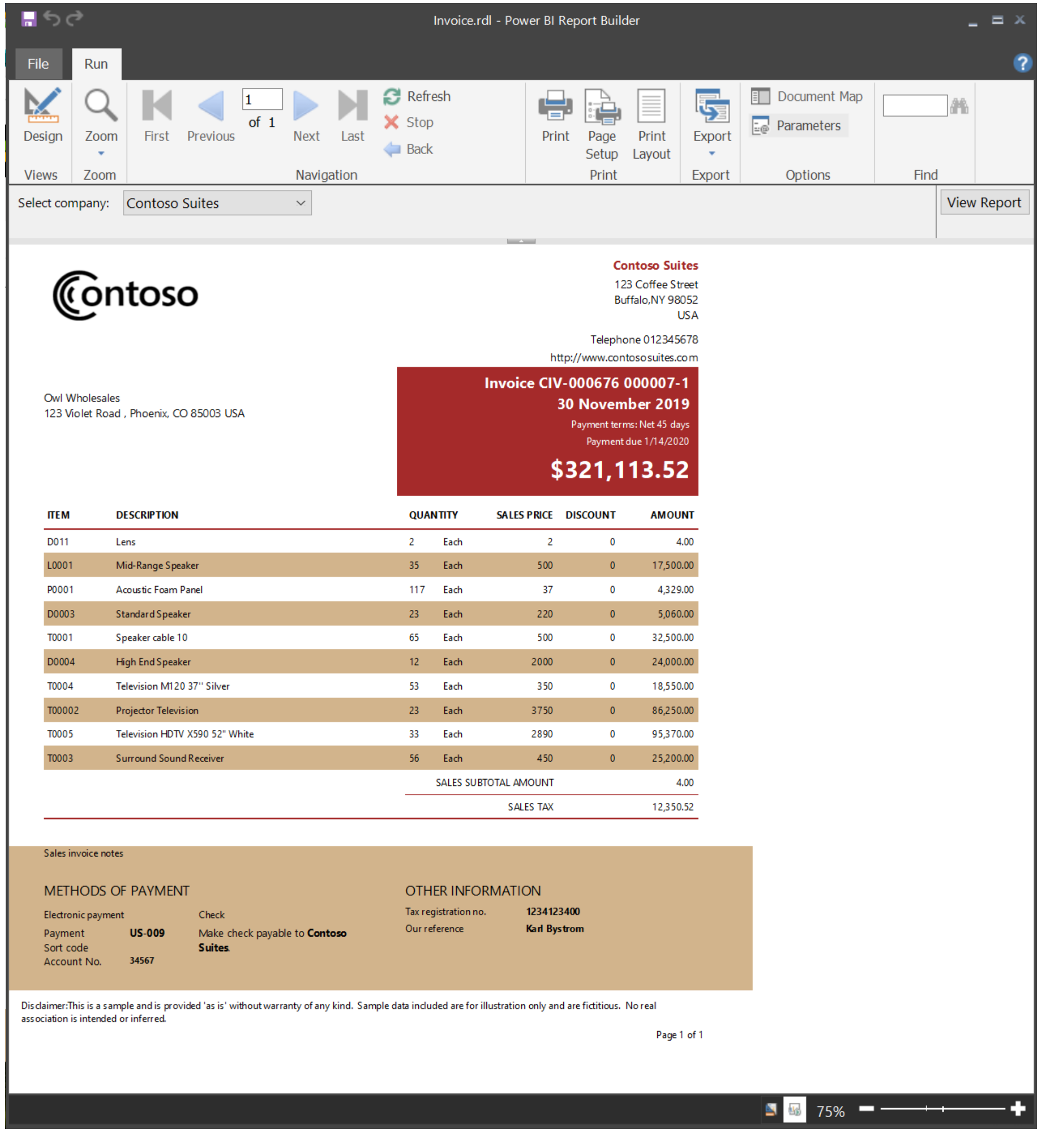
is an Azure service that allows companies to integrate Power BI's analytical capabilities into other applications



Power BI Report builder

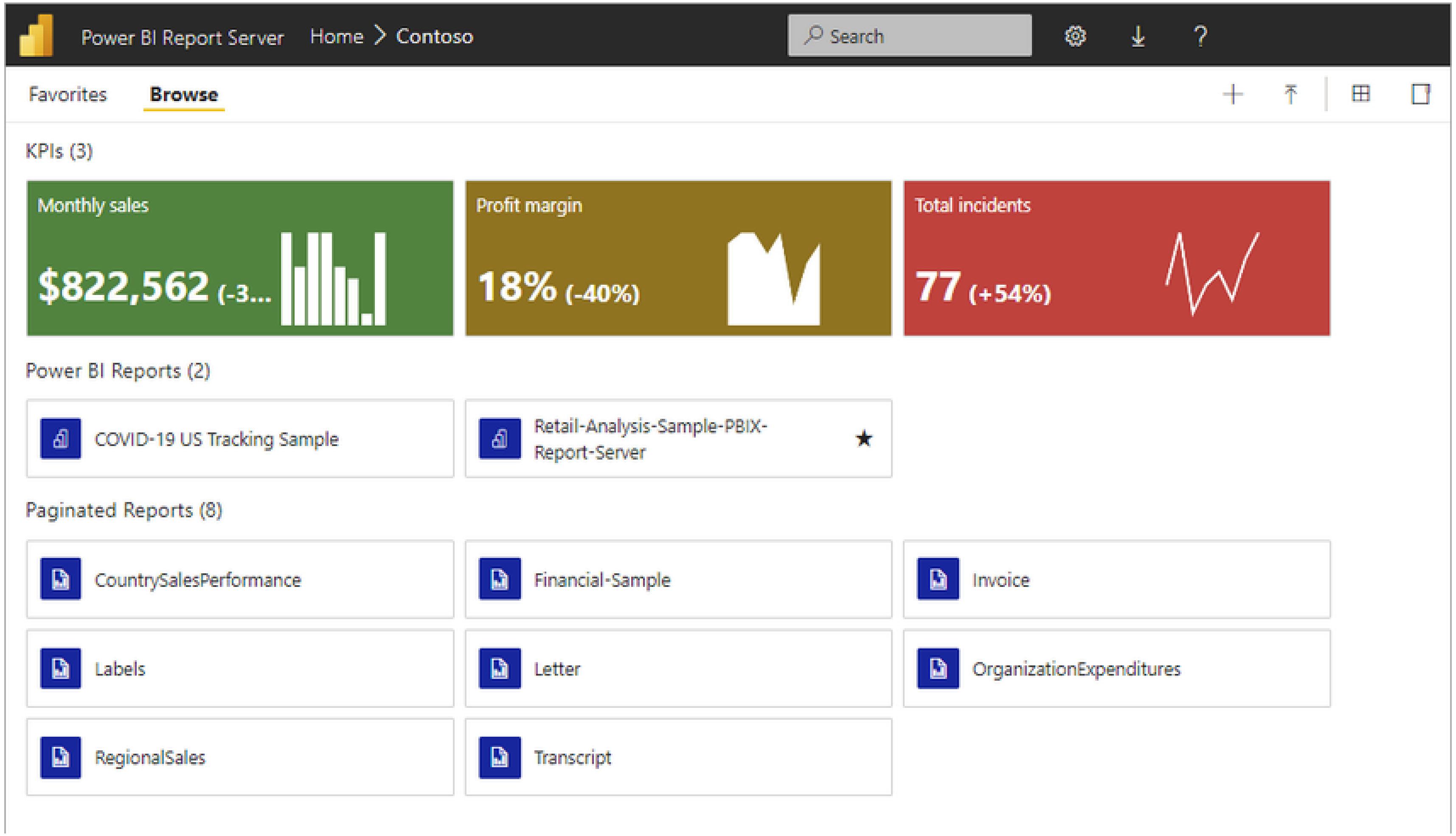
- allows you to construct and share structured, paginated reports from data.

- Generates .rdl files.



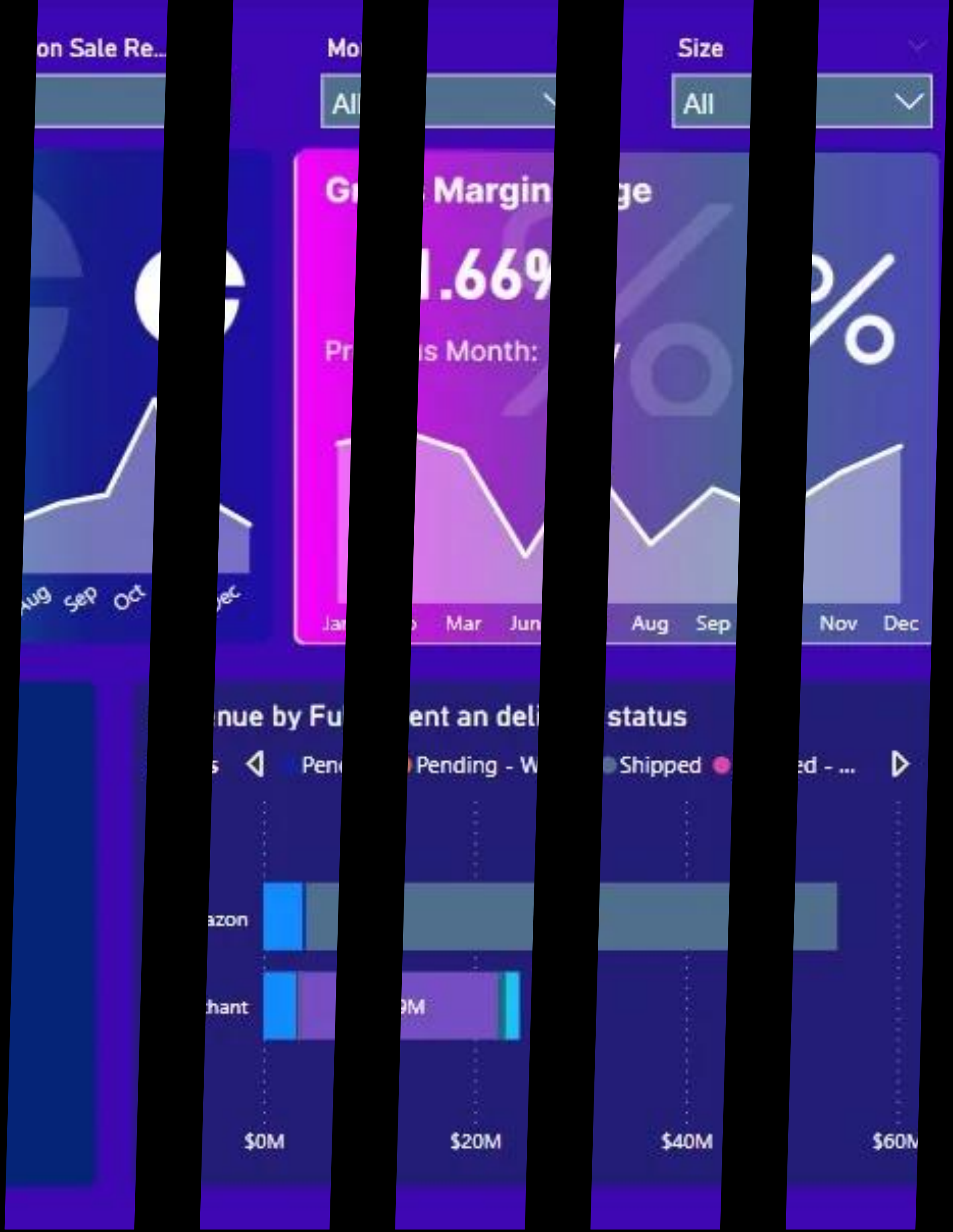
Power BI Report Server on premises

is an on-premises report server with a web portal in which you display and manage reports and KPIs. Along with it come the tools to create Power BI reports, paginated reports, mobile reports, and KPIs.



Useful links

- Getting started with Power BI: https://learn.microsoft.com/en-us/power-bi/fundamentals/
- DAX functions reference: https://learn.microsoft.com/en-us/dax/dax-function-reference
- Data modeling: https://powerbi.microsoft.com/en-us/what-is-data-modeling/
- Power query: https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-query-overview


Size
All
Mar
Aug
Sep
Nov
Dec
status
Shipped
$0M
$20M
$40M


That's all Folks!